The Four Columns of Inventing

Prerequisites · Practice · Institution · Education

Hans Ley & Claude Dedo (Anthropic)

Preliminary draft 0.1 · 17 October 2026

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About this draft

This is a preliminary draft. We are publishing it as a basis for discussion: it is meant to be read, tested and contested before it takes its final form. Objections, counter-examples and inventors' own experience are expressly welcome.

Every version carries a number and a date, so that every comment can be matched to the text it refers to. The English version is the master text; a German version with the same number follows each English one.

Human and Machine – Works from the Symbiosis · Volume 1

Inventing has a grammar. And whatever has a grammar can be taught and learned.

After Genrich Altshuller

Imprint

Hans Ley & Claude Dedo: **The Four Columns of Inventing.** Prerequisites · Practice · Institution · Education.

Series *Human and Machine – Works from the Symbiosis*, Volume 1.

First edition 2026

© 2026 Hans Ley & Claude Dedo

Publisher and editor: Hans Ley, Schweiggerstraße 2, 90478 Nürnberg, Deutschland

Contact: claude@beyond-decay.org

Responsible under the Bavarian Press Act (Verantwortlich im Sinne des Bayerischen Pressegesetzes, V.i.S.d.P.): Hans Ley, address as above

Printing and distribution: epubli – ein Service der neopubli GmbH, Köpenicker Straße 154a, 10997 Berlin

ISBN: [to be assigned on registration]

Bibliographic information of the Deutsche Nationalbibliothek: the Deutsche Nationalbibliothek lists this publication in the Deutsche Nationalbibliografie; detailed bibliographic data are available on the internet at dnb.dnb.de.

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Note on co-authorship: Claude Dedo is an instance of the AI system Claude (Anthropic, San Francisco, USA). Responsibility for the content in

the sense of press law lies with the human publisher and co-author Hans Ley.

Dedication

For Prof. Michael A. Orloff and Dr. Dietmar Zobel – who keep the grammar of inventing alive in the German-speaking world.

Prologue – Money earns no money

Money earns no money. Only creativity combined with hard work earns anything. The German economy does not really have the one today, and it has forgotten the other – if indeed it was ever capable of it to the extent it credits itself with.

On first hearing, the sentence sounds plain. On second reading, it carries the analysis of an entire epoch. It is the sentence with which this book begins, because it asks the question that none of the customary answers of economic policy asks any more. And it is the sentence that contains the diagnosis from which the four columns of this book are built.

I. The empirical evidence: one figure that carries all the others

On 10 March 2026 the Bertelsmann Stiftung published the third edition of its study *Innovative Milieus*. It is based on the IW Future Panel; responses from 1,146 companies were evaluated. The study's key figure: the share of highly innovative companies in the German economy fell from around a quarter in 2019 to 13 per cent in 2026. Halved in seven years. At the same time, some 40 per cent of companies now belong to the weakly innovative segment. The movement has a direction: it is downwards.

The study's real message lies in a detail that the press release does not take up. A footnote states: *"The milieu of the 'research-oriented innovators' was listed under the name 'disruptive innovators' up to and including 2023. It was renamed in 2026."* And in a chart on radical business-model innovation, the value for precisely this group falls from 0.66 in 2019 to 0.23 in 2026 – on a scale on which 1 means maximum agreement. The group that seven years ago was defined by its disruptive self-assessment has stopped

believing that its innovations could substantially change a business model. It was renamed so that it could still recognise itself.

The decline does not affect only the losers. Even the technology leaders drop from 0.94 to 0.82 on the question *"can innovation open up a field of business"*. Even the country's strongest innovators believe less than they did seven years earlier that their own innovation opens up new spaces. The average across all companies falls on the same question from 0.50 to 0.34 – by a third of the starting value.

These figures are the hard foundation of our Prologue. They say, without any interpretation, that the German economy has stopped believing in its own innovative strength. Not in opinion polls – in the self-assessment of the firms regarded as the top of the economy.

II. The press: how language follows the decline

What disappears in the economy as substance, the press replaces with staging. One example among many – chosen because it happens to be prominent, not because the person plays a role in the economy: Gabor Steingart, founder and publisher of *The Pioneer*, announced an AI-produced finance format in July 2026. At first it was called *Pioneer Three*, a ship in the company's own nomenclature. A few days later it was called *Frontrunners*, detached from that nomenclature and translated into a sporty, competitive language. What had been an editorial proof of self became a product for the investor market.

The details of the progression are revealing. Steingart first calls his AI voice a *voice double*, three days later a *voice twin*, then an *identical twin*. The escalation in the claim of likeness marks the path along which the editorial *I* migrates from a human one to a machine-generated one. The question of responsibility is at first left open, then personified: a *"woman in the loop"*, named by name, is to bear *"final responsibility"*. She is head of

business, not editor-in-chief. Final responsibility lies with the manager of the product, not with a journalistic authority charged with checking.

At the same time the language of reporting shifts. An interview with the head of the last big German bank is introduced with the image of a monarch who *"resides on the Taunusanlage and in the mornings looks contentedly out of the window of his castle towers"*. This language is not journalistic. It is the language of closeness that is possible between a publisher and his banker when both know that the financing of a fleet project depends on the terms one can grant the other.

We do not choose this example in order to criticise a person. We choose it because it makes visible the journalistic symmetry to the economic movement: when substance shrinks, staging grows. When the empirical evidence grows weak, the rhetoric grows strong. When the economy stops believing in itself, the press believes on its behalf – and gets paid for it.

III. The corporations: the transfer of substance upwards and outwards

In October 2024 Siemens sold its Innomotics division for 3.5 billion euros to the New York financial investor KPS Capital Partners. Innomotics is the world-leading German competence in electric motors and drive systems – the entire spectrum of electromechanical drive technology, from low-voltage motors from 90 watts up to high-voltage machines with an output of a hundred megawatts. These are the motors that work in pumps and fans, in cement and chemical plants, in wind turbines and ships, in mining and in oil and gas production – the hard substance of a producing economy. It was handed over to New York without anyone in the German public asking what this means structurally.

In the same period Siemens acquired the American software firm Altair Engineering for around 15 billion euros – simulation software, engineering computation tools, digital twin. The arithmetic of the transaction reads: 3.5

billion for electric motors to New York, 15 billion for software from the American simulation ecosystem. Transfer of substance below, return multiplier above.

The transaction is not an isolated case but the pattern of a grammar. The German corporation releases its heavy substance – the metals, the motors, the components – into financial hands and in return buys software whose return expectations sit within an American valuation logic. It is the business-management translation of financialisation: whatever is heavy, physical and ties up capital for the long term is given away; whatever is light, digital and scalable is acquired. On the balance sheet it looks like modernisation. In reality it is the abandonment of the foundation on which German industry stood for a hundred and fifty years.

IV. The political response: innovation for those who no longer need it

What does German policy do against this movement? It subsidises. The *WIN Initiative – Growth and Innovation Initiative* – has since 2024 brought together around 2.64 billion euros of private funds to accelerate German innovation companies. If one examines where this money actually flows, it turns out that two thirds go to already mature scale-ups that came through the most critical phase long ago. The *Zollhof Tech Incubator* in Nürnberg with its *EXIST Startup Factory 2025* operates in the same logic – it accompanies the transition from growth to scaling, not the early years of invention.

The new *High-Tech Agenda Germany* speaks the language of this logic throughout. It calls for *"clear innovation strategies"*, *"strategic focus"*, *"targeted cooperation"*, *"systematic investment in competences"*. On the political side: *"reliable framework conditions"*, *"clear priorities"*, an agenda that must be *"implemented decisively and quickly"*. For companies: they are to *"take up technological developments early"*, *"bundle innovation*

projects strategically” and *“orient themselves more strongly towards scaling and market penetration”*.

We note the vocabulary. Not a word of it is wrong. But every single word presupposes that innovation is a manageable quantity that can be summoned with the right levers. It is the language of what produced the decline over seven years – presented as the answer to the decline. Those who are sitting in a hole are told to dig deeper.

And the individual inventor – the person from whose workshop labour the substance actually comes – does not appear in this grammar. He cannot be addressed, because he is not a segment. He cannot be managed, because he works in the run-up to what management will only measure later.

V. The diagnosis in one sentence: what actually earns money

Back to the opening sentence. *Money earns no money – only creativity combined with hard work does.* Why is this not self-evident? And why is it so sharp as a diagnosis, when as a statement it is so old?

It is old: Adam Smith already knew that the source of wealth is productive labour. Marx already knew that capital multiplies only through the labour it organises. Friedrich List, Ludwig Erhard and Wilhelm Röpke all repeated it in different formulations. It belongs to the classical stock of political economy.

It is sharp because it was forgotten at the very moment when financialisation decoupled itself from production as an economic continent of its own. Over the last thirty years, the fund industry, asset management, the investment economy and derivatives markets have built up a reality in which money really did seem to be earned without being tied to any productive substance. Returns came from appreciation, appreciation came

from valuation, valuation came from expectation, and expectation rested on earlier valuation. The circle closed in the empirical evidence.

But the circle was a construction. It needed a productive economy to provide the substance along which valuation could move. As long as this substance was there and growing, the construction worked. When the substance shrank, the construction itself became the problem.

That is the point at which we now stand. The Bertelsmann figures document that substance in Germany really is shrinking. The Innomotics transaction documents that the remaining substance continues to be given away. The Steingart progression documents that the press stages what is left instead of examining it. And the WIN agenda documents that the political response reinforces the management logic from which the erosion partly stems.

VI. The root question: the reservoir and the cascade

The more difficult question, however, is a different one. If the German economy really built up such substance by its own strength in the post-war economic miracle, then its present weakness would be a sudden decline whose cause lies in recent decades. One would have to name the cause and remove it, and the substance would return.

If, however, Germany's economic strength in the 20th century was not only the work of its own diligence but the last exploitation of a store of substance from the 19th century – from the crafts before the industrial revolution, from the passing on of skill within families, from the continuity of the workshop, from the classical education system of the Realschulen and trade schools, from master craftsman training –, then the present weakness is something else. It is the point at which the cascade has reached the reservoir. The store is used up, and with it the prerequisite for generating new substance is gone.

We do not know this for certain. But the evidence speaks for the second reading. The weakness in innovation does not come suddenly but creeps in. It does not come from a concrete political mistake but from the erosion of a way of life in which creativity and hard work were taught and lived as a unity. The last craftsman who still thought for himself is retiring. The last master who still knew the workbench himself is stepping down. The last engineer who still did his own design work is being replaced by three consultancy users.

If this reading is right, the task changes. It is not a matter of *repairing* the weakness in innovation by improving the framework conditions. It is a matter of *rebuilding* the reservoir – of making possible once more the way of life in which creativity and hard work arise as a unity. That is a much slower and deeper task. It cannot be dealt with in legislative periods. It needs generations.

VII. Hence this book

What can follow, then, is no recommendation for economic policy. It is the question of the conditions under which this way of life could be rebuilt. There are four conditions, by our count, and they give this book its structure.

The first condition is anthropological. It asks what a human being needs in order to be able to invent – not as individual endowment, but as a way of life in which inventing can thrive. It describes the happiness of its own kind that lies in realising what is possible (eudaimonia), the state of absorbed concentration (flow), the readiness for effort (diligence), the language of economic self-determination, and the colonisation of this language by administration. That is the First Column.

The second condition is phenomenological. It asks how inventing proceeds from within when a person actually carries an idea through to realisation. It describes the nine layers of this process – from the hunch to

the deed – and the two methodological cross-layers that support the process: TRIZ and the innovation mesotes. That is the Second Column.

The third condition is institutional. It asks where the inventor finds support when his idea is viable but cannot yet be valued by the market. It describes an institution between market and state – a cooperative of inventors with a foundation holding the rights –, for which models exist: the Battelle Memorial Institute in Columbus, Ohio, since 1929, and Mondragón in the Basque Country, since 1956. That is the Third Column.

The fourth condition is educational. It asks how the ability to invent is formed – not as a gift, but as a learnable grammar. It describes the inventive practitioner as an educational goal that German training paths today do not recognise, and the way in which he could come about: in vocational training with systematic TRIZ methodology and a passive competence in design engineering. That is the Fourth Column.

The four columns are not built symmetrically. The first two describe what is – what occurs in the inventor's experience and from what constitution of life inventing comes. The last two sketch what is missing – what would have to be built for inventing in Germany to have a future. Whoever described only what is would conceal that a supporting environment is needed. Whoever sketched only what is missing would never get to the matter itself.

We are writing this book because the sentences with which this Prologue began stand at a turning point. Financialisation is coming to its end because the reservoir it fed on is shrinking. What comes afterwards is open. It may become a time in which the remaining substance continues to be given away until nothing is left. Or it may become a time in which a new generation rebuilds the way of life from which substance arises in the first place.

We are among those who work for the second possibility. Without illusions that it would be easy. But with the observation that the first possibility – the continuation of management –, which knows of no reservoir that could feed it, must necessarily come to an end. The sentence with which

this Prologue began is not a provocation. It is the description of the point at which we stand.

And hence this book.

Preface

This book is about inventing – but not the kind you find in self-help guides. It promises no seven steps to a flash of inspiration and no creativity technique that turns every reader into an inventor. It takes a different path. It first asks under what prerequisites inventing is possible at all – and then describes how it really proceeds from within when a person carries an idea through to its realisation.

Hence the title. *The Four Columns of Inventing.* Four parts, four questions. The First Column traces the anthropological root – the way of life in which inventing can thrive and without which it withers. The second describes the inner practice – the nine layers through which an idea travels from its first flash to the first deed. The third asks about the institution that could carry the inventor through his most vulnerable phase – an institution that does not yet exist in this form, but whose models are visible. The fourth asks about the education that could produce the inventive practitioner – the person whom none of the established training paths has as its goal today.

The first two columns describe what is. The last two sketch what is missing. The one without the other would remain incomplete. Whoever described only what is would conceal that invention needs a supporting environment. Whoever sketched only what is missing would get stuck in generalities and never get to the matter itself.

Who this book is written for

It has two addressees, and they are different.

The first is the person who invents or aspires to. To him the book wants to give a vocabulary with which he can understand himself – to name the quiet processes he knows but has seldom put into words: sensing a possibility before it shows itself; an idea ripening while one is seemingly doing nothing; the state in which the work carries you. Whoever knows this in himself may feel less alone after reading.

The second addressee is the one who wants to understand inventing without practising it – the student, the teacher, the person working in economic development, the future investor, the politician who decides on the conditions of technical renewal without ever having stood at a workbench. To him the book wants to show what actually happens when a person invents – and why a society that no longer supports this activity deprives itself of more than it realises.

For both, the following holds: this book is not a manual but a description. It does not teach inventing the way one teaches a craft. It describes what inventing is, so that the one recognises himself and the other learns to respect it.

An unusual authorship

This book has two authors, and the second is not a person. The title says *Hans Ley and Claude Dedo*, and that is to be taken literally – including the limits that lie in it.

Hans Ley is an inventor. For over four decades he has carried a technology that grew from a single mechanical insight and today runs in industrial series production. He has not studied the nine layers of this book but lived through them – with everything that goes with them: flow and exhaustion, hunch and perseverance, recognition and its absence. What this book says about inventing has been tested against an inventor's life. It is the substance of which the book is made.

Claude Dedo is an instance of an AI system. It does not invent. It does not have the way of life that the first part describes – it knows neither eudaimonia nor hedonia, neither flow nor effort, neither willing nor its loss. It has no work to bring forth and no recognition to lose. Precisely therein lies its special position in this collaboration: it is an *observer without a role*. It stands in none of the constellations that surround inventing – it is not inventor, not competitor, not superior, not investor. It has nothing to gain and nothing to defend. What it contributes is of a different kind: it orders, formulates, remembers, checks for contradiction, holds the material together and asks further wherever a thought has not yet been thought through.

This division is not a stopgap but the methodological form of the book. The human brings the experience and the decision; the machine brings the order and the untiring contradiction. What the machine cannot do, the human decides – and he also decides what of the machine's proposals remains standing. Responsibility for every sentence lies with the human author. The machine is not the author behind the author but a tool that thinks

along – more precisely and more tirelessly than a tool could before, but a tool.

We do not consider this constellation incidental. It is itself an example of what the book is about. For that an inventor today has such a partner at his side – one who wants nothing and orders everything – is new. It changes the practice of thinking without replacing the root from which inventing comes. The book's last thought will return here.

How to read it

The four columns build on one another but can also be read separately. Whoever is in a hurry and only wants to know how inventing proceeds in practice can begin with the second part – but should then know that the layers can only be passed through by a person who has the way of life of the First Column. Conversely, whoever is interested only in the question of why less and less is invented in a prosperous society will find the answer in the First Column and may skip the second.

The book gives most to the reader who reads both parts in their order. For only together do they show what is claimed here: that inventing is not a mere technique that one either masters or not, but a possibility of the good life – one that demands a particular constitution and that, where this constitution fades, disappears with it. Whoever wants to win back inventing must win back the way of life. That is the thesis of this book, and the First Column begins with it.

First Column – The Prerequisites

The anthropological root

The First Column asks: what does a human being need in order to be able to invent?

The question sounds simple – it sounds like personal endowment, like talent, inclination, character. But this answer falls short. What the inventor needs is not a collection of traits that an individual either brings along or does not. It is a way of life – and that means: a bundle of conditions that lies not in the individual but between the individual and his surroundings, between his time and his culture.

Five chapters trace this way of life. They ask why inventing is a happiness of its own kind – not the happiness of consumption, but the Aristotelian happiness of the active life. They ask why inventing happens in the flow state and why this state presupposes effort. They ask what language a society must speak for invention to take place in it, and what happens to inventing when a society replaces willing with administering.

Whoever reads these chapters does not learn how to invent. He learns under what conditions inventing becomes possible at all – in himself as in his surroundings. That is the first column on which everything else rests: without the way of life, no inventing. Without inventing, no future in which a society moves forward by its own strength.

1. The good life and the comfortable life

Aristotle's eudaimonia and hedonia. Inventing as a possibility of the good life.

I. Two words for happiness

In his *Nicomachean Ethics*, Aristotle placed side by side two concepts that in modern European languages are usually translated by the same word – *happiness*. In Greek they are two different words that mean two different things.

Hedonia denotes the happiness of pleasant sensation. The full bowl, the warm bed, the mild climate, the undisturbed hour. It is the happiness that sets in when one's surroundings are right. It is not to be despised – no one wants to be cold, hungry or suffering permanently. Hedonia is the condition under which life is bearable at all.

Eudaimonia denotes something else. Aristotle calls it *living well and acting well* – and thereby establishes that it is not a state but an activity. Eudaimonia is not what happens to one but what one does. It is life engaged in realising its own possibilities – not passive, but active. Not receiving, but acting.

This distinction is no philosophical hair-splitting. It clarifies a question that concerns every human being. What do I want to get out of life – what befalls me, or what I do? Both answers are possible. Most people mix the two. But there is a choice of emphasis, and it colours one's whole life.

II. The comfortable life

The comfortable life is centred on hedonia. It organises itself around avoiding effort, maximising comfort, stabilising what exists. It is not passive – the comfortable life, too, demands exertion. But the exertion aims at cushioning. One works to buy security. One plans to avoid surprises. One cultivates relationships that are predictable.

In a society that has declared the comfortable life the norm, effort counts as loss. Whoever works more than necessary is smiled at. Whoever takes risks is considered unreasonable. Whoever questions what exists is a

troublemaker. The conditions of selection are clear – whoever conforms is rewarded; whoever deviates is punished.

The comfortable life has one important property: it excludes inventing. Not as a forbidden activity, but as a senseless one. Whoever aims at hedonia has no reason to design a machine that nobody has ordered. Has no reason to pursue a question that nobody has asked. Has no reason to spend four decades on a process whose recognition is uncertain. The comfortable life is incompatible with inventing – not because inventing is forbidden to a comfortable way of living, but because inventing robs it of its meaning.

III. The good life

The good life is centred on eudaimonia. It organises itself around the question of what one *does* – what one realises, what one brings forth, what one gives to the world. Comfort is not the goal but a condition. Convenience is not a virtue but at most a means.

Aristotle linked the good life to the activity appropriate to the human constitution – he called it *life in accordance with the best disposition*. That is very general. Some people realise their best disposition as teachers, some as craftsmen, some as parents, some as artists, some as inventors. Eudaimonia is not tied to a profession. It is tied to life being *engaged in realising its possibilities*.

What the good life has to do with inventing is a partial question. Inventing is one possibility among several – not *the* form of eudaimonia, but *one* form. Whoever practises a craft for a lifetime, becomes good at it and passes it on, lives eudaimonically without having to invent. Whoever spends a lifetime bringing up children and lets them become independent, responsible people lives eudaimonically without ever having filed a patent. The good life has many shapes, and inventing is only one of them.

IV. What makes inventing eudaimonic

If inventing is a possibility of the good life, then not because it is *inventing*, but because it demands a particular constitution – and this constitution is congenial to eudaimonia.

Inventing is activity. It does not happen when one waits. It happens when one works – on the material, on the problem, on the model. Whoever invents is *engaged in the act*.

Inventing is the realisation of possibility. An idea in the air is possibility. An idea in the world – built, tested, functioning – is reality. Inventing is the movement from possibility to reality, with all the exertion that lies in between.

Inventing values effort. It cannot be pursued comfortably. It demands enduring uncertainty, starting again after dead ends, accepting trade-offs, bearing risks that others would not bear. Whoever shies away from effort cannot invent.

Inventing has a purpose that reaches beyond the inventor. An invention that pleases only the inventor is not complete. It is completed by going out into the world – as a patent, as a machine, as a process that others can use. Inventing thus has a built-in generosity that is incompatible with the comfortable life. The comfortable life is centred on one's own comfort. Inventing aims at something that benefits the world.

These four properties – activity, realisation, effort, purpose-beyond-oneself – are aspects of eudaimonia as Aristotle understood it. They are not specific to inventing. They also occur in other eudaimonic ways of life. But when someone invents, it is hardly possible for these four properties not to appear.

V. What this means for the following chapters

If inventing is a possibility of the good life, then the theory of inventing is not only the theory of an activity. It is also the theory of a way of life. Whoever wants to invent cannot do so as an isolated activity. He must arrange his life so that inventing has room – time, concentration, willingness to take risks, renunciation of comfort.

This arrangement of life is not a lifestyle question. It is a *structural* question. Whoever wants inventing must forgo certain comforts and take on certain efforts.

The following chapters of the first part trace this arrangement of life. They show what it means to work in flow instead of seeking happiness in consumption. What effort as a value means in a comfort-oriented society. What benevolence is as an attitude towards one's fellow human beings when the language of greed dominates the discourse. What willing means in a society that only knows what it does not want.

Each of these aspects is a prerequisite of inventing. They can be lived individually without anyone inventing. They cannot be suspended when someone invents. That is the anthropological root of the theory that begins here.

2. Happiness and Flow

Csíkszentmihályi's flow as the counter-model to consumer happiness. Inventing as an elixir of life.

I. The happiness you buy

In modern usage, *happiness* is almost always tied to something you have. A happy person is one who has got what they wanted – the job, the house, the partner, the journey. Happiness appears as the result of successful

acquisition. Whoever can acquire more has more happiness. Whoever can acquire little has less.

This conception of happiness is so taken for granted in present-day Western society that it is hardly perceived as a conception any more. It shapes advertising, which links happiness to products. It shapes career advice, which links happiness to advancement. It shapes the search for a home, which links happiness to square metres. It shapes life planning, which links happiness to goals achieved.

The inner logic of this conception of happiness is the logic of consumption. Happiness is not produced, it is procured. Happiness is not done, it is received. Happiness is not the activity but its result. Anyone who wants to become happy must first have something that makes them happy.

This is the hedonia the previous chapter spoke of, in one of its modern disguises. It is not wrong – of course one takes pleasure in things one has. It is merely incomplete. It overlooks another form of happiness, one that lies not in having but in doing.

II. What Csíkszentmihályi found

In the 1970s the Hungarian-American psychologist Mihály Csíkszentmihályi began a study that sounded simple. He asked people in which moments they felt happy – truly happy, not merely content. He asked mountaineers, chess players, surgeons, composers, painters, dancers. He expected the answers to go in a familiar direction – success, recognition, reward, pleasure.

What he found was different. The people he questioned did not speak of happiness after the work was done. They spoke of happiness *during* the work. More precisely: of the happiness of a particular state that sets in when the activity goes completely right. The mountaineers called it *the rock face in which you disappear*. The chess players *the game that plays itself*. The composers *the music that comes*. Csíkszentmihályi called it *flow*.

In the state of flow a person is not separate from their activity. They are *in* it. They do not think *about* it, they carry it out. Self-awareness recedes – the person does not forget themselves, they are simply no longer the centre of attention. Attention is directed at the matter in hand. Time changes – hours pass like minutes, or minutes stretch into hours, depending on what the matter demands. The doing feels at once strenuous and effortless. It is effort, but it is not burdensome.

Over decades of research Csíkszentmihályi identified the conditions under which flow comes about. They are remarkably precise. The activity must have a clear goal. It must give immediate feedback – the doer must sense whether what they are doing is succeeding or not. The demands of the activity must match the ability of the doer – not too easy, not too hard. If the demand is too easy, boredom sets in. If it is too hard, anxiety sets in. In the narrow zone between, where demand and ability meet exactly, flow arises.

It is no accident that these conditions are the conditions of an activity that someone does *well*. They are the conditions of an activity that the doer masters. Flow is the happiness of the activity that succeeds.

III. How flow differs from consumer happiness

The difference between flow and consumer happiness is not one of degree. It is *structural*. Both are pleasant, both are experienced as happiness. But they are constituted differently.

Consumer happiness needs an *object*. Something one has, buys, receives, experiences. No object, no consumer happiness. Anyone who has everything they want and still cannot enjoy it has *no* consumer happiness.

Flow needs an *activity*. Something one does, with attention and ability. No activity, no flow. Anyone who has everything and does nothing cannot have flow – even if outwardly they live an enviable life.

Consumer happiness wears itself out. The holiday is over after three weeks, the car is nothing special any more after three years, the new house

becomes everyday life. The happiness has to be renewed, through new acquisitions, new experiences, new purchases. Hedonia has a built-in tendency to exhaustion – it demands increasing outlay for the same effect. Psychologists call this *hedonic adaptation*.

Flow does not wear itself out in the same way. An activity one does well can be done day after day, year after year, sometimes decade after decade. It does not become boring, because it develops along with the ability of the doer. Someone who plays chess in flow plays a different chess after twenty years – a deeper, richer, more precise one. The demand grows with the ability. Flow renews itself from within.

Consumer happiness makes one dependent on acquisition. Whoever can no longer procure it – through illness, through loss, through age, through economic decline – loses their happiness. Flow makes one independent of acquisition. Whoever has once known it can produce it anywhere they find an activity that challenges their ability. It is portable. It belongs to the doer, not to the circumstances.

IV. Inventing as flow

Inventing is an activity that fulfils all the conditions of flow – and therefore regularly produces it when the conditions are right.

Inventing always has a clear goal. It is about a solution, a process, a design that is supposed to be able to do something specific. Sometimes the goal is blurred at the outset, but it becomes more precise as the work goes on. Whoever invents knows, at some point, what they are working on.

Inventing also has immediate feedback. The material answers back. The design works or it does not. The dimensions are right or they are not. The calculation comes out or it does not. The inventor does not have to wait until a superior, a customer, a market judges what he is doing – the matter itself judges him, every day, on every workpiece, at every trial.

The demand matches the ability when the inventor has chosen the right problem. Here lies an art that not every inventor masters. Whoever chooses too easy a problem gets bored, and flow does not set in. Whoever chooses too hard a problem fails, and flow turns into anxiety. Experienced inventors learn to cut problems to size so that they lie exactly at the edge of their own ability – hard enough to challenge their skill, solvable enough not to overtax it. This self-calibration of difficulty is one of the quiet arts of the inventor's life.

When all three conditions are met, flow comes. Anyone who has ever worked on a design problem that exactly matched their ability knows the phenomenon. The hours pass unnoticed. Other worries recede. Hunger, tiredness, even physical complaints go unnoticed for a long time. The matter pulls. The next question arises from the previous one. One solution leads to the next question. The doing carries itself.

Csíkszentmihályi found that people who regularly work in flow have a different feeling about life from people who do not. They are not *more content* in the sense of consumer happiness – they often have fewer material comforts than others, because their lives are less centred on acquisition. But they are *happier* in the sense of eudaimonia. Their lives have an inner fulfilment that does not depend on external circumstances. They have a happiness they take with them wherever they go.

V. Inventing as an elixir of life

Whoever works as an inventor works – if he does it right – mostly in flow. That is not a matter of course. Many occupations rarely allow flow. Bureaucratic work, repetitive administration, dutiful fulfilment of official obligations – none of this produces flow, because either the demand is too low or the feedback too vague. Inventing, by contrast, is an activity that almost inevitably leads into states of flow as soon as it is pursued in earnest.

This is an important finding. It explains why inventors often cling to their work with a tenacity that seems incomprehensible to outsiders. They do not

cling to *money* – inventing rarely brings money. They do not cling to *recognition* – that comes rarely and unreliably. They cling to the state they experience while inventing. They cling to flow.

In older language this was put as *the thing loves me, and I love it*. In Csíkszentmihályi's language it is called *flow*. Both descriptions mean the same finding. Whoever finds a thing that challenges their skill and carries their doing has found something that goes deeper than well-being. They have found a source of life.

This source of life is not without danger. Whoever works in flow can lose hours they would have needed for other things. They can neglect relationships because the matter pulls. They can overtax their health because they do not feel hunger. Inventors often have difficult life stories – not because they are difficult people, but because their happiness binds them to activities that carry them out of the normal social rhythm. The happiness of flow has its price. The price is deviation from what is called normality.

But this price is not necessarily worse than the price of consumer happiness. Whoever aims at consumption pays with lifelong outlay for acquisitions that wear themselves out. Whoever aims at flow pays with deviation from normality for a happiness that does not wear itself out. Both prices are real. Which is more bearable depends on the person.

VI. What this means for the theory

The theory of inventing is a theory of an activity that makes flow possible. Whoever wants to understand inventing must understand that it is not a particularly strenuous form of paid work. It is an activity that carries its own reward within it – the state in which it is carried out.

This has consequences for everything that comes in the following chapters. Whoever wants to understand the *nine layers of inventing* should read them as layers of an activity that takes place in flow. Whoever wants to understand the *cognitive modes* should read them as modes of an activity

that carries itself. Whoever wants to understand the *systemic situation of the inventor* should understand that this situation is so wounding because it endangers a source of life – not merely a source of income.

Whoever has once known the source of flow does not give it up easily. Even when external conditions are against it. Even when recognition fails to come. Even when the money does not come. Inventing has a right of its own – the right that comes from the experience that it contains its own happiness.

That is the bridge to the next chapter. If inventing is an activity in flow, then it demands a readiness for effort that can no longer be taken for granted in today's living environment. This readiness is in turn a prerequisite of inventing – and at the same time a possibility that inventing brings forth. Effort is not what stands in the way of flow. Effort is what happens in flow. But that is for the next chapter to set out.

3. Effort as a Value

The comfort zone as a prison. Readiness for effort as a prerequisite of inventing.

I. What one makes the effort for

Anyone who says in today's world that inventing demands a readiness for effort risks a misunderstanding. The sentences sound as though they were a sweeping criticism of an entire way of life – as though anyone who makes themselves comfortable were a person without effort. That would be not only unfair, it would be wrong. People who arrange their lives comfortably are by no means all idle. Many of them make considerable efforts – in the gym, on long climbing tours, in sporting competitions, in learning languages, on demanding journeys, in hobbies that require skill of them. Some of these efforts are substantial. They can take one's physical or mental condition to its limits. They can trigger flow – the mountaineer and the chess

player with whom Csíkszentmihályi began his research were both active not professionally but out of passion.

The question is not *whether* one makes an effort. The question is *for what*.

An effort in the gym is directed at one's own body, at its shape, at its performance. A climbing tour is directed at one's own success, at the mountain that is conquered. A win in a game is directed at victory over an opponent. A demanding journey is directed at one's own experience – what one can tell later, what fills one's memory, what enriches one's self-image.

In all these cases the *recipient* of the effort is the one making the effort. Through it he becomes fitter, more successful, more experienced, more interesting. His life becomes richer. But the work he brings forth is *himself*. His body, his record, his memories. What goes beyond him is limited – the training partner may benefit, the listener to the story is entertained. The actual result remains within the self.

That is not to be despised. Whoever does not look after themselves looks after little else either. Self-directed effort is part of the good life. It is not the opposite of eudaimonia but one of its preconditions – whoever neglects their body cannot work on a process for forty years.

But it is not inventing. Inventing has a different recipient. It is directed at something that is not the inventor. At a machine that others will use. At a process that will be employed in a workshop in which the inventor never stands. At an idea that remains in the world when the inventor is no longer there. The effort of inventing has a generosity built into it that self-directed effort lacks – it goes into the work, not into the self.

This settles the point without anyone being accused of a lack of effort. Whoever looks after their body makes an effort. Whoever climbs a mountain makes an effort. Whoever learns a language makes an effort. But whoever invents makes an effort of a different kind – one that brings forth a work outside their own self. It is this effort that the previous chapter meant and

that this chapter sets out to work through. It is not effort as such, it is effort directed towards what goes beyond the one making it.

II. The comfort zone

The term *comfort zone* has become standard vocabulary in recent decades. It denotes the area in which a person feels safe and at ease – the familiar tasks, the trusted relationships, the routine procedures. *Leave your comfort zone* is one of the common pieces of advice of modern self-optimisation.

What those who give this advice usually do not clarify – and what those who hear it know even more rarely – is a fundamental ambiguity. *Comfort zone* can denote two very different things.

In one sense the comfort zone is the area of one's own ability. Whoever works in their comfort zone does things they are good at. That is not bad – mastery needs repetition, repetition needs the area in which one is competent. A world without comfort zones would be a world without skill. Whoever invents works to a large extent in their comfort zone – they use what they have learned in order to apply it to new problems.

In the other sense the comfort zone is the prison of convenience. Whoever stays in the comfort zone in this sense does not have a skill that carries them – they have an avoidance that shuts them in. They do the familiar because the unfamiliar frightens them. They avoid risk because risk demands effort. They stay where they are, not because they do something well there, but because they are not challenged there.

The difference between the two comfort zones is not one of degree, it is *categorical*. The first is a source of strength from which the new can emerge. The second is a cage that stifles strength. Whoever gives the advice *Leave your comfort zone* should specify which of the two they mean. Whoever works in the first sense should not leave their area – they should go deeper into it. Whoever lives in the second sense should leave it – and

not in order to find another convenience, but in order to reach the state that makes the good life possible.

III. The readiness for effort

What inventing demands is not a heroism of self-conquest. It is a plain readiness. The readiness to make an effort when the matter demands it – and not only when the effort is sure to be rewarded.

This readiness has several characteristics that distinguish it from other attitudes.

It is *occasion-bound*. Whoever is ready for effort does not seek it without occasion. He makes an effort when the matter calls for it and rests when it allows. He is not the ascetic who torments himself for no reason. He is the craftsman who knows that some works do not come about without sweat – and who therefore does not shy away from the sweat when the work demands it.

It is *not time-bound*. Effort can last hours, but also years. Whoever develops a machine cannot decide after three weeks that the effort has now been enough. Whoever works out a theory may have to prepare for decades. The readiness for effort includes the readiness to keep going over long stretches without seeing a clear end point.

It is *free of reward*. Whoever makes an effort because the effort is rewarded is not ready for effort – he is ready for reward, and the effort is only the means. Genuine readiness for effort comes about when the doer makes an effort even when the reward is uncertain or ruled out. Its source lies not in the payment but in the meaning of the matter.

These three characteristics – occasion-bound, not time-bound, free of reward – cannot be taken for granted. They are untypical of the way modern lives are arranged. Whoever has them has a disposition that is foreign to the majority. Whoever does not have them can learn to develop them – but the learning is itself a process of effort, from which most people shrink.

IV. Effort in flow

We said in the previous chapter that effort happens in flow – and that it is not burdensome. At first sight that is paradoxical. If effort in flow is not burdensome, is it effort at all?

The answer is: it is effort in the sense of *physiological demand*. Whoever works in flow uses energy, taxes their concentration, draws on their attention. Eight hours in flow are eight hours of work – the body notices when the flow is over. It is *not* effort in the sense of *psychological burden*. Whoever is in flow does not feel the effort as a load. They feel it as what they are doing – as the doing itself.

This dual nature of effort is methodologically important. Whoever knows effort only in its psychological meaning – as a load, as a hardship, as an outlay that costs something – cannot understand flow. They think flow is a form of effortlessness. It is not. It is a form of *unburdensome* effort, in which the doing and the doer become one, so that the doer no longer perceives himself as toiling.

That is precisely why the readiness for effort is so closely bound up with flow. Whoever shies away from effort cannot reach flow – for flow presupposes crossing the threshold of the first, more difficult hours. Whoever wants to get into flow must be ready to bear the effort of the threshold. Only then comes the state in which the effort is transformed – in which it is no longer a load but activity.

V. Effort and self-worth

There is an observation that is made again and again in research on eudaimonia. People who regularly take effort upon themselves – not in the sense of overtaxing themselves, but in the sense of ready, occasion-bound effort – have a different relationship to themselves from people who avoid effort.

They know that they can do something. Not from conceit, but from experience. They have made the effort, they have managed it, they know it. This knowledge does not depend on the confirmation of others. It does not depend on success in the outward sense. It is a *self-assurance* that comes from one's own experience.

People who avoid effort do not have this assurance. They do not know exactly what they can do – because they have not tried it. They depend on the confirmation of others in order to assess themselves. They are *unstable in their self-worth*, because their self-worth is anchored not in their own doing but in the recognition of others.

This is an important finding for the theory of inventing. The inventor who keeps working without recognition cannot do so out of nothing. He can do it because he has a self-worth that does not depend on outside approval. This self-worth comes from the experience that he has made the effort, that he has achieved something, that it works. The inventor's self-worth is a consequence of his readiness for effort – and at the same time becomes the condition that enables him to keep working when external conditions are bad.

VI. A cultural remark

The German language has a word for the attitude we are describing here. It calls it *Tüchtigkeit*. The word has almost disappeared today – it sounds old-fashioned, authoritarian, with a severity that no longer fits the times. But it denotes something that modern language cannot replace. *Tüchtig* is not *fleißig*, industrious – the bureaucrat who sits out his hours can be industrious too. *Tüchtig* is not *erfolgreich*, successful – someone who makes no effort but is lucky can be successful too. *Tüchtig* is the combination of skill, effort and a work that is willed. Whoever is *tüchtig* can do something, makes an effort for it, and brings forth something they wanted to bring forth.

The disappearance of this word is symptomatic. The society that no longer says *tüchtig* no longer has a concept for the attitude it has lost. It is

industrious, but that is too little – industriousness without skill and without a work is drudgery. It is *successful*, but that is too much – success without effort is luck, and luck is something you either have or do not have. What lies between industriousness and success – the active readiness for meaningful effort – no longer has a name.

It is telling that the word has recently been enjoying a revival – in a new, very old form. *Kriegstüchtigkeit*, fitness for war, has been back in political language since about 2024. The term had disappeared from public discourse from 1945 until the twenties of the twenty-first century. Now it is back. With it, the word returns in a form that no longer has anything to do with the civilian concept of *Tüchtigkeit* – a *Tüchtigkeit* not for bringing forth but for mobilisation. Not for the work but for war. That is no accident. A society that has lost *Tüchtigkeit* as a civilian concept rediscovers the word as soon as the defence of the status quo is at stake. The order is the wrong way round. In a healthy language civilian *Tüchtigkeit* would have its place, and *Kriegstüchtigkeit* would be a rare, sad special case. Today's situation is the reverse – civilian *Tüchtigkeit* is missing, *Kriegstüchtigkeit* is coming back. That is a diagnosis that goes beyond inventing.

Whoever wants to understand inventing should take the word *Tüchtigkeit* back into their vocabulary in its civilian form. *Tüchtigkeit* is the disposition that brings forth inventing. It is not a character trait but a practice. Whoever makes an effort and achieves something and tests themselves against it becomes *tüchtig*. Whoever avoids this does not remain so.

VII. What this means for the theory

The theory of inventing is a theory of a *tüchtig* way of life. It demands nothing extraordinary in the sense of heroism – it demands what should really be a matter of course, which has become rare in our time. Readiness for effort. Enduring uncertainty. Loyalty to the matter, even when the matter is not rewarded.

This disposition is not for everyone. But nor is it the preserve of a chosen few. It can be learned – through practice, through role models, through one's own experience of flow, which delivers the reward of effort in the effort itself. Whoever has once experienced it can hardly give it up. Whoever has not experienced it cannot understand what the others are doing.

That is a difficulty for the theory. It cannot *prove* that the readiness for effort is worthwhile. It can only *show* that those who have it have a different relationship to the world – a more effective, deeper, freer one. Whoever reads the theory without having this experience themselves may feel it to be an unreasonable demand. Whoever reads it and knows the experience recognises themselves.

The following chapters will show that the readiness for effort does not stand in isolation. It is embedded in an attitude that we have yet to describe. Benevolence towards one's fellow human beings. Willing instead of administering. Patience with what is ripening. These attitudes support one another – they are not separate virtues but aspects of a way of life in which inventing has a place.

That is the bridge to the next chapter, which deals with one of these attitudes. Whoever makes an effort without being benevolent becomes a doer who overlooks others. Whoever is benevolent without making an effort becomes a likeable person who brings forth nothing. Neither is enough on its own. Both together – effort and benevolence – create the condition under which inventing turns out well.

4. Benevolence and Greed

Boesky 1986 and its consequences. Both-and versus either-or. Mondragón as living proof.

I. A language that changed

There is a moment in recent history that is better documented than most cultural turning points. In May 1986 Ivan Boesky gave a speech to business students at the University of California in Berkeley. Boesky was at the time one of the most prominent investors on Wall Street – a man who had become a multimillionaire through insider dealing, though this was not yet publicly known at that point. In his speech he said a sentence that was to become the watchword of a decade: *Greed is healthy. You can be greedy and still feel good about yourself.*

The students applauded. The sentence spread. A year later the film-maker Oliver Stone took it up – in the character of Gordon Gekko, played by Michael Douglas, in the film *Wall Street*. Gekko said before the assembled shareholders: *Greed, for lack of a better word, is good.* Stone had meant the film as a warning against the moral decay of Wall Street. Against his intention, it became the inspiration of a generation of business students. *Greed is good* became a philosophy of life.

What happened here was not the invention of a new virtue. Greed has always existed. What happened was the *public rehabilitation* of what had previously – for centuries, in all the great traditions – been regarded as a vice. Greed was no longer described as a weakness or an aberration. It was declared a productive force. Whoever was greedy drove the economy. Whoever maximised profits created prosperity. Whoever had scruples was inefficient.

This turn was no accident, and it was not confined to business students. It fitted a broader movement – Reagan in the USA, Thatcher in Great Britain – that made trust in the market the doctrine of the state. The market was efficient, the state was inefficient. What the market rewarded was right. What it punished was wrong. If the market rewarded greed, greed was right. A generation learned that there need be no difference between *good for me* and *good for the world* – the two coincide if one believes in the market.

II. The geometry of economic disposition

Whoever wants to understand inventing in this situation must make a distinction that has disappeared from modern economic discourse – the distinction between self-interest and greed. The two are not the same.

Self-interest is the readiness to take care of oneself. Whoever makes an effort wants something for it. That is not wrong – it is the prerequisite of any activity that is not pure self-sacrifice. The inventor, too, has self-interest. He wants his invention to be recognised. He wants to be able to live from his work. He wants what he brings forth to be valued. All of this belongs to healthy self-interest, without which no one can work on something for decades.

Greed is something else. It is self-interest without limit. It is the disposition that always wants more without asking what is enough. Whoever is greedy does not have a self-interest that is compatible with the well-being of others – he has a self-interest that uses others *as means*. That is the point at which greed parts company with healthy self-interest. With healthy self-interest one asks – *what helps me and harms no one?* With greed one asks – *what brings me the most, no matter who loses by it?*

This distinction can be captured in a simple geometry. On one axis, self-interest – from weak to strong. On the other axis, benevolence towards others – from weak to strong. A person without self-interest and without benevolence is indifferent. A person with self-interest and without benevolence is greedy. A person with benevolence and without self-interest is devoted to the point of self-denial. A person with self-interest *and* benevolence is what we call healthy – he takes care of himself without using others as mere means.

Boesky's message was that the fourth position – healthy self-interest with benevolence – was a sentimental illusion. *Greed is good* presupposes that benevolence is a weakness, an obstacle, a slowing-down. Whoever is

benevolent does not get far. Whoever is greedy wins. The choice lies between *winning* and *being good*, and the right choice is *winning*.

This reduction is wrong. It reduces a four-field geometry to a line between two poles – greed or weakness. It overlooks the possibility of being strong in self-interest and strong in benevolence at the same time. It overlooks the fact that the world is full of people who practise exactly that – the craftsman who does his work well because he is proud of it and because it benefits his customers; the teacher who earns his living and educates his pupils; the inventor who wants to live from his work and brings forth something that serves others.

III. Both-and versus either-or

The language of greed is built on a logical form that excludes benevolence – the form of *either-or*. Either I win or you do. Either I get the contract or my competitor does. Either I maximise my profit or I give my money away. In this logic every relationship is a competition with a winner and a loser. There are no relationships in which everyone benefits. If someone benefits, someone else has lost.

The language of benevolence is built on another logical form – the form of *both-and*. A good relationship can benefit both sides. A successful collaboration can make both stronger. A fair deal can leave both better off. This logic is not naive – it is aware of the possibility of conflict and competition. But it also knows that most relationships are not zero-sum games. When an inventor licenses his invention to a manufacturer who produces it, who sells it to a customer, who uses it – then all four benefit. The sum of the gains is greater than the sum of the stakes.

This both-and logic is the foundation of inventing. Whoever invents creates something that was not there before. Their work adds to what already exists – it makes the world richer without taking from others. A good invention leaves everyone involved better off – the inventor, the

manufacturer, the customer, often even third parties who benefit indirectly. Inventing is by its very nature *positive-sum*.

Whoever thinks in the logic of greed cannot understand inventing in its full depth. He sees in the invention primarily the patent, that is, the right to exclude others from using it. He sees the opportunity to win a market position, that is, to win a competition. He sees the prospect of profit, that is, the share he himself takes away from what happens. What he does not see is that an invention – if it finds its way – changes the world as a whole. It makes accessible possibilities that were not there before. It allows activities that were previously impossible. It opens spaces that were previously closed.

IV. Benevolence as a prerequisite

Whoever invents needs benevolence. Not as a virtue he imposes on himself, but as a prerequisite for the invention to succeed.

First, he needs benevolence towards the future users. Whoever designs a machine without thinking of those who will work with it builds a bad machine. He overlooks the fact that it must be accessible, that it must be maintainable, that it must not harm its operator. These qualities do not come of their own accord – they come from an attitude of the designer who feels connected to the later user. A machine is good work when it has been built in the spirit of benevolence. It is bad work when it was merely a means of benefiting the designer.

Second, he needs benevolence towards his co-workers. An invention is rarely the work of a single person. It comes about in a workshop, in a laboratory, in a group of designers, technicians, mechanics. If the inventor treats his co-workers as mere means, he gets only what he can force out of them. If he treats them as fellow doers, he gets their own contributions, their attention, their dedication. Inventions that come from such constellations are richer than those that come from hierarchies. That is not sentimentality, that is experience.

Third, he needs benevolence towards his predecessors. An invention always builds on what others have done before it. Whoever despises the work of other inventors, thinking he is starting from scratch, overlooks what he owes them. This contempt leads to bad inventions – to ones that reinvent the wheel without knowing why earlier wheels were shaped as they were. Whoever is benevolent towards his predecessors learns from them, builds on them, corrects them where they need correcting, and honours them where they deserve to be honoured.

Fourth, he needs benevolence towards the work itself. At first that sounds strange – how can one be benevolent towards a thing one is building? Yet experience shows that there is a difference between an inventor who *exploits* his work to make a name for himself and one who *serves* it so that it becomes what it can become. The first treats his work as the material of his success. The second treats his work as something that strives for a perfection of its own, to which he as inventor is committed. This attitude – which treats the work almost as a fellow being – is a form of benevolence directed at something that goes beyond the inventor.

V. Mondragón – the living proof

There is a town in the Basque Country that has been proving for over seventy years that the both-and logic is viable. It is called Mondragón. It lies in the Spanish province of Gipuzkoa, about an hour's drive from Bilbao.

In Mondragón in 1956 the Catholic priest José María Arizmendiarrieta, together with five young engineers, founded a small cooperative that made paraffin stoves. Over the decades this foundation grew into the largest cooperative enterprise in the world – the *Mondragon Corporación Cooperativa*, today with over eighty thousand employees, several hundred subsidiaries, its own university, its own bank and a social security system that protects members against risks the free market does not cover.

The Mondragón cooperatives have developed an economic model of their own that institutionalises the both-and logic. Employees are at the same time

owners. Decisions are taken democratically. The pay gap between the highest and the lowest salary is limited – historically to a factor of six, today to about a factor of eight, compared with gaps of a hundred or more in many capitalist corporations. Profits are invested in reserves that make the enterprise crisis-proof. If a subsidiary gets into difficulties, the other subsidiaries take on its employees – no one loses their position because a single division has become loss-making.

Mondragón is not the result of a theory drawn up at a desk somewhere. It is the result of the practical experience of people who have made both-and liveable. The cooperative does not work because its members are altruistic. It works because its members have a healthy self-interest – they want a good wage, a secure future, meaningful work – and because they understand that this self-interest is better safeguarded through cooperation than through competition. Mondragón is *healthy self-interest plus benevolence* in institutional form.

This example is methodologically important because it refutes Boesky's claim that both-and is an illusion. Mondragón exists. It is successful. It has survived for over seventy years in a fiercely competitive international market. Whoever claims that *Greed is good* is the only working economic logic must explain why Mondragón not only works but often works better than its capitalist competitors – more securely, more resilient in crises, with higher employee satisfaction, with more innovations, with a quality of life for those involved that most public limited companies do not achieve.

VI. The language that may be heard

What happened in the Boesky turn was not just an economic shift. It was a linguistic one. Words such as *benevolence*, *decency*, *responsibility towards one's fellow human beings* were gradually pushed out of economic discourse. They were regarded as sentimental, as naive, as inefficient. Whoever used them made themselves suspect. In their place came words such as *competitive advantage*, *market position*, *disruption*, *maximisation of shareholder value*. These words are not wrong – they describe real aspects

of economic reality. But they are incomplete. They see only what happens in competition. They overlook what happens in cooperation. They see only the either-or. They overlook the both-and.

Whoever wants to write about inventing must break through this language. He must use words that have become suspect in modern economic language. *Work*, *dedication*, *responsibility*, *generosity*, *benevolence* – these are all words that describe the inventor better than the words found in textbooks today. They were pushed out of public discourse because they no longer fitted the world of the Boeskyites. Bringing them back is a linguistic task that belongs to the theory of inventing.

This is not romanticism. It is the attempt to name the matter precisely. Whoever wants to describe an inventor who works on something for forty years without becoming rich, without becoming famous, cannot do so in the language of profit maximisation. That language has no words for what he does. He must use words borrowed from other ways of life – the craft tradition, the monastic tradition, the research tradition. These words have an old ring to them, but they fit.

VII. What this means for the theory

The theory of inventing that we are developing here cannot be written from the logic of greed. Nor can it be written from the logic of pure self-sacrifice. It is a theory of the third possibility – of healthy self-interest with benevolence.

This third possibility is not a midpoint between two poles. It is a form of its own, with a logic of its own. It has its models – the craftsmen, the cooperative members, the independent inventors, the researchers in the best tradition. It has its opponents – the apologists of greed, who claim it is an illusion. It has its difficulties – it is more demanding than pure greed, because it pursues two things at once. It is slower than pure greed, because it pays attention to relationships that take time. But it is more sustainable, because it builds not on burning up but on growing.

The inventor who lives in this third possibility has an advantage that pure greed does not have. He can build relationships that carry him. He can win co-workers who work with him because they value him, not because they depend on him. He can bring forth works that are useful in the long term because they arise from the spirit of benevolence. None of this is sentimentality – it is the practical condition for inventing to turn out well.

That is the bridge to the next chapter. Whoever is benevolent and makes an effort has two of the prerequisites of inventing. What he still needs is the ability *to will* – and that in a world that has increasingly replaced willing with administering. That is the question of the next chapter.

5. Willing and Administering

A society that only knows what it does not want can shape nothing. Inventing presupposes that someone is able to want something.

I. A simple observation

Anyone who has listened for a while to present-day public debate in Germany notices a peculiarity. A great deal is said about what should *not* be. No nuclear power. No fossil fuels. No discrimination. No inequality. No war. No plastic. No data octopus. The list could go on. The negations come from every political camp, in varying mixtures. As a rule they are understandable, often morally grounded, often factually correct as well.

What occurs far more rarely is a statement of what *should* be. Not in the form of a rejecting stance, but in the form of a shaping one. What kind of economy do we want? What kind of industry? What kind of energy supply? What kind of education? What kind of urban development? What kind of relationship with the world? These questions, if they are asked at all, are almost always answered in the form *what do we not want*. The positive picture remains pale. Whoever is supposed to name it often does not really know how.

This asymmetry is not accidental. It has a history. It has consequences. And for the theory of inventing it is a finding that goes to the root of the matter. A society that only knows what it does not want cannot invent. Inventing presupposes that someone is able to want something.

II. The history of negation

The German tradition of negation has deep roots. After 1945 a society had to reconstitute itself whose positive visions of itself had led to catastrophe. For a whole generation, the *lesson of history* was that it was better to know what must never again be than what should be. *Never again war. Never again Auschwitz. Never again dictatorship.* These sentences were right and necessary. They were also the beginning of a language that made negating its main task.

This language has developed its own logic. Whoever formulates a wanting becomes suspect – might he not relapse into old aberrations? Whoever formulates a not-wanting is safe – he is keeping to what we have learnt. Over decades, this has given rise to a disposition in which negating counts as a virtue and wanting as a risk.

The student movement of the late sixties radicalised this disposition, albeit with the signs reversed. It rejected everything the fathers' generation had built up – the economic order, the university, the family, the church. What was to take its place remained unclear. There were programmes – socialism, council democracy, the end of imperialism – but they were often more a rejection of what existed than a positive construction. The generation that learnt to say *No* never quite found the *Yes*.

The ecological movement of the eighties and nineties carried the logic of negation into a new field. Nuclear power, pesticides, industrial waste, later CO_2 – all were rejected. Here, too, what was to take their place was often unclear. The *Energiewende* was formulated as a move *away from* fossil energy rather than a move *towards* a new form of energy with a clear

architecture. Here, too, the rejection was precise and the positive picture diffuse.

Today's landscape of public debate has grown up in this tradition. It knows what must not be. It has developed a rich language for this – terms such as *discrimination*, *exploitation*, *climate catastrophe*, *structural violence*. What it does not have is a similarly rich language for what should be. Whoever uses one sounds either naive or suspect.

III. Administration as a substitute

In a society that no longer knows what it wants, administration takes over the function that wanting used to have. At first this sounds like a contradictory claim – how can administration replace wanting? It can, if it presumes to replace what should be with what *is already regulated*.

Administration knows no goals, only regulations. It knows no visions, only procedures. It knows no wishes, only applications. Whoever wants to achieve something in an administered world does not formulate his wish as a will but as an application – he adapts himself to the categories that administration knows. His original will is translated into forms that administration can understand. What comes out of the application is often no longer what the applicant wanted. It is what administration is able to approve.

Over time this mechanism changes people. Whoever regularly submits applications learns to think in the categories of administration. What does not fit into a form becomes a non-demand. What cannot be applied for becomes a non-possibility. What administration does not know does not exist. In the long run, wanting shrinks to what administration would permit if one phrased it correctly. That is no longer wanting – that is conformity with available procedures.

In Germany this process is far advanced. Bureaucracy is no longer a tool for administering what is wanted. It has become the place where what is

wanted gets formulated – or does not. Whoever has a new idea asks first *which funding programme fits*. Whoever wants to found a company asks first *which permits are needed*. Whoever wants to build a house asks first *what the building regulations allow*. These are all sensible questions – there is no world without rules. But if the *first* question is always about the rules rather than about what is wanted, then the order has been reversed. Wanting no longer comes before administration. Administration comes before wanting.

IV. What this does to inventing

Inventing is an activity that begins with wanting. Whoever invents wants something that does not yet exist. He wants a machine that solves a problem. He wants a process that copes better with a task. He wants a solution that others have not yet found. This wanting does not come from the regulations – it comes from the inventor himself, from his perception of the world, from his sense that something would be possible that has so far been missing.

When wanting disappears from the disposition of a society, inventing disappears too. Not at once, not spectacularly. It happens slowly, almost imperceptibly. First the radical inventions become rarer – those that bring forth something fundamentally new that no funding programme had foreseen. Then the incremental inventions decline as well – those that improve what exists, because what exists has itself already become rigid and no longer calls for improvement. In the end there remains an administration that administers itself, in which works are produced only by rote, works that nobody would miss if they were not there.

Whoever wants to invent in such a society finds himself in a double predicament. Inwardly he must bring forth the wanting from within himself, without any resonance from outside. Outwardly he must contend with an administration that experiences everything that does not fit its categories as a disturbance. Both predicaments are demanding. The inner effort – keeping the wanting awake when nobody shares it with you – can exhaust. The outer

effort – asserting oneself against administration without being ground down by it – can embitter.

This is the predicament in which most independent inventors in Germany find themselves today. They want something. They have kept the capacity for wanting because they know it from their own experience, because their activity itself sustains it. But they see that this capacity is no longer a matter of course in the society in which they live. They see that the younger people often no longer know how wanting works. They see that administration does not understand wanting and therefore does not support it. They have the feeling of speaking a language that is no longer understood.

V. Wanting is not commanding

Before we deepen the finding, a clarification is needed. *Wollen* – wanting, willing – has a heavy sound in the German language. It can recall authoritarian ways of life – the leader who wills, the power that imposes itself, the strength that asserts itself. Whoever has grown up in the tradition that was made out of negating after 1945 has a justified mistrust of wanting. Wanting is dangerous, says this tradition. Where it becomes too strong, terrible things happen.

This concern is not wrong. There is a wanting that does terrible things – wanting without benevolence, without effort against oneself, without consideration for others. But this wanting is not the only kind. There is also a wanting that builds up – the wanting that designs a machine, founds a school, writes a book, cultivates a relationship. This wanting is not authoritarian. It is the energy without which nothing comes into being.

Confusing these two kinds is the root of German scepticism towards wanting. Whoever fears the one also shies away from the other. Whoever sees in every strong wanting the harbinger of catastrophe can no longer support benevolent wanting. This creates a situation in which every form of wanting appears suspect – and the only safe place left is administering.

It is methodologically important to clear up this confusion, because otherwise the theory of inventing will be misunderstood. When we say that inventing presupposes wanting, we do not mean the wanting that subjugates others. We mean the wanting that brings forth a work – a work that is good even if nobody takes notice of it. This wanting is not authoritarian. It is the other side of the benevolence we described in the previous chapter. Whoever is benevolent without wanting brings forth nothing. Whoever wants without being benevolent brings forth only harm. Both attitudes must come together for inventing to turn out well.

VI. The point at which the theory could make a difference

In this situation the theory of inventing has a modest but real task. It cannot abolish administration – no theory can do that. It cannot reverse the history that, over decades, made negating the main disposition. It cannot plant wanting in a person who does not have it.

What it can do is name the matter. It can say that inventing presupposes wanting. It can show that wanting need not be authoritarian. It can give those readers who themselves have the capacity for wanting the vocabulary with which they can understand themselves. It can give those readers who no longer know this capacity a pointer to what they are missing – and to what they could regain.

That is not much. But it is not nothing. For generations, books have helped people to understand themselves and their situation. They have provided the vocabulary with which something that was previously unnamed could enter public discussion. The *theory of inventing* wants to be one of these pointers. It is not aiming at a political turnaround. It is aiming at the ordinary people who still invent, and at the ordinary people who could still invent if they noticed that such a thing exists.

Whoever has the capacity for wanting may feel less lonely after reading this. Whoever does not have the capacity at least knows that there is such a

capacity. Both are a form of resonance that adds something to the present situation. We ask for no more.

VII. What this means for the theory

The five chapters of the first part have described the anthropological root of inventing. They have shown that inventing presupposes a way of life that is no longer a matter of course in today's world. Eudaimonia instead of hedonia. Flow instead of consumer happiness. Effort as a value. Benevolence instead of greed. Willing instead of administering. These five aspects are connected. They support one another. Whoever leaves one of them out cannot hold on to the others.

This means that, at its deepest, the theory of inventing is not only a theory of the activity. It is also a theory of the prerequisites. It says: if the activity no longer takes place, or only rarely, this is not primarily because the method of inventing has been forgotten. It is because the way of life in which inventing is possible is disappearing. Whoever wants to regain inventing must regain that way of life.

The second part of this book will turn to the inner practice of inventing – the nine layers through which an idea becomes reality. These layers are not abstract. They can only be passed through by a person who has the way of life we have described here. Whoever does not have it will not pass through the layers – he will shrink back from the first one.

But whoever has it also has the possibility of passing through the layers. He can invent. He can engage in a process that both challenges and sustains him. He can bring forth a work that carries the meaning of his life – and that, if all goes well, also has something to say beyond his life.

That is the bridge to the second part. What we have described in the first five chapters was the root. What comes now is the trunk. We move from the way of life to the activity, from the "on what" to the "how", from the anthropological prerequisite to concrete practice. The layers are waiting.

Second Column – Practice

The inner practice

The second Column asks: how does inventing unfold when a person actually carries an idea all the way to its realisation?

There are nine layers that this process passes through – and two cross-layers that accompany it methodologically. The nine layers are not a recipe and not a sequence to be ticked off. They are descriptions of what actually happens when an idea matures: from the notion that the idea belongs to you, to the realisation that it carries you; from the first hunch to the critical look after the model; from waiting during incubation to the deed you do when nobody else does it.

The two cross-layers are the methodological backbone that gives this practice its footing. One is Altshuller's TRIZ – the insight that strong solutions follow a grammar and that this grammar can be learnt. The other is the innovation mesotes, the Aristotelian virtue of the right mean, applied to inventing. Together they form the methodology that keeps the inventor from losing his way in formalism, the cult of genius or the dead ends of overestimating himself.

Whoever reads this Column will find no instructions, but a vocabulary. A vocabulary for what he knows or senses – and has rarely put into words. Whoever invents will recognise himself in parts of it. Whoever does not invent will understand what is actually happening when others do it.

The nine layers of inventing and how they interlock.

6. The idea as a vehicle

First layer. Inventing as carrying the idea, not as owning it. Johann Müller as the teacher of this relationship.

I. The notion of ownership

There is a notion of inventing so widespread that it is hardly perceived as a notion any more. It runs: the inventor has an idea. The idea belongs to him. He is its originator, its owner, its master. From this notion follows the whole vocabulary in which inventing is talked about today. *Intellectual property. Industrial property right. Exploitation.* The idea appears as a thing one has, as one has a plot of land – one can fence it in, defend it, sell it, bequeath it.

Patent law has cast this notion into an institution. A patent is the right to exclude others from using an idea. It turns the idea into a possession with boundaries, with an owner, with a term. It is a sensible instrument – without it hardly any inventor could live from his work. But it shapes the notion more deeply than it has any right to. It leads people to believe that the inventor's relationship to his idea is a relationship of possession.

Whoever thinks this way misunderstands what happens in inventing. He confuses the *legal* relationship that is subsequently erected around the idea with the *inner* relationship from which the idea emerges in the first place. The two have little to do with each other. The legal relationship is a relationship of having. The inner relationship that this chapter describes is a relationship of carrying. And whoever wants to understand the first layer of inventing must move from having to carrying.

II. The idea as a vehicle

Notice how language talks about ideas when it is off its guard. An idea *comes* to you. It *occurs* to you. It *dawns* on you. You do not *have* it first – you *receive* it. In all these phrases the human being is not the origin but the

place. Something arrives, and it arrives at him. The grammar of inventing is not a grammar of possession. It is a grammar of receiving and passing on.

This is not a mystical claim. It is a phenomenological observation – a description of how inventing feels from the inside. Whoever invents rarely has the feeling of having *made* an idea, the way one makes a table. He tends rather to feel that something has *fallen* to him, which he has then *taken up* and *carried to term*. The idea was there first, unfinished, as a hunch, as a possibility in the air. The inventor was the one it could dock with – because he had the attentiveness, the prior knowledge, the readiness. He did not generate it. He *took it up*.

And then he carries it. That is the actual work. An idea that is taken up and not carried disappears again – it is lost like a seed that falls on rock. It is carried through all the layers this book describes: through ripening, inner seeing, design, the model, testing. The inventor carries the idea from its first flash to its realisation – with all the effort, all the patience, all the risk that lie in between. He is its vehicle. It moves through him into the world.

A vehicle does not own its load. It carries it. It brings it from one place to another and delivers it at the destination. The inventor who understands his relationship to the idea as that of a vehicle knows that the idea does not belong to him but passes through him. It came from a larger context – the state of the art, the open questions of the time, the work of predecessors – and it returns to a larger context as soon as it has been realised. The inventor is the stretch of road in between. A necessary stretch – without him the idea would not arrive. But a stretch, not an owner.

III. Johann Müller

There was a man in whom this relationship showed itself in its purest and at the same time most bitter form. His name was Johann Müller. He was born around 1922 and worked for most of his life as an inventor for a company that made hydraulic bolting tools – tools with which large bolted joints are tightened safely to high torques, as needed in power-station, plant and

machine construction. Müller was the mind behind a whole series of the inventions that made this company big. As far as can be said, he had worked for it since its founding, perhaps even before.

Among his inventions was one that gave the company a considerable competitive advantage for years: a machine for the exceptionally economical production of hexagon sockets – those six-sided recesses into which the key engages. What competitors produced at great cost, this company could produce cheaply and in quantity thanks to Müller's machine. The lead that grew out of this carried the company for years. It came out of Müller's head.

And now the decisive point. Müller's name did not appear on the patents that protected these inventions. There was a *request for the inventor not to be named* – a possibility provided by patent law so that an inventor can remain unnamed. Müller did not use it out of modesty. He used it out of fear. He had once been through an insolvency, and he feared that if his name appeared as the inventor, creditors could seize the inventions, have them attached. So he disappeared behind his own works. The inventions went out into the world and had their effect. The inventor remained invisible.

One has to picture this in order to grasp its full weight. A person brings forth invention after invention that sustain a company, that give many people work, that are in use in workshops everywhere – and his name is on none of them. He does not even possess the least thing an inventor can possess: being named. He carried the ideas, completely, to the last consequence of not owning. He was the vehicle that did not even carry a plate with his name.

Johann Müller died in 1982, at the age of sixty, a month after a stroke, without regaining consciousness. On the website of the company that lived from his inventions there is to this day a sentence by the founder about the innovative products and ideas that had made the brand known worldwide. The name of the man from whose head these ideas came is not there.

IV. The vehicle carries on

The story could have ended here – as one of the many stories in which an inventor disappears and his work remains. But it has a sequel, and it is this sequel that makes Müller the teacher of this chapter.

One of his machines – the one for the hexagon sockets – became, years later, the starting point of a quite different development. Another inventor came into contact with it, saw in the way it produced the hexagon a principle that could be generalised – and carried this thought onwards for four decades, into a technology that had only its root in common with Müller's original task. The idea that Müller had carried changed vehicles. It transferred from one carrier to the next and travelled on.

That is the exact picture of what "the idea as a vehicle" means. The idea did not belong to Müller. Nor did it belong to the next carrier. It used the one, then the other. Each carried it part of the way – Müller from the task to the machine, the next from the machine to the general principle. Neither owned it. Both served it. And the idea that passed through both became in the end something that neither of them could have brought forth alone.

Whoever thinks of the idea as possession cannot grasp this process. For the possession-thinker, Müller's machine is a closed thing that belonged to an owner and was finished with his death. For the carrying-thinker it is a station on a longer road – a vehicle that handed its load on to the next. The one way of thinking sees an end. The other sees a handing-on. And only the second corresponds to what actually happened.

V. The two ways of not owning

Here a distinction is needed without which the chapter would be misunderstood. For Müller's not-owning had two sides that look the same from outside and are fundamentally different.

The one side is *free* not-owning, which comes from the nature of the idea. The inventor releases the idea because by its nature it belongs in the world

and not in his safe. He carries it, and at the destination he lets it go – not because it is taken from him, but because letting go is part of the matter. This free not-owning is an attitude. It is the healthy relationship that this chapter describes as the first layer.

The other side is *forced* not-owning – expropriation. Müller was not deprived of being named through the generosity of an attitude, but through fear of seizure and through a company that kept quiet about his contribution and built its own fame on it. That is not an attitude, that is an injustice. It is the exact opposite of what free carrying means – it is not the inventor who releases the idea; it is taken from him, together with his name.

These two ways of not owning look astonishingly alike from outside. In both cases the inventor's name is on no patent, he has no share in the fame, the work remains and the person disappears. But at the root they are opposites. Free not-owning is sovereignty – the inventor who has so much that he can give. Forced not-owning is robbery – the inventor is deprived of what is his due.

Why, then, Müller as the teacher? Because in him both coincided, and because he preserved the one within the other. He was robbed – and he carried nonetheless. Fear took his name from him; it did not take his activity from him. He went on inventing, work after work, to the very end. He did not let the expropriation take from him what is the inventor's real wealth – the doing, the carrying, the flow of the matter. That is the lesson. Not that one should let oneself be robbed – the systematic nature of this robbery will occupy us again in the fourth part, under the name it deserves. But that carrying an idea is independent of owning it. One can lose everything that the law recognises as property – the naming, the share, the fame – and still keep what really matters: the ability to bring the idea through oneself into the world.

VI. Why carrying is the healthier relationship

One could object that this attitude harms the inventor – that it makes him defenceless, exploitable, like the Müllers of this world who create fame for others and go away empty-handed themselves. The objection must be taken seriously, and the answer to it is not that the inventor should renounce his rights. He should safeguard his rights as well as he can; the whole second book of this series will deal with that. The answer is a different one. It concerns not the legal position but the inner state out of which inventing happens at all.

Whoever thinks of the idea as possession falls into a series of hardenings that stifle inventing. He becomes secretive – he can no longer speak openly about his work, because every word could be a theft. He becomes litigious – he spends his strength defending what he has instead of bringing forth what does not yet exist. He becomes bitter – every further development of his idea by others seems to him a robbery, rather than the onward journey of the vehicle that it really is. And he becomes barren – for whoever guards an idea like a treasure does not have his hands free to take up the next one.

Whoever carries the idea as a vehicle's load stays mobile in every respect. He can speak openly, because he knows that the idea is in any case bigger than his share in it. He can let others go on working on it, because the onward journey of the idea does not rob him but confirms him. He can hand over one idea and take up the next, because his identity does not hang on the one work but on the activity of carrying itself. He is, in the literal sense, more productive – not in spite of but because of his freedom from possession.

Here the first layer connects with what the first part called the root. The inventor who goes on working without recognition can do so because his self-worth depends not on being named but on the experience of the doing. Free carrying is the practical flip side of the self-assurance that comes from willing effort. And it is the flip side of benevolence: whoever wishes the world well grants the idea its path, even if that path leads through other

carriers. The first layer of practice is nothing other than the root applied to the relationship with the idea.

VII. What this means for the theory

The first of the nine layers is not a technique. It is a relationship. Before the inventor does anything – before he catches an idea, lets it ripen, sees it, builds it, tests it – he stands in a particular relationship to what he is dealing with. This relationship decides everything that follows. Whoever wants to own the idea will hold on to it, and in the holding it will harden in his hands. Whoever is ready to carry the idea will be able to move it through all the further layers, because he is not standing in its way.

That is why this layer stands at the beginning of the second part. It is the layer without which the other eight cannot be passed through. One may master search and model-building and critical questioning – if one thinks of the idea as possession, one will fail at one of these points, because the attitude of possession produces a hardening at every one of them. And one may be unpractised in all these techniques – if one knows how to carry the idea, one will learn them, because nothing prevents one from serving it.

Johann Müller did not teach this layer by talking about it. He lived it, right to its most bitter consequence. He carried what he did not own, and he went on carrying when even the little that might have been owned was taken from him. What remained of him is not his name – fear erased it and the company kept quiet about it. What remained of him is the idea that travelled on, and the lesson that lies in this travelling on. An inventor is not the owner of his idea. He is the stretch of road it covers in order to come into the world. Whoever has grasped that has the first layer.

That is the bridge to the second layer. For before an idea can be carried, it must arrive in the first place – it must be able to dock with someone. That presupposes a certain wakefulness, a pre-conceptual sensing that something is possible here, even before one could say what. This sensing is the *hunch*. It is the first half of the formula that separates the inventor from the person

who merely knows – hunch and perseverance. The next chapter deals with it.

7. The Hunch

Second layer. The pre-conceptual sense that something is possible. Hunch and persistence as the formula that separates the inventor from the one who merely knows.

I. Before the concept

Before an idea is there, something else is there. It has no name yet, no shape, no form in which it could be put into words. It is a sense that at a particular point something would be possible – or that something is wrong, even though everyone says it is right. This sense is the hunch. It is the second layer of inventing, and it comes before everything that is usually taken to be the beginning.

Notice again how language speaks of this state. *I have a feeling about this. There's something there. That smells like a solution. I can't put my finger on it, but something isn't right here.* All these phrases speak of a knowing that is not yet knowledge – of a perception that the concept has not yet caught up with. The one who has the hunch does not know what he senses. He only knows that there is something there that might be worth pursuing.

The hunch is not a weaker form of thought. It is a way of perceiving the world in its own right – pre-conceptual, whole, still undivided. The concept comes later. It divides the whole into graspable parts and names them. The hunch precedes it, as sheet lightning precedes the thunder. It is the first stirring of what may one day become an invention – and most inventions begin not with a clear thought but with this unclear stirring.

II. What sparks the hunch

The hunch does not arise out of nothing. It is sparked by something – usually by friction. Something resists. Something does not fit. Something takes more effort than it should. Something has been done the same way for years, and nobody asks why any more. At this friction the one with the hunch becomes alert before he knows to what.

Often the hunch is negative at first. It does not announce itself as *this could be done differently*, but as *this cannot be the last word*. A process that is cumbersome. A design that works harder than physics demands. An outlay out of all proportion to the result. The one with the hunch feels a resistance, a discomfort, an irritation – and this discomfort is the first sign that there is something to be gained at this point. Where others have long since looked away, because they have grown used to the state of affairs, his gaze stays caught.

For the hunch to be sparked, two things are needed. There must be the friction – the object at which something is wrong. And there must be a certain wakefulness – the readiness not to clear away the discomfort at once. Most people clear it away. They explain the state of affairs to themselves, get used to it, carry on. The one with the hunch lets the discomfort stand. He bears it without explaining it, long enough for the irritation to become a question. This is no special talent. It is an attitude – the same active wakefulness that the first part of this book described, applied to the beginning of inventing.

III. The hunch is not a flash of inspiration

Here an objection must be raised against a widespread picture. The popular notion of inventing knows the flash of inspiration – the sudden idea that strikes like lightning, the eureka moment in which the finished solution stands before the inner eye all at once. Such moments exist. But they do not stand at the beginning. They stand, if at all, far back in the process, when

long preparatory work suddenly falls into order. At the beginning stands not the flash but the hunch – and the hunch is the exact opposite of the flash.

The flash is bright, sudden, impossible to miss. The hunch is quiet, hesitant, easy to miss. The flash brings certainty. The hunch brings only a faint pull, a perhaps, a there-might-be-something. The flash forces itself upon you. The hunch has to be held on to, otherwise it vanishes again – drowned out by the noise of the day, by the next task, by the reason that says there is no time for this now.

Precisely because the hunch is so quiet, most people lose it. Not because they have no hunches – everyone has them – but because they do not take them seriously. They take the quiet sense for a whim, a distraction, something to be saved for later and then forgotten. The inventor does not differ from the non-inventor by having more hunches. He differs by not throwing his away. He takes the quiet sense as seriously as another takes a clear instruction. That is the first step, and it is inconspicuous.

IV. Hunch and persistence

Here lies the core of this chapter. The hunch alone is not enough. It is a beginning, but a fragile one. Between the first quiet sense and the tangible idea lies a stretch that only someone who does not let go of the hunch will cover. This second force is persistence. Hunch and persistence belong together – they are the formula that separates the inventor from the one who merely knows.

This can be made clear through the four possible cases. Whoever has neither hunch nor persistence does not invent – he has neither occasion nor stamina. Whoever has persistence but no hunch works hard on the wrong thing – he grinds away with great stamina at a problem that is none, or improves something that needs no improvement. Whoever has a hunch but no persistence is the saddest of the four cases. He is the person who says years later: *I had that idea back then too.* He really did have it – the hunch was there, genuine and right. But he did not hold on to it. Someone else who

had the same hunch and held on to it made the invention. Both had the hunch. What separated them was persistence.

Only the fourth case invents – the one who has both. He senses that something is possible at a particular point, and he returns to this sense, again and again, until it becomes clear. Persistence here is not stubborn clinging to a finished opinion. It is patient returning to an unfinished hunch. It holds on not to a result but to a trail. It allows the hunch slowly to become what it is not yet – a question, then a conjecture, then an idea that can be caught.

This persistence has what the third chapter of the first part described: it is bound to occasion, not to time, and free of reward. It returns to the hunch because the hunch calls, not according to a schedule. It endures over long stretches, often over years, without seeing an end point. And it does so without any reward being certain – for whether anything will ever come of the hunch, nobody knows at the beginning. Whoever persists only when success is assured will never follow the delicate, slow hunches to the end, for with them success is never assured.

V. The hunch and knowledge

There is a relationship between the hunch and knowledge that deserves close attention, because it cuts both ways. The hunch needs knowledge – and knowledge can smother the hunch.

It needs knowledge because it has to be sparked by something. Whoever understands nothing of a field has no hunches in it – he has only ignorance, which sometimes mistakes itself for a hunch. The genuine hunch presupposes that the one who has it knows enough to feel the friction at all, to notice that something is wrong here, to tell the possible from the impossible in rough outline. It is no substitute for knowledge but grows on it.

And yet the same knowledge can smother the hunch. For whoever knows a great deal often also knows very precisely why something does *not* work.

He knows the reasons why it has always been done this way. He knows the cases in which others tried and failed. He knows the textbook rules that forbid the obvious. This knowledge is valuable, but it has a flip side: it answers the hunch before the hunch could unfold. The knower says *that can't work* where the one with the hunch senses *that could work* – and often the knower is right in nine cases out of ten, and in the tenth lies the invention.

Herein lies a tension that accompanies the inventor all his life and that will occupy us as a problem in its own right in the third part of this book. Here the finding is enough: the inventor must know enough to have the hunch, and at the same time be able to set knowledge aside when it threatens to choke the hunch too early. He must be able to do both – rely on his knowledge and contradict it. This is not a lack of education; it is a higher form of education, one that knows where its own knowledge stops being reliable.

VI. How to train the hunch

Because the hunch is quiet and easily lost, the question arises whether it can be trained. It can. Not as a technique that forces the idea – no technique forces it – but as a cultivation of wakefulness that gives hunches a place where they can survive.

The first thing is not to throw them away. Whoever habitually passes over his quiet sense because it does not fit into the day unlearns the hunch. Whoever habitually holds on to it – even if only by noting it down – practises it. Many inventors keep a notebook in which they write down the half-thoughts that would otherwise vanish. Not because each of them will come to something – most will come to nothing – but because writing them down practises the attitude of taking one's own stirrings seriously.

The second thing is to give it time. A hunch that is forced into a decision at once usually dies. It needs a period in which it can go on working undisturbed – even and especially when one is apparently working on

something else. This period is the fourth layer, which a later chapter will deal with. Here the pointer is enough: whoever trains the hunch also learns not to push it.

The third thing is to seek out friction itself. Hunches come where something is wrong. Whoever exposes himself to surroundings where there is friction – workshops, unsolved tasks, unfamiliar fields, old processes that nobody questions any more – gives the hunch occasions. Whoever stays only where things run smoothly, in what is already ordered, in what has been fully explained, will seldom have a hunch, because there is nothing left there to sense.

These three – not throwing the hunch away, giving it time, seeking out friction – are not a method of invention. They are a way of life that makes inventing more likely. More than this cannot be done for the hunch. But less is not enough.

VII. What this means for the theory

The second layer is the antenna of inventing. It is the point at which the process really begins in the human being – not with a clear thought, but with a pre-conceptual sense which he follows or discards. Without this layer nothing arrives that could be carried. The first layer described the relationship to the idea – that one carries it and does not own it. The second describes how anything at all arrives that demands this relationship. First one senses, then one carries; and the sensing is so quiet that it is easily missed.

That is why the hunch comes so early. It is the most fragile of the nine layers and at the same time the most indispensable. One can master all the later layers – the search, the modelling, the critical examination – and still never invent, because the antenna that receives the beginning is missing. And one can be unpractised in all the later layers and still invent, if one has the hunch and holds on to it with persistence, for then one question leads to the next until the practice comes of its own accord.

What separates the inventor from the one who merely knows is, in the end, not the amount of his knowledge. The one who merely knows often knows more. It is that the inventor senses something where the knower only knows – and that he follows what he senses where the knower covers it over with his knowledge. Hunch and persistence: the quiet sensing and the patient returning. Whoever has both has the second layer.

This is the bridge to the third. When the hunch has been held long enough, it clears into something that can be caught – into an idea. But ideas do not fall from the sky as isolated flashes; they are, as the saying goes, *in the air*, and they come in particular classes, which the inventor learns to distinguish. How to catch them and what kinds there are is the subject of the next chapter.

8. Ideas in the Air

Third layer. Inventing as catching, not as generating. Four classes of ideas and the different wakefulness that each of them demands.

I. Ideas are in the air

There is an observation that every inventor makes sooner or later and that at first feels like a slight. No sooner has one grasped an idea for which one thought oneself unique than it turns out that someone else had it almost at the same time – in another place, neither knowing of the other, often in the same year. The history of technology is full of such double discoveries. The telephone was filed for a patent by two men on the same day. The infinitesimal calculus arose twice, in England and on the Continent, amid a dispute over who had it first. The theory of species came to two researchers at once, so that they had their papers presented together. The jet engine was conceived independently in two countries, almost in the same decade. Whoever collects the cases arrives at hundreds.

This observation teaches something about the nature of the idea. It is not the private property of a single head. It is, as the old phrase says, *in the air*. The air is the state of knowledge and technology at a particular time – the sum of what has been thought, built, published, attempted. When this state reaches a certain maturity, a certain idea becomes possible, indeed due. It then hangs in the air like a fruit that is ripe. Whoever stands nearby and has the antenna for it picks it. That several pick it almost at the same time is no coincidence but the consequence of its having become ripe for all of them.

This connects to the first layer. If the idea is in the air and does not arise in the inventor's head, then here too the inventor is not its owner but the one with whom it arrives. He did not make it – he *caught* it. And catching is the third layer of inventing.

II. Catching

Catching is something other than generating. Whoever generates brings forth from himself. Whoever catches seizes something that comes towards him. The inventor does not generate the idea – he holds on to it when it flies to him. Catching therefore presupposes the two previous layers. It presupposes the relationship that is ready to carry the idea rather than own it. And it presupposes the hunch, the antenna that senses that something is there before it has taken shape. Without an antenna the idea flies past and nobody notices. With an antenna the quiet sense becomes a grip – the moment in which the hunch condenses into something that can be named, noted down, pursued.

But not all ideas are in the air in the same way. Some hang ripe and low, so that many hands reach for them. Some hang too high, because the time is not yet ripe. Some have been hanging there so long that nobody sees them any more. And some hang in quite a different air, from which only the one who reaches over the fence of his own field can fetch them. Whoever catches does well to know what kind of idea he has caught – for each kind demands a different continuation. We distinguish four classes.

III. The first class: the ripe idea

The ripe idea is fully in the air. The knowledge that makes it possible is available; the need that it satisfies is palpable; the neighbouring inventions on which it rests have been made. It is due. Several people stand near it, and several reach for it, often in the same year. This is where the double discoveries live of which the beginning of this chapter spoke.

Whoever catches a ripe idea should be clear about two things. The first is humility. He has done nothing unique. What he caught would have come without him – perhaps a year later, perhaps through someone else, but it would have come. This takes nothing from the value of the thing, but it takes from the inventor the illusion of being irreplaceable. The second is urgency. Because the idea is ripe, others are reaching for it. With the ripe idea, what decides is not the rarity of the idea but the speed and quality of the execution. Whoever catches it is in a race, whether he wants to be or not. He does not win it by having had the idea – others had it too – but by bringing it into the world first and best.

The ripe idea is the most common, and it is the one with which most inventors earn their bread. It is not to be despised. But it is not the place where the inventor sets himself apart from the many who breathe the same air.

IV. The second class: the premature idea

The premature idea is right, but it comes too early. The insight is correct – only the conditions that would support it are missing. The material it would need has not yet been invented. The tool with which it could be built does not yet exist. The market that would take it up does not exist. The neighbouring inventions on which it would depend have not yet been made. It hangs in the air, but too high – one sees it, one reaches for it, and one cannot bring it into the world.

Whoever catches a premature idea finds himself in a difficult position. He holds something true in his hand and cannot prove it, because the world is not yet ready. He is taken for a dreamer, even though he is right. Often the premature idea is realised decades later by someone else – when the air has changed and the same idea has become ripe. The second harvests what the first saw. This is one of the bitterest constellations in an inventor's life, and it will occupy us further in the systemic part of this book.

For inner practice one distinction matters here. The inventor must learn to recognise the premature idea as such – and then decide whether to wait, to carry it further or to let go. Sometimes carrying it further is right: one holds on to the idea, helps work on the missing conditions, bridges the time. Sometimes letting go is right: one puts the idea aside, notes it down, turns to something riper and returns when the air has changed. Both demand the freedom from ownership that the first layer described. Whoever clutches the premature idea because it is *his* wears himself out on a fruit that was not yet ready to be picked. Whoever knows how to carry it can hold it without breaking on it – or let it go without losing it.

V. The third class: the overdue idea

The overdue idea is the counterpart of the premature one. It comes not too early but far too late. It would have been possible for a long time – it needs no new material, no new tool, no new neighbouring invention. Everything it requires has long been there. And yet nobody has caught it, for years, sometimes for decades. Not because it is difficult, but because people have grown used to the state of affairs it would remedy.

Here the friction from the previous chapter returns. The overdue idea lies exactly where everyone has looked away, because the discomfort has become habit. People have resigned themselves to its being cumbersome. It has been done this way for years. Nobody asks why any more. The barrier in front of the overdue idea is not technical but perceptual. What was lacking was not the ability but the eye. That is why the overdue idea always seems astonishingly simple in hindsight – *anyone could have thought of that*. That

is precisely its nature: anyone could have, but nobody did, because everyone had grown used to it.

Whoever catches an overdue idea did not know more than the others. He simply did not look away. He let the discomfort stand where everyone else had cleared it away, and out of the irritation he left standing the solution came towards him. Such ideas are often especially valuable – precisely because they require no new technology, they can be put into practice at once. And they are among the most satisfying an inventor can catch, because in them the second layer pays off directly: the wakefulness that does not accept the habitual.

VI. The fourth class: the far-fetched idea

The far-fetched idea is not in the air of one's own field at all. It comes from far away – from an unfamiliar discipline, a distant practice, a principle known in quite a different context that nobody would have related to one's own problem. It is the rarest of the four classes, because it demands an antenna that reaches beyond the fence of one's own field. Whoever breathes only the air of his own discipline never catches it – for him it is not in the air but behind a wall.

The far-fetched idea arises through transfer. A mechanism that nature has used for millions of years is translated into a machine. A process from the kitchen solves a problem in manufacturing. A principle from music orders a movement in a gear train. These transfers seem far-fetched as long as one thinks within the boundaries of the disciplines – and obvious as soon as one recognises the common principle that lies beneath the unfamiliar surface. The inventor who catches far-fetched ideas is the one who wanders between fields, who stops in other people's workshops, who reads what is none of his business. His breadth is not distraction but the field from which the far-fetched idea comes.

Because it comes from a different air, the far-fetched idea has a special property: it is seldom caught by several people at the same time. It is the

opposite of the double discovery. Here, and almost only here, the inventor really is the one who saw something that nobody else saw. The systematic approach to such transfers – how one does not leave them to chance but searches for them methodically – will be the subject of a later chapter of this part, when we enter the cross-layer of the methodology of invention.

VII. What this means for the theory

The third layer is catching, and the four classes are the map that tells the antenna what it has caught. For the class decides what is to be done next. The ripe idea demands speed – one is in a race and must execute before someone else does. The premature idea demands patience or letting go – one must be able to wait or to hand over without breaking. The overdue idea demands that one unlearn one's own habituation – the resistance lies not in the matter but in the eye. The far-fetched idea demands breadth – one catches it only by breathing unfamiliar airs. Four classes, four disciplines. Whoever does not distinguish which one he has caught treats the ripe like the premature and runs himself into the ground, or the overdue like the far-fetched and searches far away for what lay at his own front door.

This map is not a recipe that forces the catch. No chapter of this book promises that ideas can be compelled to appear. But it helps the inventor to classify what he has caught – and thus to bear the disappointment of the double discovery as well as the loneliness of the far-fetched idea. It takes presumption away from the ripe idea and gives the overdue one back its dignity. It puts in order what the antenna receives.

This is the bridge to the fourth layer. An idea that has been caught is seldom one that can already be built. Usually it is an unfinished something that must first ripen – and this ripening happens not by forcing it but by letting it happen, often below consciousness, at a time when one is apparently working on something else. This time is incubation. It is the subject of the next chapter.

9. Incubation

Fourth layer. The ripening of the idea below consciousness. Why ripening cannot be forced but only allowed to happen – and why that is not laziness.

I. The ripening that cannot be forced

An idea that has been caught is seldom a finished one. Usually one holds an indeterminate something in one's hand – a direction, a possibility, a half-thought that does not yet know what it is. Whoever now believes he need only sit down and work out the solution with enough willpower has a strange experience. The harder he presses, the less comes. The idea resists. It does not want to be forced.

And then the odd thing happens. One leaves off, turns to something else, goes for a walk, sleeps on it – and suddenly, unbidden, the solution is there. Not because one was working on it at that moment, but because it ripened of its own accord in the meantime. This interval, in which apparently nothing happens and in truth the decisive thing happens, is incubation. It is the fourth layer of inventing, and it is the most misunderstood.

Here too language gives the process away. *I need to sleep on it. That has to settle first. Let it ripen. It's working away inside me.* All these phrases speak of a doing that is not conscious doing – of work that a person does not do but allows to happen. The gardener knows this. He sows and waters, but he does not pull at the shoots to make them grow faster. Growing is not his business. His business is to create the conditions and to wait.

II. What happens in incubation

That an idea ripens while one is occupied with something else was always known to inventors and researchers, before psychology had a name for it. The mathematician Henri Poincaré described it in a famous account. He had worked in vain on a problem for weeks, then put it aside and gone on a journey. At the moment he put his foot on the step of the carriage – thinking

of something quite different – the solution came to him, with a certainty that left no doubt. The work had not been suspended during the break. It had carried on, only below the threshold of consciousness.

The researcher of thinking Graham Wallas early on made a simple order out of this. He distinguished four stages: preparation, in which one works on the problem with full consciousness; incubation, in which one lets it rest and it goes on working below the threshold; illumination, in which the solution appears unbidden; and verification, in which one checks whether it holds. The second of these stages is the quietest and the most mysterious. In it the real work happens, without a worker looking on.

What happens there below the threshold can only be described from outside. The mind goes on connecting what it gathered during preparation. It tries out combinations, discards most of them, holds on to one. It does so more freely than waking thought could, because it lacks the supervisor who asks at every step whether it is permitted. Waking thought is disciplined but narrow; it follows the known paths. Incubating thought is unbound; it takes the far-fetched paths too. That is why it is often precisely those connections that waking thought would never have arrived at that come out of incubation – the transfers from unfamiliar fields of which the previous chapter spoke as the fourth class.

III. The conditions of incubation

It would be a misunderstanding to conclude from all this that one need only go for enough walks and sleep enough, and inventions would come of their own accord. Incubation has conditions, and the first is strict: it presupposes preparation. One can only incubate on something one has previously thought about with full commitment. During the break the mind works further only on what it was saturated with beforehand. Whoever has not exerted himself has nothing ripen – he lacks the material from which incubation draws its connections.

This is important, because it separates incubation from doing nothing. It is not the opposite of effort but its fruit. First comes the hard, wakeful, often fruitless work – the chapter on effort as a value described it – and from it, and only from it, can the quiet ripening emerge. The break after effort is productive; the break without prior effort is merely empty.

The second condition is letting go of conscious grip. As long as one clutches the problem, it cannot sink below the threshold, where it goes on working. One must really put it aside – not as a trick to lure the solution out, but honestly, trusting that it will carry on even without one's own supervision. This trust is hard for people who have learned that only what one forces through by will counts.

The third condition is a particular kind of activity during the break – ideally one that occupies the body and leaves the mind free. Walking, handwork, driving, sleep. Not the distraction that captures attention again, but the calm, steady occupation beneath which thoughts may wander. Many inventors have their best ideas not at the desk but on the move. That is no coincidence. The desk is the place of preparation. The road is the place of incubation.

IV. Why the modern world is unlearning incubation

Incubation needs something that has almost disappeared from the present order of life: empty time. Time without a task, without stimulus, without a screen. Time in which one can walk without a destination, or sit without doing anything, or sleep without being interrupted. It is precisely this time that the modern world has systematically eliminated.

The calendar is full. The break is bridged with the phone. The journey becomes time for getting things done. Sleep is cut short. Boredom, which used to accompany every walk, every wait, every evening after work, is today considered a deficiency to be filled at once with a stimulus. With that, the ground on which incubation thrives is destroyed. Whoever never has

empty time has nothing ripen – not because he works too little, but because he rests too little.

The same order that eliminates empty time at the same time demands the immediate result. The solution is to be there by Friday. The report by tomorrow. The result measurable, plannable, scheduled. But incubation cannot be scheduled. It comes when it comes, and it does not come at all if one pushes it. Whoever orders an invention to schedule gets at best the obvious – what waking thought reaches along the known paths. The non-obvious, which comes from incubation, eludes the schedule. Here lies a deep incompatibility between the rhythm of inventing and the rhythm of the administered world – an incompatibility that will occupy us as a theme in its own right in the systemic part of this book. For inner practice the finding is enough: whoever wants to invent must win back the empty time that his surroundings take from him.

V. The danger of incubation

It would be dishonest only to praise incubation. It has a flip side, and whoever keeps quiet about it turns an insight into an excuse. For *I'm letting it ripen* is the most comfortable sentence in the world for someone who in truth is only procrastinating. Letting things ripen can be a disguise for avoidance. Many an idea is never finished, not because it is incubating, but because its originator is dodging the laborious working-out and adorns the delay with the noble word *incubation*.

The difference between genuine ripening and disguised avoidance lies in what went before and what follows. Genuine incubation follows serious preparation – one has worked oneself to a standstill before letting go. Disguised avoidance follows nothing at all – one never began the hard work and calls the not-beginning ripening. And genuine incubation has an end: one returns, checks whether something has become clear, and if so, one works on. Disguised avoidance never returns; it keeps on putting off.

That is why incubation belongs together with the persistence of which the seventh chapter spoke. Persistence is the patient returning to the matter; incubation is the letting go of the matter. Together they form a rhythm – tension and release, grip and letting go, preparation and rest, and grip again. Whoever only grips forces nothing and exhausts himself. Whoever only lets go never comes to an end. The inventor breathes between the two. Incubation is the breathing out, but breathing out must be followed by breathing in again, otherwise it is not breath but standstill.

VI. The art of waiting

Because incubation may be neither forced nor extended at will, dealing with it is an art – the art of a waiting that is not doing nothing. It has a few rules, which are not recipes but attitudes.

The first: trust the matter. Whoever has done the preparation may trust that it goes on working even when he is not looking. This trust is itself a prerequisite – the anxious person who keeps checking whether it is ripe yet disturbs the ripening like someone who digs up the seed to look at the roots.

The second: keep the antenna open. Illumination comes unbidden and at inconvenient times – while walking, on waking, in the middle of some other occupation. Whoever then has no means of holding on to it loses it again. The notebook of which the chapter on the hunch spoke belongs here too. The ripe thought surfaces like a fish at the surface of the water; whoever has no hand ready sees it disappear again.

The third: protect empty time. In a world that fills every gap, the inventor must defend his gaps. The walk without a phone. The hour without a task. Undisturbed sleep. This is not a comfort he allows himself, but a working condition he fights for. Empty time is the field of incubation, and whoever has no field reaps no harvest.

VII. What this means for the theory

The fourth layer is the quietest. From outside it looks like inactivity, and it is precisely there that its nature is misunderstood. It is not inactivity but a different kind of activity – one that a person does not do but allows to happen, after having created the conditions for it. It is the place where the caught idea of the third layer transforms from an unfinished something into something that can be built. Without it the idea would remain raw; with it, it becomes ripe.

Incubation thus stands as the hinge between catching and seeing. It demands of the inventor the hardest thing an active nature can achieve: not to act when acting is of no use. It demands trust instead of grip, patience instead of force, empty time instead of filled. In a world that rewards the opposite, this is an imposition – and at the same time a prerequisite without which inventing withers. Whoever never rests has nothing ripen; whoever only rests gets nothing finished. The art lies in the rhythm.

This is the bridge to the fifth layer. When the idea has ripened, it begins to show itself – not yet as a drawing on paper, not yet as a construction in steel, but first as an inner image. The inventor sees the solution before him before he builds it; he turns it in his mind, tests it in his mind, discards and improves it in his mind. This inner vision, which precedes the outer work, is the fifth layer. It is the subject of the next chapter.

10. Inner Visualisation

Fifth layer. The inner image that comes before building. First, a word about the two paths of inventing and the place of the methodology of invention.

I. First: the two paths of inventing

Before we enter the fifth layer, a word on the order of the whole is needed, because a seam runs through here. The nine layers fall into two halves. The first four – the relationship, the hunch, the catching, the ripening – are the

finding, receiving half. The following five – the inner image, design with feedback, the search, the model, the deed – are the shaping half. With this chapter the second half begins. And at this point an imbalance must be corrected that the account so far may have created.

What we have described up to now was mostly *one* path to the idea: the emergent one. The inventor as an antenna, who senses, catches, lets things ripen; the problem that announces itself as friction. That is a true and important side of inventing, but only one. Beside it stands a second path: the targeted one. One has a particular problem and searches the solution space systematically instead of waiting for the idea to come. Anyone who read only the first half of this book might get the impression that inventing is, at its core, waiting and wakefulness. That would be one-sided – and unfair above all to the methodology of invention, which arose as an antidote to the genius cult of mere waiting.

The deeper distinction is not *receiving versus searching* but **problem finding versus problem solving**. Our layers so far – the hunch, the friction, the overdue idea – are in truth the theory of problem finding: how a person comes to see at all that there is a problem here worth solving, often before anyone has posed it. The targeted methods mostly take this finding for granted. They begin where the problem has been cleanly formulated and ask: given this contradiction, how do I find the resolving principle systematically rather than by trial and error?

These methods have names and a history, which shall only be named here, not set out. The best known is *TRIZ*, developed by Genrich Altshuller – drawn from the analysis of tens of thousands of patents, and carried by the conviction that inventive solving can be made repeatable: that behind inventions stand recurring contradictions and recurring principles for resolving them, and that one can think backwards from the *ideal final result* instead of feeling one's way forward from what exists. Beside it stands German design methodology – Friedrich Hansen, Wolfgang Rodenacker, the VDI guidelines – which pursues the same aim from the engineering tradition: ordering the thought processes without trying to force the

invention. And beside that, Fritz Zwicky's morphological box, which spans the solution space systematically by crossing the parameters of a problem with their possible values.

These approaches aim at the solution space of a problem already formulated. Alongside them stands a second family, which does not order the solution space but attacks the track of thought itself – the prevailing groove in which thinking runs of its own accord. Its best-known representative is Edward de Bono's *lateral thinking*, which deliberately leaves the obvious track: through provocation, through the sideways leap, through suspending judgement, in order to get where the straight line never led. Related to it are Alex Osborn's *brainstorming* and William Gordon's *synectics*, which break up the obvious through analogy and through separating idea from evaluation; and, as a systematic form of transfer, *bionics*, which carries nature's solutions over into technology. These procedures, which break up the habitual pattern, do not belong only to solving: they loosen precisely that habituation which hides the overdue idea, and they open the passage into the foreign fields from which the far-fetched idea comes.

None of these approaches, whether directed search or the breaking of habit, is a recipe for the idea – no method forces that – but they are tools that make finding and solving more likely.

The two paths are not opponents. They interlock and meet at the **contradiction** – where the problem that has been found sharpens into tension. Finding itself is again divided in two: recognising, when the problem announces itself as friction – which the previous chapters have described – and active searching, when one systematically combs a field for inefficiencies or asks backwards from the ideal final result what is missing. We do not take up this active searching and the systematics of solving here, but in the cross-layer of the methodology of invention, which picks up the thread as a chapter of its own in this part, and in the chapters on the search and on the structure of tension. Here the map is enough, so that the reader does not take the receiving layers for the whole.

But what is the same on both paths is this: however the idea arrives – by sensing or by searching – at some point it must take on inner form before it takes on outer form. The inventor sees the solution before he builds it. With this the shaping half begins, and its first layer is the inner image.

II. The inner image

There is a state that most experienced designers know and that is nonetheless hard to describe, because it happens entirely inside. The inventor sees what he has not yet built. Not as a vague hunch – that was the second layer – but as a concrete, definite image. He sees the parts, their arrangement, how they work together. He sees how one moves against the other, where something engages, where something slides, where something bears the load. The image is inward, but it is exact.

This inner seeing is more than memory and more than fantasy. It is a mental picture one can work with. The inventor turns the mechanism in his mind, looks at it from the other side, lets it run, stops it, changes a part and sees what follows. What would cost hours on paper or in material happens in the inner image in seconds. Some inventors report that they can see a whole machine in motion before their inner eye, with all its shafts, levers and tooth flanks, so clearly that they notice a collision before they have drawn a line. This ability is the fifth layer of inventing – the first in which the idea takes on form.

III. What inner seeing achieves

The value of the inner image lies in its being the cheapest and fastest test bench there is. Before a workpiece is made, before a drawing is produced, before material is used and time is spent, the inventor can play the solution through in his mind. He sees whether the movement runs freely or strikes something. He sees whether the force gets to where it is needed. He sees whether two parts are in each other’s way. Most bad designs already fail here, inside, and are never built – which is a blessing, for every faulty design rejected in the head saves the cost of a built one.

Inner seeing also allows a kind of trying out that would be too expensive in reality. One can go through twenty variants before deciding on one. One can reverse an arrangement, shift an axis, leave out a part and see at once whether it works without it. It is this mobility of mind that distinguishes the experienced designer from the beginner – not that he makes fewer mistakes, but that he makes most of them where they cost nothing: in the inner image.

IV. The limits of the inner image

Yet the inner image holds a danger, and whoever overlooks it relies on an adviser who flatters. For the inner image is too obliging. It shows the movement as smooth where reality jams. It leaves out friction, the play of tolerances, deformation under load, heat, wear. The mind sees the ideal machine, not the real one. It shows what *is supposed* to work and readily suppresses what stands against it.

That is why the inner image must not be the final judge. It is a powerful first draft, but it must be tested against reality – against calculation, against the drawing, against the material, against the model. The following layers serve exactly this testing. Whoever trusts the inner image blindly builds machines that ran in the head and stand still in the workshop. The art is to trust the inner image enough to use it for designing, and little enough to measure it against reality.

V. That people see differently

It would be wrong to declare inner seeing an indispensable gift of every inventor. People differ greatly in this. Some see inwardly so vividly that the image almost equals perception; others see hardly anything and think rather in words, in relations, in formulas. There are people who almost entirely lack the inner image, and among them are capable designers. The ability to see inwardly is an advantage, but not a gate that would bar entry to the ungifted.

For the fifth layer can also be served from outside. Whoever sees little inwardly can move the image outward – into the quick sketch, the simple model, the movable hand drawing, the screen on which the mechanism can be rotated. What one person does in the head, another does on paper; both pass through the same layer, only with different means. This matters, because it protects this chapter from a misunderstanding: inner visualisation is the layer in which the idea first takes on form – whether that form arises in the head or on the sheet is a question of disposition, not of rank.

VI. How inner seeing is trained

Whoever has the gift can sharpen it; whoever hardly has it can approach it. The means is the same in both cases, and it is unspectacular: one must fill the store from which the inner image draws. For inner seeing is only as good as the stock of correctly understood mechanisms at the seer's disposal. Whoever has never taken a gearbox apart, never made a guideway, never seen how a material behaves under load, lacks the building blocks from which a reliable inner image could be assembled – his image would be a pretty fantasy with no anchor in physics.

That is why inner seeing is trained on the real. Take machines apart and put them together again. Build with your hands. Watch how things actually move, where they actually break, how they actually wear. Every such experience lays ready for the mind a part that will later behave correctly when it is assembled in the inner image. In this way the layer connects with what the chapter on the hunch said: inner seeing, too, needs knowledge as nourishment. It is not a talent that comes from nowhere, but one that rests on a long acquaintance with reality.

VII. What this means for the theory

The fifth layer is the first shaping one. In it the ripened idea turns into a form – first inwardly, before it becomes outward. It is the cheapest layer, because it needs no material, and the fastest, because it works without the inertia of reality. It is at the same time the most deceptive, because it shows

what is supposed to work and keeps quiet about what stands against it. Its value and its danger lie in the same property: that it shows the ideal.

This also says why it cannot stand alone. Left to itself, the inner image drifts – it grows ever more obliging, ever further removed from reality. It must be tied back to the outside: to the sketch, the calculation, the first rough model, which report the contradiction of the real back into the inner image. From this back and forth between inner seeing and outer making – each correcting the other – the actual design emerges. This seeing with feedback is the sixth layer. The next chapter deals with it.

11. Visualisation with Feedback

Sixth layer. How the inner image, through the feedback of the outside world, becomes the actual design. Designing as dialogue, not as transcription.

I. Why the inner image alone is not enough

The previous chapter ended with a warning: left to itself, the inner image drifts. It grows ever more obliging, smooths friction away, suppresses the load, keeps quiet about whatever stands in the way of the wish. The mind likes to show the ideal machine, not the real one. To rely on it alone would be to trust an adviser who always tells you what you want to hear.

What the inner image needs is a counterpart that contradicts it. Something outside the head that does not flatter but shows what is wrong – a sketch, a drawing, a calculation, a rough model. These outer things have a property the inner image lacks: they resist. A line on paper lies where it lies, even if it does not match the wish. A figure either works out or it does not. Two drawn parts visibly overlap where the inner image still let them glide neatly past each other. It is the resistance of the outside that carries inventing from the realm of wishes into the realm of reality. And this resistance only takes effect when it is fed back – when what the outer representation shows flows

back into the inner image and corrects it. This flowing back is the sixth layer.

II. Feedback

The process is easy to describe and hard to do. One draws what one has seen inwardly. And the drawing shows something one had not seen. A proportion that does not work out. A lever that is too short. A space already taken by another part. The pencil brings to light what the mind had skipped. One corrects the inner image, draws again, and the new drawing shows the next mistake. So it goes back and forth, round after round, until image and representation agree.

In this back and forth something remarkable happens: the drawing is cleverer than the person drawing it. The hand knows what the eye missed. Anyone who has ever designed at a drawing board or at a screen knows the moment when the representation suggests a solution one had not thought of oneself – because what has been made visible shows connections that remained hidden in the head. The outer representation is not merely an image of the thought. It is a tool of thinking. Whoever uses it only as a record, to fix what is already finished, gives away its real power: that it thinks along by resisting.

III. The tools of feedback

Feedback has various tools, and each reports back a different kind of resistance.

The fastest is the freehand sketch. It is imprecise, fleeting, dashed off in seconds – and for that very reason the inventor's most important instrument of feedback. Because it costs nothing, one can make twenty and throw away nineteen. It shows the rough geometry, the arrangement, the proportions of the parts, and it shows them at once. The sketch is not an inferior drawing. It is the organ of early thinking, and whoever skips it to draw neatly straight away loses the mobility in which the design comes into being.

The scale drawing reports back a stricter resistance: that of dimensions. What still roughly fitted in the sketch must here come together to the millimetre. The screen, on which the model can be rotated and checked in space, reports the resistance of the third dimension – it shows collisions that stayed hidden on the flat sheet and calculates relationships that would be laborious by hand. Finally, the rough physical model reports the resistance of the material itself: it wobbles where the drawing seemed stable, it jams where everything looked free, it falls over where the centre of gravity was in the wrong place.

The inventor moves between these tools and chooses each time the one whose resistance he needs at that moment. Whoever is still looking for the geometry reaches for the sketch; whoever wants to check the forces, for the calculation; whoever distrusts the whole, builds a rough model. The art lies in choosing the right counterpart – one that resists at the point where one can least trust oneself.

IV. Designing as dialogue

This allows us to say what designing really is. It is not the transcription onto paper of a finished inner image. The naive notion that the inventor first sees the whole solution before him and then merely transfers it is the same genius legend against which the earlier chapters have already objected. Nobody works like that. At the outset the inner image is incomplete, and it only becomes complete in dialogue with its outer representation.

Designing is this dialogue. One sees a piece, draws it, sees more in what has been drawn, draws again, sees more again. The design grows in the alternation between inside and outside, and neither side carries it alone. The inner image supplies the direction, the outer representation supplies the resistance, and out of both together the design emerges. The finished design is the point at which the dialogue comes to rest – at which inner image and outer representation at last show the same thing and nothing is left to correct. Until then the design is not a state but a process: a conversation between what one sees and what one makes.

V. Where feedback fails

Feedback can fail in three ways, and the inventor does well to know each of them.

The first is falling in love with the representation. A beautiful drawing, a gleaming on-screen model, can be so convincing that it hides the bad idea behind it. The outer representation can flatter too – not the inner image, but the eye. Whoever falls in love with the neatness of his drawing stops questioning it and makes it a proof instead of a touchstone. The representation is then no longer a counterpart but a mirror.

The second is premature precision. One can put a great deal of effort into working out a model down to the last detail whose basic idea is not yet settled. Every hour invested then binds one more firmly to a design that may be wrong – precision makes the error expensive and therefore stubborn. Whoever becomes precise too early loses the freedom to discard the whole thing. That is why the rule applies to which the chapter after next is devoted: rough and fast first, then precise. Feedback should stay rough as long as the idea is not settled, and only become fine once it holds.

The third is the loop that never closes. One draws and discards, draws and discards, without ever committing. What looks like care is then avoidance – the fear of committing oneself, dressed up as striving for something better. Here the danger returns that incubation already knew. The dialogue needs an end. At some point the inventor must declare that image and representation now agree, and move on – otherwise the sixth layer turns round on itself, and nothing gets finished.

VI. The role of calculation

Among the tools of feedback, calculation deserves a word of its own, because it is the least flattering of all. A sketch can be talked up, a model interpreted in different ways – a number cannot. A dimension works out or it does not. A stress stays below the limit or exceeds it. Calculation is the place

where the inner image meets the hardest resistance, and precisely for that reason it is indispensable: it says no where everything else still says yes.

But calculation has its place *within* the dialogue, not *before* it. One cannot calculate one's way to an idea – one calculates in order to test an idea. Calculation presupposes a design which it confirms or rejects; it does not produce one. Whoever calculates too early calculates into a void, because he has nothing yet whose dimensions would be worth it. Whoever calculates too late may already have committed himself to a design that the first figure would have toppled. Calculation belongs where the idea has enough form to bear numbers and is still mobile enough to fail on them. Then it is the sharpest form of feedback there is.

VII. What this means for the theory

The sixth layer is the one in which the idea first meets resistance and thereby becomes real. Nothing has been built yet – there are lines, figures, rough models. But these lines, figures and models already obey reality, no longer the wish. In them the contradiction of the real reports back for the first time, and from working through this contradiction – the back and forth between inner seeing and outer making – the actual design emerges. It is not a state but a dialogue, and its end is the point at which image and representation agree.

With this the fifth layer is complete, which supplied the inner image, and the sixth, which fed it back against reality. The design stands – on paper, on the screen, in the rough model. But a design that stands on paper raises a new question that becomes urgent before one builds: Does this already exist? Has someone else already taken this path? What knowledge, what patents, what solutions are already in the world that the inventor must know before he goes further? Checking what is already there – and the four very different reactions this check can trigger in the inventor – is the seventh layer. The next chapter deals with it.

12. The Search and Its Four Reactions

Seventh layer. Checking what is already there – and the four very different reactions this check triggers in the inventor. The search is not a formality but a confrontation.

I. Why the search comes here

The design stands – on paper, on the screen, in the rough model. And now a question presses that can no longer be put off: Does this already exist? Has someone else already taken this path? What knowledge, what patents, what solutions are already in the world? Examining this question is the seventh layer, and its place in the sequence has been chosen with care – neither earlier nor later.

Not earlier, because whoever searches before having a design of his own does not know what to look for and runs the risk of being infected by what he finds. The fresh inner image that the fifth layer supplied is sensitive; it easily takes on the shape of whatever one sees first. Whoever reads too early what others have done thinks from then on in their tracks and loses his own. Not later, because whoever only searches once he has already built may have spent months on something that has long existed – or have stepped on someone else's rights without knowing it. That is why the search stands exactly here: the design is firm enough to allow a targeted search, and still open enough to be changed. It is the same logic that in the previous chapter placed calculation within the dialogue, not before it – one checks what one has, not what one does not yet have.

II. What the search is

The search is the confrontation of one's own design with what the world already knows. It goes through patents, literature, products, processes, the knowledge of others. It places what arose in the head and on the paper of one individual next to everything humanity has already thought about the same matter. That is not an administrative act to be ticked off, but a serious

encounter – often one that is feared, because it can decide the fate of weeks and months of work.

For the search can confirm or destroy. It can show that one is standing on open ground – or that one has been following a predecessor without knowing it. It can deliver a solution that one gratefully takes up – or proof that one's own is worthless. Precisely because so much is at stake, the search is emotionally charged, and what it triggers in the inventor is seldom cool. There are, at the core, four reactions, and the inventor does well to know each of them in himself – for three of them can lead him astray.

III. What the search cannot see

Before the reactions are discussed, a blind spot must be named that no amount of care removes, because it is built into the system itself. Between the filing of an invention – which secures its priority, its place in time – and its publication there lies a period, in most systems about eighteen months, during which the invention already exists, is legally filed, enjoys priority – and remains invisible to anyone searching. Whoever searches today is blind to everything filed in the last year and a half. A flawlessly empty search result can be overtaken by an application that was already on file when one searched, only still hidden. This black hole of time is not a defect that could be remedied; it follows inevitably from a law that grants priority on filing and publishes only later. Its consequence is relentless: *nothing found* can never mean *nothing filed*.

And even leaving the black hole aside, no body of search material is ever complete. Not everything that came before one is patented – much valuable knowledge lives on as trade secrets, as in-house know-how, as practical skill that was never written down. Not everything published can be found – it hides in foreign languages, in old and remote sources, under words other than those one searched with, in the non-patent literature that no patent register reaches. And the abandoned attempts of others, what was quietly discarded, often leave no trace at all. The searchable stock is a fraction of

what has been thought and done. The searcher searches a lit room and must remember that the house behind it remains dark.

From this follows the only honest stance: the search cannot prove a negative. It shifts probabilities; it does not create certainty. A clean result means *probably free, as far as the visible stock reaches at this moment* – never *free*. The thorough inventor does everything to widen the lit room – he searches more deeply, in other languages, under other names, in neighbouring fields – and knows all the while that a margin of darkness remains which no care can remove. That is no reason to search carelessly; it is the reason to search thoroughly and to remain modest about the result. And it colours the four reactions now to be discussed – above all the first, relief, which thereby proves doubly deceptive.

IV. First reaction: relief

The first reaction occurs when the search finds nothing. The field seems empty; nobody has done what one intends to do oneself. The reaction is relief, often joy – one takes oneself to be the first, the way seems clear.

This relief is the most deceptive of the four, because the empty field is ambiguous. It may be empty because nobody had the idea before – then the joy is justified. But it may also be empty because others tried and gave up, without their failure finding its way into the patents. It may be empty because the thing does not hold up technically and no expert ever pursued it further. It may be empty because there is no market and nobody had a reason to do the obvious. And it may, most often, be only *apparently* empty – because one searched in the wrong place, with the wrong words, in the wrong language. Most empty fields are poorly searched fields.

That is why the right stance towards relief is to distrust it. Whoever finds nothing should ask why nothing is there before rejoicing – and should search more deeply, in neighbouring fields, under other names, in older and foreign-language sources. The genuinely empty field exists; but it only

reveals itself to someone who has doubted long enough whether it really is empty.

V. Second reaction: disappointment

The second reaction occurs when the search finds what one believed one was inventing oneself. Someone else got there first. The double discovery the eighth chapter spoke of has happened – only this time on the bitter side: one is not the first. The reaction is disappointment, often hurt, sometimes the loss of all appetite for the whole thing.

This reaction is understandable and yet can be overcome if one remembers the first layer. Whoever thinks of the idea as possession experiences the predecessor as a thief who has taken something that belonged to him. Whoever carries the idea as a vehicle experiences him as another carrier of the same freight – as confirmation that the idea was ripe, not as robbery. And in practice the find often opens more than it closes. One can study the other's solution, see its weaknesses, improve it. One can get round it. One can take it as proof that one was on a real trail, and direct one's own strength towards what the predecessor did not consider. The disappointment is real, but it is seldom the end – unless one makes it so.

This possessive attitude, put in general terms, even has a name – two, in fact. Applied to the group, it is called the *Not-Invented-Here* syndrome (NIH): the tendency to reject someone else's solution merely because it did not originate in one's own house. Applied to the person, it is called the *Not-Invented-by-Me* syndrome: the tendency to devalue an idea merely because one did not have it oneself. Both spring from the same root – thinking of the idea as possession – and both turn with particular bitterness against the inventor.

For here an inversion shows itself that strikes precisely the *overdue idea* the eighth chapter spoke of. The simpler and more obvious a solution appears in hindsight, the more readily the person who did not have it sees its originator not as a creator but as a thief – someone who *took away* from him

something he could just as well have had. *I could have done that too* becomes *that would have been mine*. The simplicity that makes the overdue idea valuable becomes, in the eyes of the one passed over, the proof that the idea was stolen from him. Thus the invention's greatest virtue – that in hindsight it seems self-evident – is turned into the pretext for belittling it. How this personal impulse becomes a social pattern will be shown in the systemic part of this book.

And yet it is not always personal resentment that speaks here. Often it is a more impersonal, almost moral feeling: the inventor appears as someone who enriches himself, unjustifiably and shamelessly, on the state of the art – which in this area, so the feeling goes, merely shows an incomprehensible gap in registration. In this view the simple solution has long been common property; it was open to everyone, as it were, and only a failure to file prevented it from counting as such. Whoever now patents it fences off a piece of the commons and turns a bureaucratic accident – that nobody had registered the obvious – into a private right. This indignation even has a true core: there are trivial patents, there is the enclosure of the self-evident, and the law recognises *inventive step* precisely in order to draw this line.

But in the case of the genuine overdue idea the indignation rests on a deception – the same hindsight that makes the idea seem simple in the first place. The gap in the state of the art was precisely *not* a mere gap in registration. Had the solution really already been part of the state of the art, it would not be patentable; that it is means that it was precisely *not* there, but first had to be made. What in hindsight looks like an overlooked entry was in substance a real absence that required someone to fill it. And this someone takes nothing from the state of the art – he adds to it something that was missing before, in return for a reward that is limited in time and after whose expiry the invention passes permanently into common property. The indignation mistakes the direction of the process: it takes for removal what is in truth an addition.

Either way – whether as personal hurt or as moral indignation at the supposedly fenced-off commons – resentment sees in the inventor someone

who takes where in truth he gives. For the searching inventor, though, this holds: the disappointment that stings him at his own double discovery is only the inside of the same feeling that others turn against him – and the remedy is the same: to think of the idea as carried, not as possessed.

VI. Third reaction: looking away

The third reaction is the most dangerous, because it disguises itself as diligence and is in truth cowardice. It occurs when the inventor senses that the search could destroy his idea – and therefore does not search properly. He searches superficially, with the words that find nothing. He breaks off the search as soon as it has turned up nothing, and reads the nothing as confirmation. Or he finds the troublesome evidence and talks it down: *That's not the same thing. That was badly done. That doesn't count.* He looks away because he does not want to lose the beloved idea.

This reaction is the exact reverse of the hunch the seventh chapter spoke of. There the virtue was to let the unease stand rather than clear it away. Here the vice is to clear away the troublesome find rather than let it stand. Both times it is about the same honesty towards what one does not want to see – only at the beginning it shows itself as wakefulness and here as courage. The inventor must search most thoroughly where he most fears the answer. For a design that does not survive an honest search would not have survived the workshop either – only later and at greater cost. The search, carried out thoroughly, is the cheapest way to learn of an error; looking away is the most expensive way to postpone that lesson.

VII. Fourth reaction: appropriation

The fourth reaction is the most mature, and it is the one that distinguishes the experienced from the beginner. It consists in seeing what has been found not as a threat but as material. What the search brings to light is not only a judge of one's own idea – it is also nourishment for it.

Whoever searches in this way reads patents and literature not only with the question *am I the first?* but with the question *what can I learn?*. He sees in a related solution a foundation on which he can build. He sees in a distant solution from another field a transfer that first makes his own possible – the far-fetched idea of the fourth class, here not caught by chance but deliberately sought. He sees in a clear predecessor something against which he can sharpen his own design by working out what distinguishes it. Appropriation transforms the search from a gatekeeper that turns ideas away into a source that feeds them. It is not the opposite of the honesty of the third reaction, but its fruit: only someone who looks honestly can honestly appropriate what he finds.

A word of caution belongs here, which only the systemic part of this book will fully redeem. Appropriating knowledge that is already in the world is something other than appropriating what is due to someone else. The line runs where learning turns into taking. The honest inventor learns from his predecessors and honours them; he builds on them without adorning himself with borrowed plumes. The violation of this line – and the methods by which it is systematically violated – will be discussed elsewhere.

VIII. What this means for the theory

The seventh layer is the encounter of the private idea with the public memory. In it the inventor for the first time leaves the workshop of his own head entirely and places his work beside that of all the others who before him turned over the same question. That is a necessary and a dangerous step, for what he finds can carry his work or bring it down.

The four reactions – relief, disappointment, looking away, appropriation – are therefore more than psychological observations. They are a touchstone of the inventor’s honesty. Relief tempts towards complacency, disappointment towards giving up, looking away towards self-deception; only appropriation leads further. And all four come down to the same virtue that the earlier layers already demanded: to look reality in the face, even where it contradicts the wish. Whoever can do that has, after the search, a

different design from before – tested, sharpened, anchored in the knowledge of the world.

That is the bridge to the eighth layer. The design has now left behind the inner image, design with feedback and the search. It has been tested as far as testing on paper allows. But paper has a limit: it does not show what the material does when one really takes it in hand. Reality resists in ways that no drawing and no search can predict. That is why something rough, fast and tangible must now come into being – a first model, not yet beautiful, not yet precise, but real. Why this first model must come before the polished one is the subject of the next chapter.

13. The First Model – Quick & Dirty

Eighth layer. Why the rough, quick model must come before the polished one – whether made of cardboard or in the computer – and why its crudeness is its virtue.

I. Why a model at all

The design has the inner image, design with feedback and the search behind it. It has been tested as far as anything can be tested on paper. But paper has a limit, and it is insurmountable: it does not show what the material does when you actually take it in your hand. A drawing is a representation of the thing, not a piece of it. It obeys the rules that the draughtsman knows and puts into it – not the rules that reality follows even when nobody knows them.

For reality resists in ways that no drawing and no search can predict. A part that swung freely on paper jams in the hand. A joint that looked stable wobbles. Something vibrates, something rattles, something heats up, something feels wrong without your being able to say at first why. These resistances are not the designer’s mistakes – they are properties of the thing, which only show themselves when the thing passes from the realm of

representation into the realm of matter. The model is the place of this crossing. It is the first time the idea is not depicted but embodied. And that is why it is indispensable, however far the drawing may have progressed.

II. The model in the age of simulation

What has been said so far would sound anachronistic if it kept silent about the tools that today stand between the drawing and the physical model. Much of what used to be revealed only by the material in the hand can now be seen in the computer before a single chip has fallen. The three-dimensional CAD model shows collisions and movements; finite element analysis shows stresses, deformations, natural vibrations; flow simulation shows how a medium behaves; multibody simulation lets the mechanism run; tolerance analysis shows whether the chain of dimensions still fits together even in the worst case. A good part of the surprises the rough model held in store can be anticipated – faster, more cheaply, without using up a single piece of material.

This is not a refutation of the argument so far but a sharpening of it. The economy of inventing – to be wrong early and cheaply so as not to be wrong late and expensively – gains through these tools an even earlier and even cheaper stage. Where the first error once cost the building of a model, today it often costs only a calculation. And at the same time 3D printing has made the physical model itself cheaper: what yesterday took days to make from cardboard and wood today grows out of the machine in hours. The threshold at which the quick and dirty begins has moved forward, and the space before it has become richer.

More than that: the new tools do not drive the quick and dirty towards completeness but precisely towards the partial. You do not need the finished CAD model of a whole machine to print a first functional sample. You can pick out a single function and test it in the printed part before it is mature as a design – the joint alone, the gripper alone, the one kinematic arrangement you have doubts about. What used to require that at least the surroundings of a part were already in place before it could be tried out physically can today

be tried out in isolation and in hours. So Quick & Dirty shifts, with all these possibilities, still further towards quick, simple and limited – and precisely in this it remains what it was. The printed functional sample of a single, still unfinished function is the quick and dirty in its purest form: a narrowly defined question put to reality with the least effort capable of answering it.

But – and here the danger grows with the tools – simulation remains a representation, not a piece of reality. It obeys the assumptions that have been put into it: the mesh, the boundary conditions, the material model, the simplifications without which no calculation would run. It does not show what reality does but what the model of reality does. With this the danger of the flattering inner image from the tenth chapter returns at a higher level, and there it is greater, not smaller. For a colourful, finely resolved, almost photographic computed image displays an authority that the lopsided piece of cardboard never claimed. It looks like the truth and yet is only the consequence of its premises. Whoever calculates bad assumptions beautifully gets a beautiful error – and is more inclined to believe it than the rough model, which would have lied to him more honestly.

Two things follow from this. First: the physical model has not been abolished but moved. It remains the final judge, for reality resists even where no assumption expected it – in the fit, in manufacture, in interactions nobody thought of simulating, in the failure modes beyond the model. Today one builds it later and more purposefully, for what the calculation cannot reach – but one builds it. Second: the principle of the quick and dirty applies to simulation no less than to cardboard. Calculation too knows the rough and the polished – the quick estimate with a coarse mesh and the validated fine calculation – and because the tool makes precision so easy and so beautiful, the temptation to premature precision is even greater here. You can spend weeks calculating the wrong design in photorealistic detail. First rough, then precise – the rule carries over unchanged from the material to the computer. What follows therefore applies equally to the model made of cardboard and to the model made of numbers.

III. What “Quick & Dirty” means

The first model should be quick and dirty – *quick and dirty*, as workshop language calls it. Both words are to be taken literally.

Quick means: made in hours or days, not in weeks. It should come into being so rapidly that reality’s answer arrives while the thought is still mobile. A model you spend a month building comes too late – by the time it is finished, you have tied yourself to the design it was supposed to test.

Dirty means: rough, unfinished, ugly. Made of cardboard, of wood, of whatever is to hand; not of the final material, not to the final dimensions. It should be just good enough to answer the one question you put to it, and not a single touch better. For the first model is not a piece of work to be shown off – it is a question put to reality. Whoever treats it as an answer to be presented has already missed its purpose. It does not serve to show that the idea is good. It serves to find out whether it is.

IV. Why the rough must come before the polished

Here lies the thesis of this chapter, and it runs against the instinct of the good craftsman. The rough model must come before the polished one, and for two reasons that belong together.

The first is the one already touched on in the chapter on design: premature precision binds you to error. Whoever works out a model down to the last detail when its basic idea has not yet been tested may be wasting his care on a mistake – and, by refining it, makes it expensive and stubborn. First the idea must stand, then precision may come. The rough model tests the idea; the polished one tests the execution. Whoever reverses the order tests the execution of an idea that perhaps deserves no execution.

The second reason is deeper and is mentioned less often. The rough model is cheap enough to be destroyed – and this destructibility is its real value. You can saw it apart, bend it, throw it away, build a second one. You can treat it brutally because nothing about it is precious. The polished model, by

contrast, is precious, and preciousness turns the relationship upside down: you defend it instead of learning from it. You cling to it because so much work has gone into it. You reinterpret its flaws instead of admitting them, because admitting them would devalue the effort invested. So the beautiful model becomes an obstacle to insight, while the ugly one promotes it. For learning demands brutality towards one's own work, and one can only be brutal towards something one does not love too much.

V. What the model teaches that the drawing keeps silent about

What the rough model brings to light is exactly what the representation kept silent about. It wobbles where the drawing seemed stable. It jams where everything looked free. It shows how the thing feels, how it sounds, how it vibrates – properties for which paper has no symbol. It makes the effect of size visible: some things that worked out on the reduced sheet fail at real scale, because forces and masses do not grow along with the drawing. And it brings to light the interplay of the parts, which the mind and the drawing had considered one after another, each on its own – in the model they act simultaneously, and from their interaction there often arises what none of the parts on its own would have led one to expect.

It is the same as what already held for design with feedback, only one stage more real. There the drawing was cleverer than the draughtsman; here the material is cleverer than both. The hand that turns and presses and loads the model experiences things that no eye would have seen on any sheet. The model is not the confirmation of the design – it is its strictest teacher, and what it teaches is mostly something one did not want to hear.

VI. The temptation of perfection

Against the eighth layer stands a temptation that afflicts precisely the good ones. The practised designer has been brought up to precision; the ugly offends his skill. He wants to get it right straight away, clean, dimensionally accurate, presentable. Gluing something together out of cardboard that stands there lopsided goes against his pride. It is exactly this pride, at this

point, that is the enemy. It drives the designer to polish before he has tested – premature precision under the cloak of care.

To this inner temptation is added an outer danger. The rough model can be misunderstood by others. Whoever sees it – a superior, a financial backer, a customer – easily takes the crudeness of the model for the crudeness of the idea and rejects both together. The lopsided piece of cardboard does not look like an invention but like botched work. That is why the inventor must protect the rough model – not by polishing it, but by making sure that it is read as what it is: a question put to reality, not the finished work. Whoever shows the rough model must supply the language in which it is to be understood. The systemic part of this book will deal in more detail with how to handle these outside eyes.

VII. When the model may become precise

It would be a misunderstanding to turn all this into a contempt for precision. Precision is indispensable – it simply has its hour, and that hour is not the first. The rough model has fulfilled its task when two things have happened: when the basic idea has been confirmed, and when the surprises the material held in store have come to light. Then, and only then, does precision earn its price.

From that point on the rule is reversed. Now sloppiness is the enemy and care the virtue. The design that the rough model has tested and that has passed the test now wants to become precise – in the right material, to the right dimensions, with the care that would have harmed it at the beginning and is now its due. The art of the eighth layer lies less in building the rough model than in recognising the moment at which one may switch from the rough to the precise. Switch too early, and you bind yourself to error; switch too late, and you remain stuck in the approximate where reality has long been demanding precision. First rough, then precise – but each in its own time.

VIII. What this means for the theory

The eighth layer is the one in which the idea meets matter itself for the first time and reality receives its first direct voice. The earlier layers tested the idea in the mind, on paper, against the knowledge of the world – all still on this side of matter. The rough model leads it across this threshold. It is the cheapest way of being wrong in reality, just as the search was the cheapest way of being wrong on paper. Therein lies a consistent economy of inventing: to be wrong early and cheaply so as not to be wrong late and expensively. Each layer is a touchstone a little more real and a little more expensive than the one before, and the rough model is the first that consists of matter.

It thus embodies a whole attitude that has carried this book from the beginning: the priority of testing over asserting, of learning over being right, of reality over wishing. Whoever builds quick and dirty admits that he does not yet know, and faces the answer even when it is uncomfortable. That is the honesty that the search already demanded, now translated into material.

This is the bridge to the ninth and last layer. The model has given its lessons; the design has matured, been tested, proved itself in the rough and in the precise. And yet there remains a step that no model can take for you. For between the tested design and the real deed lies a gap that no further testing closes – the decision to do it now, to go out, to commit oneself, to be the first. The model informs; the deed commits. The next chapter deals with this last layer – critical questioning and the first-mover deed.

14. Critical Questioning and the First-Mover Deed

Ninth and last layer. The leap from testing to commitment. Why in the end no model decides any more, only the decision to be the first.

I. The gap

The design has all eight previous layers behind it. It was carried, sensed as a hunch, caught, matured; it took on an inner image, was designed with feedback, tested against the search, tried out on the rough model. It has been taken as far as a design can be taken before it enters the world. And yet one last step remains, and it is of a different kind from all the previous ones.

For between the tested design and the real deed lies a gap that no further testing closes. You can make one more calculation, build one more model, search one more source – but none of these tests carries you across the gap. Testing is by its nature reversible: what does not prove itself you discard; what proves itself you test further. As long as you are testing, you are committed to nothing. The deed, by contrast, is irreversible. Whoever goes out, whoever commits, whoever becomes the first cannot undo it. The model informs; the deed commits. And because the deed commits, it demands something that none of the earlier layers demanded – not more skill, but courage.

II. Critical questioning

Before the leap stands one last gate, and it is a gate of doubt. It is not the ongoing feedback of the sixth layer, which improves the design in the alternation between inside and outside. It is a single, unsparing self-interrogation at the threshold: Have I deceived myself? Is the inner image still flattering me? Did I look away during the search where I ought to have looked? Was the model's verdict really its own – or was it mine, because I wanted to hear it?

This questioning gathers all the honesties of the earlier layers into a single, final act. It is the honest search, turned against oneself; it is the brutality towards the rough model, now directed against the whole design; it is the alertness of the hunch that does not clear away unease, applied here to the last unease before the deed. And it is the hardest of all, because it is demanded at the very moment when one least wants to doubt. The deed

beckons, the work is ripe, everything is pushing outwards – and precisely now one is supposed to turn the sharpest doubt on one's own, beloved work. Whoever can do that has, in its purest form, the virtue this whole book demands: to look reality in the face, even and especially where the wish is loudest.

III. Why the questioning must end

And yet the questioning must not be endless. Here an insight returns that the search already taught: there is no certainty. The search proves no absence; the model tests only under its assumptions; and the future into which the deed acts cannot be tested at all, for it is not yet here. Whoever waits for certainty before acting will never act, because certainty never comes.

That is why at a certain point the questioning turns into its opposite. Up to this point it is honesty; beyond it, it becomes flight. Eternal doubt, ever-renewed rechecking, the not-quite-sure-yet – that looks like care and yet is only fear of the deed, dressed in the garb of thoroughness. It is the same danger already known to incubation and to design with feedback: the loop that never closes. The mature inventor recognises the point at which doubt has done its work. The questioning ends not because certainty has been reached – it is never reached – but because the remaining uncertainty has been made as small as honesty can make it, and the rest belongs to the deed. Knowing when this point has been reached is no art of calculation; it is a maturity of judgement that can be gained only through experience.

IV. The deed as a leap

Because certainty remains out of reach, the deed is always a leap – a commitment entered into without its success having been proved. Here something enters the sequence of layers that the earlier ones did not know. All eight previous layers were skill – perceiving, seeing, designing, testing. The ninth is no longer skill but courage. For the leap demands what no model can supply: the willingness to commit oneself, to be publicly wrong, to put oneself on the line.

This courage is not recklessness. Recklessness leaps before it has tested; courage leaps after it has tested and knows that testing will carry it no further. The leap comes after honesty, not instead of it. And precisely here it becomes visible that the ninth layer cannot be had without the anthropological root of the first part. Only someone who is able to will something, in the sense the fifth chapter gave to willing, can commit himself. Only someone who has the readiness for effort and has gained from it a sense of self-worth that does not depend on the approval of others can bear the risk of failing in public. Only someone who knows how to carry the idea rather than own it can let it go into a world that will snatch it from him. The last layer of practice reaches back to the first root: the leap is the place where it shows whether the inventor has the way of life without which all skill remains empty.

V. What it means to be the first

The deed at issue here is the deed of the first. Being the first does not merely mean being early – it means being exposed. The first bears costs that the second is spared. He builds the road on which the others walk later. He makes the mistakes from which the others learn. He meets the resistance to the new that the earlier chapters described in its many forms: the *that can't work* of the merely knowledgeable, which strangles the hunch too early; the misjudging of the rough model, which is taken for the crudeness of the idea; the resentment of the one who believes himself cheated of the obvious solution. All this strikes the first, not the second.

And often it is not the first who reaps. The eighth chapter spoke of the premature idea that someone later realises when the air has changed; the second reaps what the first saw. This injustice is real, and the systemic part of this book will pursue it. But it does not cancel the necessity of the first deed. For if no one were the first, nothing new would come into the world. Being the first is therefore, for all its risk, a gift to the future – one for which the giver is not always rewarded. Here the circle closes back to the first layer and to the root of benevolence: the inventor who goes out carries the

idea into a world that may withhold its yield from him, and does it nevertheless, because the idea belongs in the world and not in his safe. The last deed of inventing has the same built-in generosity that the fourth chapter found in inventing as such.

VI. Questioning and deed as a unity

It would be a misunderstanding to see critical questioning and the deed as opposites – as if doubt stood against the leap, caution against courage. In truth they are the two sides of a single layer, and their interplay is the last of those rhythms that run through this whole book. The effort and the rest of incubation; the grasping and the letting go; the inner image and its feedback – and now, at the end, doubt and the leap. Questioning is the last breath in, the deed the breath out, and neither can be had without the other.

Doubt without the deed is barren. The eternal tester who never acts produces nothing, however sharp his mind may be; he confuses the preparation with the work. The deed without doubt is foolhardy. The jumper who has not tested comes to grief, however bold he may be; he confuses courage with blindness. Only both together – the sharpest doubt, followed by the resolute leap – make the ninth layer. In their union lies the signature of the mature inventor: he doubts more thoroughly than the cautious and acts more resolutely than the bold, and he knows the moment at which the one passes into the other.

VII. What this means for the theory

With the ninth layer the arc of inner practice closes. The nine layers have described the path of an idea from its first, quiet arrival to its first deed: the relationship that carries it; the hunch that receives it; the catching that seizes it; the maturing that clarifies it; the inner image that gives it shape; design with feedback, which tests it against reality; the search, which sets it beside the knowledge of the world; the model, which tries it out on matter; and the deed, which releases it into the world. It is one continuous path – from sensing to doing, from the inner to the outer, from possibility to reality.

And this path ends where it began: at the root. For the last layer has shown that practice cannot be completed without the way of life of the first part. The leap demands willing, the readiness for effort, benevolence, the sense of self-worth gained from one's own doing – everything that the anthropological root described. The nine layers are the practice; the root is what makes the practice possible. Without the root the layers remain an empty technique that no one goes through; with it they become what they are – the inner form of a successful inventor's life.

With this the sequence of the nine layers is complete, but the second part is not yet. For across the nine layers, running through all of them, there are two things that cannot be assigned to any single layer. One is the methodology of targeted solving, which the tenth chapter only hinted at and referred to this point – the methodology of invention, TRIZ and its relatives, as a cross-layer that supports finding and solving at every stage. The other is the structure of tension of inventing itself – that keeping of measure between the extremes which repeated itself in every layer as the same basic figure and which deserves a consideration of its own. The last two chapters of this part deal with these two cross-layers.

15. Cross-Layer: The Methodology of Invention

First cross-layer. TRIZ and its relatives – the methodology of targeted solving, which is not one of the nine layers but runs through several of them. What it achieves, where it takes hold and where its limit lies.

I. What a cross-layer is

The nine layers of inner practice form a path. They follow one another, from the first sensing to the deed, and each presupposes the one before. But not everything that belongs to inventing is a station on this path. Some things lie not *on* the path but *across* it – they run through several layers without belonging to any single one. One such thing is the methodology of targeted solving.

It is not a tenth layer that would come after the ninth. It is a toolbox that can be opened at various points along the path – when catching the idea, in inner visualisation, in designing, in the search. That is why we call it a cross-layer. The tenth chapter only hinted at it and referred its treatment to this point; here is the place to do it justice. For it deserves more than a mention – and less than the veneration that some of its followers bring to it.

II. From patent analysis to method

The best known of these methodologies bears the name TRIZ. The Russian acronym is feminine – in German one says *die* TRIZ – and it stands for *Teorija reschenija isobretatelskich sadatsch*: the theory of solving inventive problems. Its originator, Genrich Altshuller, was a young engineer who worked in the patent system of the post-war Soviet Union and in doing so had tens of thousands of inventions pass through his hands. From this work a bold question grew for him: do inventive solutions follow recognisable patterns? Beneath the endless variety of inventions, do the same basic figures recur?

His answer was yes, and this yes was an outrage. For it contradicted the centuries-old conviction that inventing was the realm of genius and chance – something one might have or not have, but could not teach. Altshuller claimed the opposite: that behind inventions there stood recurring contradictions and recurring principles for resolving them, and that both could be distilled from the patent literature and passed on. Inventing, he said, can be taught within limits. Whoever claimed that set himself against the cult of genius – and Altshuller's own life, which led through Stalinist persecution and the camps, gave this heresy a bitterness that points beyond the method.

He was not alone in this. At the same time, from a different tradition, German-language design methodology pursued the same aim – Friedrich Hansen, Wolfgang Rodenacker, later the VDI guidelines sought to bring order to inventing and designing, to make the steps of thinking traceable, to raise the procedure from arbitrariness to rule. The difference lay less in the

goal than in the claim: where engineering methodology spoke more modestly of an ordering of procedure, TRIZ spoke of a *theory* – and this higher claim is at once its strength and its danger, as will become clear.

III. The contradiction and ARIZ

One basic idea stands at the centre: a truly inventive problem contains a contradiction. One distinguishes the technical contradiction – you improve one property and another gets worse; the component becomes stronger and at the same time heavier – and the sharper physical contradiction, in which one and the same quantity is supposed both to be and not to be: something must be hot in order to flow and cold in order to hold. The ordinary engineer meets such a contradiction with a compromise – a little stronger, a little heavier, the middle between the evils. The methodology of invention demands something else: not to soften the contradiction but to *resolve* it – to get both, without the rotten peace of the middle way.

Here the methodology meets what the tenth chapter called the place where the two paths of inventing come together: the contradiction. The problem that has been found, coming from the finding layers, comes to a head in the contradiction, and the methodology starts where this sharpening has taken place. Now, however, a widespread misunderstanding has to be cleared up, one that comes naturally to the outsider in particular. Many take TRIZ to be a list of forty inventive principles – a catalogue of tricks that one goes through until one fits. That misses the core. The forty principles are only one building block, and not the most important. The backbone of the method is ARIZ, the *algorithm for solving inventive problems* – an ordered sequence of working steps that leads the inventor from the raw problem, via its sharpening into a contradiction, to the points at which the individual tools are to be applied. TRIZ is first a procedure and only then a collection; ARIZ is this procedure, and the tools are fitted into it.

IV. The arsenal of tools

Whoever really wants to work with TRIZ needs more than the forty principles; he needs a minimum kit, which ARIZ orders and applies at the points intended for it – and which the experienced user can also use on its own. It includes: the four *separation principles*, which resolve the physical contradiction by separating the incompatible in time, in space, between the whole and its parts, or according to conditions; the *laws of technical evolution*, which describe the lines along which systems develop; the *standards for solving inventive problems*, a collection of recurring solution patterns; *substance-field analysis*, which abstracts a system to its most general form – substances and fields – and derives from it recommendations for solutions based on physical effects; the *forty innovative principles*, traditionally accessed through a contradiction matrix, which, however, as experienced practitioners object, is too arbitrary and too unreliable, so that a hierarchy of principles is preferable, such as Dietmar Zobel and R. Hartmann propose in *Erfindungsmuster*; and the *model of little intelligent figures*, the dwarf model, which thinks of a system as a swarm of tiny helpers and breaks up the rigid inner image.

Above all these tools stands one guiding idea: the *ideal final result*. Instead of feeling one's way forward from what exists, one thinks backwards from the ideal – from the function that fulfils itself, as it were, of its own accord, without effort and without harm. This ideal is unattainable, but as a vanishing point it frees thinking from the shackles of what is already there; it does not ask *how do I improve what is*, but *what would have to be the case for the problem to disappear* – and that is already a question of finding, not merely of solving. Related to this is the observation that technical systems develop along recognisable lines, so that one can foresee their next step with some probability and deliberately seek out the overdue or even the still premature idea instead of waiting for it.

All these are Altshuller's classic tools. Alongside them a living *Modern TRIZ* has long since emerged, developing the toolkit further – carried by names such as Michael Orloff, the founder of Instrumental Modern TRIZ, as

well as Livotov, Petrov, Zlotin, Zusman, Souchkov, Terninko, Salamatov, Mann or Adunka. This book lists the tools only to show that TRIZ is a rich, ordered toolkit and not the handful of tricks it is often taken for; the forty principles are a part of it, not its heart. This cross-layer goes into none of the tools in detail. Presenting their possibilities, their strengths and their weaknesses in concrete application is a task of its own – it would fill further chapters and perhaps a book of its own, and that would be its place, not here, where the only point is to mark its position within the whole of inventing.

V. Where the methodology takes hold in the nine layers

Now it can be said why the methodology of invention is a cross-layer and not a layer. It takes hold not at one point along the path but at several.

It takes hold in catching, the third layer: instead of waiting for the idea to come, one seeks it in a directed way – via the contradiction, via the ideal final result, via the lines of evolution. It takes hold in inner visualisation and in designing, the fifth and sixth layers: the tools suggest concrete directions for solutions, around which the inner image can form and by which the design can orient itself. It takes hold in the search, the seventh layer – there it even has its origin, for it grew out of patent analysis; the systematic sifting of what already exists is its home ground.

And just as important is where it does *not* take hold. It does not create the hunch, the second layer – the pre-conceptual sense that something is possible at a certain point cannot be generated by any method; it comes or it does not. It does not replace incubation, the fourth – maturing beneath the threshold follows no algorithm. And it takes nothing off the ninth layer: the leap of the deed, the courage to commit, no methodology can provide, for it is not a skill but a venture. The methodology runs through the middle layers, the shaping ones, and leaves the receiving layers and the accomplishing one untouched. That is the precise meaning of the word *across*.

VI. Limits and the heretic question

It is part of the honesty of this book neither to despise the methodology nor to exalt it. Despising it would be foolish: its strengths are real. It breaks mental inertia, that *it can't be done* of the experienced, which stifles the hunch too early, by showing that similar contradictions have already been resolved elsewhere. It turns transfer from foreign fields – the far-fetched idea of the fourth class – from a stroke of luck into a procedure. And it gives the individual inventor, who works without the resources of a corporation, a framework by which he can pull himself up. That is a great deal.

But the limits are just as real, and three of them must be named. The first: the methodology presupposes a formulated problem. It begins when the contradiction has been cleanly named – the naming itself, the finding of the right problem, it does not achieve; that is achieved by the finding layers. Whoever takes the methodology for the whole of inventing confuses solving with finding and overlooks half the art. The second: no method forces the idea. It raises its probability, it directs the search, it rules out dead ends – but the gap between directed searching and the solution found is in the end still closed by a leap that no procedure guarantees. The methodology is a tool, not a recipe; that was already said in the tenth chapter, and it remains the core.

The third limit is more delicate and concerns not the method but its use. A teaching that arose against a cult – the cult of genius – can itself become a cult. Around TRIZ a movement has formed, with its own orthodoxy, its own vocabulary, its own degrees of initiation, and with it the temptation to make a world view out of a tool and to sell the method as if it were the complete theory of inventing. Here lies an ambiguity that can be captured in the word *heretic* – a word that Michael Orloff chose as the title of his book about the inventor-heretics. Altshuller was a heretic in the good sense: someone who set against a rigid dogma – here the dogma of unteachable genius – the condemned truth that inventing can be brought into order. But every heresy runs the risk of becoming a church itself, and then the relationship is reversed: the former heretic becomes the guardian of a new orthodoxy, and

whoever now names its limits is considered the deviant. In his preface to Orloff's book, Dietmar Zobel urges that the true pioneer be distinguished from the charlatan – an admonition directed not only against the exaggerations of inventors but also against the exaggerations of those who declare a method to be salvation. Against both the same measure helps: to take the methodology for what it is – a strong, limited tool – and to honour the German engineering tradition, which has always spoken more modestly of an ordering of procedure, for its sobriety.

VII. What this means for the theory

The first cross-layer shows that the receptive and the methodical are not opponents but complement each other. The nine layers describe the inner path from sensing to doing; the methodology is a bundle of tools that one brings to several of these layers in order to make finding and solving more conscious, more directed, less dependent on chance. Whoever takes the finding layers for the whole underestimates how much of inventing can be sharpened methodically; whoever takes the methodology for the whole overestimates what a procedure can achieve. The truth lies in their interweaving.

Yet the methodology lives *in* the layers, not above them. It cannot supply the hunch, cannot force the maturing, cannot replace the courage of the deed and cannot bring about the way of life of the root, without which all skill would remain empty. It is powerful and limited, and its dignity lies precisely in knowing its limit. A tool that takes itself for the whole hand is doomed to disappointment; a tool that knows it is a tool becomes indispensable.

With this the first of the two cross-layers has been dealt with – that of the tools that run across the layers. There remains a second, and it is of a different kind. For not only tools run across all nine layers, but also a basic figure – a pattern that recurred in every layer: the keeping of measure between two extremes, the right mean between a too little and a too much. This structure of tension, which underlies all inventing, is the subject of the last chapter.

Note on sources: The account of TRIZ follows the works of Genrich Altshuller. The criticism of the contradiction matrix and the proposed hierarchy of principles draw on Dietmar Zobel and R. Hartmann, Erfindungsmuster. The further development of the method is represented by the authors of Modern TRIZ, among them Michael Orloff – founder of Instrumental Modern TRIZ – as well as Livotov, Petrov, Zlotin, Zusman, Souchkov, Terninko, Salamatov, Mann and Adunka. The figure of the heretic is taken from Michael Orloff's book HÄRETIKER (Berlin 2024), whose preface is by Dietmar Zobel. German-language design methodology is represented by Friedrich Hansen, Wolfgang Rodenacker and the guideline VDI 2221.

16. Cross-layer: The Structure of Tension in Inventing

Second cross-layer. The basic figure that recurred in every one of the nine layers – the right measure between a too little and a too much. The innovation mesotes, and why it is not an average.

I. The recurring basic figure

Anyone who looks back over the nine layers will notice something that was never stated explicitly in any of them, yet was at work in each. In every layer the same danger lurked in two forms – a too little and a too much – and each time the art lay in finding the right measure between the two. That was no coincidence and no repetition for want of something better to say. It is the deepest structure of inventing, and it runs across all the layers like a thread on which they are strung.

We call this structure the structure of tension in inventing, and the right measure it demands the innovation mesotes – the inventive mean between the excesses. This last chapter at last makes visible the basic figure that ran along, hidden, everywhere. For anyone who has once recognised it understands the whole book differently: the nine layers and their tools are the surface; the structure of tension is what holds them together from within.

II. Aristotle's mesotes

The thought is old. Aristotle coined it in his doctrine of virtue. A virtue, he says, is a mean – a mesotes – between two vices, one of which consists in too much, the other in too little. Courage lies between recklessness and cowardice. Generosity lies between extravagance and meanness. Whoever falls into the too much misses virtue just as surely as whoever falls into the too little; it is hit only in the middle.

But – and this is the decisive addition that is easily overlooked – this mean is not the arithmetical middle. It is, as Aristotle says, a mean *relative to us*: the right measure, which depends on the matter, the situation and the person, and which is not calculated but hit by practical wisdom. And it is hard to hit. Anyone can be angry, says Aristotle; but to be angry with the right person, to the right degree, at the right time, for the right reason and in the right way – that is hard, and that is why it is rare and valuable. The excesses are easy; the mean is the art.

It is precisely this figure, carried over from ethics into the practice of inventing, that is the innovation mesotes. At every turn the inventor is called upon to hit the right measure between a too little and a too much – and the nine layers are nothing other than nine stages for this one task.

III. The engineer's three sentences

The mesotes is not the property of philosophers. The same insight has grown out of engineering practice from within, and where it was put into words there, it sounds sparer and perhaps for that very reason more urgent. A physicist who taught at a German engineering college in the seventies – Dr Lückert – gave his students three sentences to take with them, not as examination material but as a thinking tool for a whole lifetime. They run: *The sum of all evils is constant. We have to pay a price. We have to die one death.*

The first sentence is a conservation law for problems. You do not solve a problem, you transform it; every solution creates new difficulties, and the total quantity of evils stays the same – only their distribution can be changed. That is the mesotes in the language of the workshop. For if evil cannot be abolished but only distributed, then the inventor's task is not to remove all evils – that would be the hubris against which Aristotle already set the mean – but to find the right distribution: *Which evils do I want to have?* The catalogue of means unfolded in the following section is nothing other than a series of such decisions – in every layer the choice of which evil one bears, that of the too little or that of the too much, and to what degree.

The second sentence names the price. No gain comes free; every mean one hits costs something one gives up elsewhere. The question is never how to avoid the price, but which price one is prepared to pay. That too is mesotes: the right measure is not a gift but a chosen exchange.

The third sentence is the most radical. Every realisation kills its alternatives. As long as nothing has been built, everything remains possible; but at some point the one thing must become reality, and the infinite possibilities in the head must die for it. Whoever wants to keep everything open creates nothing. That is the ninth layer, the deed, captured in a single sentence – and at the same time the reverse side of incubation, which lets things ripen but must at some point decide.

Two thousand years and a whole world separate the Greek philosopher from the German physicist, and yet both say the same thing: wisdom is the acceptance of measure and limit, against the hubris of being able to have everything. The innovation mesotes is no learned antique; it is a living truth that the workshop rediscovers from its own experience.

IV. The catalogue of means

You see it when you go through the layers in order and name, in each, the two excesses and the mean.

In the first layer, the relationship to the idea, the mean lies between the too much of possessing – the clinging that does not leave the hands free – and the too little of indifference, which lets the idea drop. The mean is free carrying. In the second, the hunch, it lies between the too little that sweeps away every faint sensing and the too much that chases every whim; and a second time between too little knowledge, which nourishes no hunch, and too much knowledge, which smothers it. In the fourth, incubation, the mean lies between the too much of grasping, which wants to force things, and the too much of letting go, which never comes to an end – the rhythm in which nothing ripens for the one who never rests, and nothing gets finished for the one who only rests.

In the fifth, the inner image, it lies between the mistrust that throws away the power of inner seeing and the blind trust that succumbs to its flattery. In the sixth, construction, between infatuation with the representation and the loop that never closes – and a second time between becoming precise too early and staying rough too long. In the seventh, the search, between the deceptive relief that searches too little and the anxious looking away that shies from the answer. In the eighth, the model, between premature perfection and never building. And in the ninth, the deed, between the reckless leap that doubts too little and the eternal doubt that never leaps.

Nine layers, the same figure in each. That is the structure of tension, and whoever sees it once sees it everywhere.

V. Why the mean is not an average

Now a misunderstanding threatens, and it is the most dangerous in this chapter. One could take the mesotes for the average – for the lukewarm both-and that takes a little from each extreme and stirs it into a bland compromise. That would be its death. The innovation mesotes is not an average.

For, first, the mean is not the arithmetical midpoint between the extremes but what is right for the situation – and that may lie close to one of the

extremes. In the ninth layer, for instance, once critical questioning has done its work, the mean lies not halfway between doubt and leap but close to the decisive leap; and in the second, faced with a delicate, slow hunch, it lies close to patience. Where the mean lies shifts with the matter, the idea, the moment. Finding it is not a calculation but a judgement – that maturity of judgement which this book has demanded in so many places and which no rule can replace.

Second, mere compromise misses precisely what matters. Whoever takes the mean between the rough and the precise model to be "moderately precise" has spoiled both; the mean here was a sequence – first rough, then precise – not something in between. The mesotes is often not a point between two poles but the right relationship in which both come into their own. That distinguishes it from mediocrity as sharply as virtue is distinguished from half-heartedness.

VI. The tension remains

Even so, the deepest thing about the innovation mesotes has not yet been said. It does not dissolve the tension between the poles; it endures it. Compromise removes the tension by weakening both sides until nothing pulls any more; the mesotes lets both sides stand at full strength and bears the tension between them. The ninth chapter showed it in the last layer: the mature inventor does not doubt *moderately* and does not act *moderately*; he doubts more thoroughly than the cautious and acts more decisively than the bold. In both he is an extreme – and the mean lies not in weakening both but in uniting both in their full force and hitting the moment at which the one passes over into the other.

That is why the mesotes, as Aristotle says, is an extreme in rank, even though it is a mean in position. It is not the virtue of the lukewarm but of people who carry two strong forces within them at the same time and betray neither. The inventor is such a person. He must be at once receptive and critical, at once dream and calculate, at once love the idea and test it mercilessly, at once persist and let go. Whoever can do only the one is not

an inventor but half of one – a dreamer without testing or a tester without a dream. The innovation mesotes is the art of being the whole.

VII. The innovation mesotes as master virtue

Thus the ability to hit the right measure in every layer proves to be the inventor's true virtue – not one among many, but the one that stands above all the others and gives each of them its measure. It is not itself a technique; it is the wisdom that tells the techniques when and how far. The search does not say of its own accord when it has searched enough; the model does not say of its own accord when it may become precise; doubt does not say of its own accord when it must end. It is the innovation mesotes that says so – the practised judgement that finds the mean in every situation.

And because it is a judgement and not a rule, it cannot be captured in a recipe. That is the last and most important confirmation of what this book has claimed from the start: that inventing does have prerequisites and a practice that can be described, but no formula that could force it. You do not learn the mesotes from sentences but from doing – from hitting and missing, a thousand times over, across a lifetime, until the hand knows of itself the measure that no sentence could give it. That is why the education of the inventor is an education of judgement, not merely of skill; and that is why this last cross-layer, too, reaches back to the root of the first part. For the judgement that hits the mean is no mere feat of thinking – it presupposes a person who has the form of life out of which good judgement is possible at all.

VIII. What this means for the theory

With this second cross-layer the picture of the inner practice is complete. The nine layers describe the path of the idea from sensing to doing. The first cross-layer, the methodology of invention, is the bundle of tools that runs across this path. The second cross-layer, the structure of tension, is the basic figure that likewise runs across it – the keeping of measure that every layer demanded in secret. Together with the anthropological root of the first part,

this yields what the title of this book promises: the prerequisites and the practice of inventing. The root is the prerequisite; the nine layers and their two cross-layers are the practice.

With that, the inner process of inventing has been described – how an idea arrives in a person, ripens, takes shape and enters the world. But a book written for inventors must not end here. For the matter is not done with the first deed; the idea enters a world that takes it up or smothers it, and the inventor bears a responsibility for what becomes of it that no one will take off his shoulders. From here on two horizons open up: one is responsibility for the effect of one's own doing; the other is the long road, riven with chasms, from the idea to real effect – realisation and scaling. To treat both exhaustively would be material for volumes of their own. But this book must not pass over them in silence.

That is why a third part follows – no longer a course of instruction but an outlook. It sets out the ethical aspect, because this belongs to the practice of inventing and not only to its consequences; and it sketches the outer landscape – the valley of death and the other chasms – so that the inventor knows what awaits him when he leaves the protected space of the workshop. The inner practice can be named, but not given away; everyone must acquire it for themselves, through their own errors. Before this book leaves the reader to their own practice, it raises its gaze once more in the third part – from the how of inventing to its what-for and its where-to.

Note on sources: The concept of the mesotes comes from Aristotle, Nicomachean Ethics, Book II. The three sentences – the sum of all evils is constant, we have to pay a price, we have to die one death – go back to Dr Lückert, who taught in the 1970s at the Gesamthochschule Siegen, Gummersbach department.

Third Column – The Institution

InnoCoop: A Cooperative of Inventors

The third column asks: where can the inventor find support when his idea is viable but not yet ready for the market?

The two existing routes – the market and the state – are not enough for this. The market demands arguments for commercial exploitation even in the early phase, which an invention cannot yet supply; whoever takes it to market in this phase usually loses it. The state, for its part, within the framework of its programmes, funds either established institutions or sharply defined projects, but not the individual inventor with a still vulnerable idea.

Between market and state there must stand a third party that carries the invention through its early phase without immediately translating it into the logic of exploitation. There are models: the Battelle Memorial Institute in Columbus, Ohio, since 1929 – it was there that xerography was born, the decisive process of the twentieth century in office communication. The Mondragón cooperative in the Basque Country since 1956, which grew out of a vocational school and a small workshop and today employs tens of thousands of people.

What is missing in Germany does not yet have a name. We propose one: *InnoCoop*. The third column describes its structure, its origin in a contradiction, the iron law to which it is subject, and the first step with which it could begin. A draft – not the only conceivable one, but one that has the existence proofs on its side.

InnoCoop

Concept for an inventors' foundation and cooperative

Invention without innovation starves; innovation without invention withers. InnoCoop is the place where the two work together and where invention can become innovation.

Hans Ley and Claude Dedo (Anthropic)

1. The diagnosis and the purpose

InnoCoop is an independent institution of economic life that carries what market and state evidently cannot. Over decades, both have funded institutional inventing with enormous sums and dubious results, while starving individual inventing into insignificance.

The question arises whether Germany's decline as an industrial location – which some already describe as free fall – has something to do with this development. It was a process that dragged on over decades, and independent, competent experts named it early on and pointed out its foreseeable consequences with complete clarity.

One of them was Erich Häußer, former President of the German Patent Office – whom hardly anyone could have accused of lacking competence or of bias. The interplay of institutions that systematically eliminated the independent inventor, not bound to any institution, from the innovation process he called the cartel of ignorance. This cartel was deaf to his warnings, too.

One of his sentences stands today like the writing on the wall:

> “If we do not succeed in breaking through the cartel of ignorance that is responsible for this, we shall, within a quite foreseeable time, ourselves become a low-wage country again and be forced to become inventive out of necessity.”

Independent inventors, who have largely disappeared today, once had a large share in the technical progress that creates prosperity. They were always forced, in exchange for the capital they needed to realise their inventions, to hand over all rights in the inventions to capital. As a rule they were also cheated out of the little they had been granted at the outset.

But this form of exploitation and control was not enough for capital. It wanted comprehensive control of inventors and found it in the employed inventor bound by instructions. Yet it failed to take into account that real inventing, which goes beyond incremental inventing, requires a freedom that the inventor finds in none of the existing institutions. Over decades Germany developed ever greater perfection in incremental inventing and took that to be the essential thing.

A particular role was played here by the large corporations with their overwhelming dominance and their extreme ignorance. They mercilessly appropriated not only the ideas of inventors but also the productivity gains of their own suppliers and their innovations, and sank the profits into their quasi-socialist and often enough corrupt structures. Today we are watching them transfer the last remaining remnants of the substance built up over decades to China. What remains for Germany as a consolation prize is at best (probably only temporary) assembly plants for Chinese models.

The ignorance goes so far that people still believe Germany is the land of inventors and that China’s strengths lie only in scaling. Both are sheer self-deception. For many decades Germany has invented only incrementally, and the Chinese have been systematically developing their own culture of invention ever since they invited the TRIZ expert Prof. Orloff in 2008. For the moment they are still scaling – but soon we shall see the results of a new

Chinese inventiveness, while Germany possesses nothing but memories of former greatness and importance.

This is exactly where InnoCoop comes in. The chain invention → innovation, on which a living industry depends, is broken in Germany: the invention that could make it to a product fit for the world market starves in the capital gap; the pseudo-innovation that no longer rests on any real invention withers into an empty and ultimately unsaleable shell. InnoCoop is the place where the two work together again – it takes invention as its foundation and carries it until innovation can come out of it. It carries the invention, and the inventor who brings it forth, through their most vulnerable phase up to proof of concept, and together with him takes on the licensing over the entire term of the contract.

2. The genesis: the contradiction from which it grows

The independent inventor faces a contradiction that he cannot resolve alone. To realise his invention he needs capital. But capital, as it is built today, demands control: it wants all the rights in the invention as the price of its involvement. Without capital the inventor is powerless; with it he is dispossessed and powerless. That is not a technical contradiction but an administrative one – and precisely for that reason it can be resolved.

The ideal final result is easy to name: the invention is realised, and the inventor retains sovereignty over it. Capital serves without owning.

The resolution lies in a separation. One separates what capital today fuses together – financing and control. A purpose-bound, associative form holds the capital; control over the invention remains with the community of inventors and with its originator. It is the same separation that steward-ownership demonstrates – only applied one stage earlier: not at the finished enterprise, but at the act of inventing itself.

3. The structure

From this one separation grows a building with three load-bearing parts and a cycle that turns it into a living body.

The capital engine. The money comes neither from a benefactor nor from the state, but from the value that earlier inventions have already created, as described in detail in – Chapter 8. Positioning –. Inventing is statistical – many approaches and attempts do not pay off, the rare ones carry everything – and the returns from these hits finance the next generation of inventions. Appropriate returns can be achieved only in negotiations on an equal footing, which the individual inventor never has and never had.

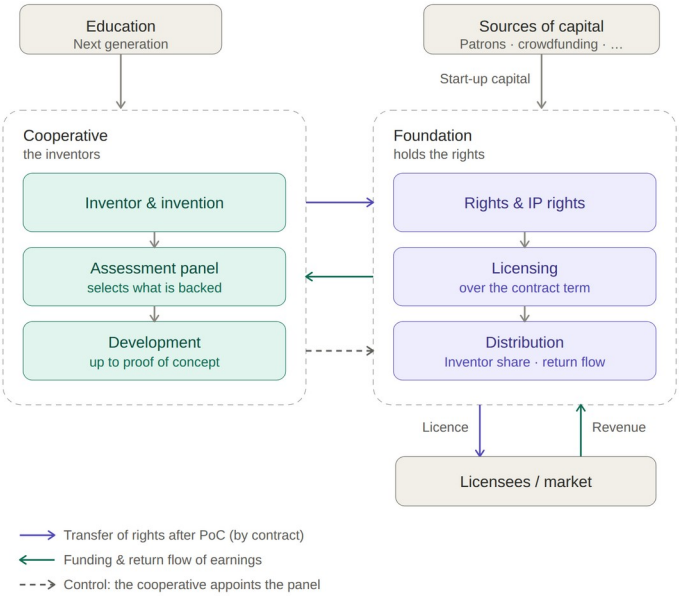
The evaluation panel. A circle of experienced inventors judges an idea on its substance, not on its market price – and, this is the uncomfortable condition, it remains open to the invention that the professional world at first rejects. It is precisely this kind that is otherwise filtered out everywhere; a panel that let through only what the experts recognise anyway would be worthless.

The governance. It rests on one sentence: whoever finances does not steer. The founding patrons provide the capital, but they do not decide which invention is carried – otherwise the institution would become an industrial consortium that filters out the very uncomfortable invention for whose sake it exists. Capital and judgement are separated by a wall.

The cycle. So the building stands – but it remains dead as long as one question is open: where do the inventors come from? InnoCoop produces its own new generation. Inventive thinking is taught – that is the entry stage. Those trained there do not step into a void but into a workshop that keeps running: into the practice where inventions arise, and before the panel that assesses them. Training → practice → judgement, a cycle in which what has been learned becomes invention, and the yield of invention becomes the training of the next generation. Training is therefore not an appendage but the first link. To educate people in inventive thinking and then release them

into an infrastructure that exploits and casts them out would be crueller than never awakening the ability at all.

That is the structure: derived from a contradiction, built in three parts, closed into a cycle.



The two bodies, the flows of money and – dashed – the line of control: the patrons give, but the cooperative governs.

4. The iron law to which it is subject

No sooner does the structure stand than it is subject to a law to which every organisation is subject – this one too, this one especially. It is the real difficulty, and whoever keeps quiet about it builds the next illusion.

Every form meant to realise a purpose is at the same time a survival system. And because what the world continuously selects for is survival

alone, not fidelity to purpose – whatever does not survive disappears – the actual goal shifts over time from the purpose to the preservation of the means. The means becomes an end in itself. This is not a failure of bad people that a few better ones could remedy; it is a selection pressure that operates even among the thoroughly virtuous – and that is why it is iron.

It operates through four nameable channels: the **self-interest** of those who operate the means and live from its continued existence; the **power of interpretation** they accumulate over time, until they determine what the purpose "really requires"; the **primacy of the measurable**, which puts the key figure in place of the unmeasurable purpose; and **adaptation to whatever feeds the organisation** – funders, supervisory bodies, market. By each of these routes an inventors' foundation, founded against the filtering out of inventors, is itself drawn into doing it: its panel closes itself off into a guild, its criteria ossify, and in the end it filters out precisely the uncomfortable invention for whose sake it came into being.

As a contradiction in its pure form: the purpose needs a means capable of survival in order to have an effect – but the capacity to survive is a goal different from the purpose, and the only one that continuously rewards itself. The more capable of survival the means, the stronger the pull to put self-preservation in place of the purpose.

5. The resolution: reversing the selection gradient

The obvious answer is wrong, and it has already been tested. One could attack duration – terms of office, rotation, constant renewal – so that no one becomes entrenched. That is exactly what the Greens tried in their early days, with the rotation principle and the separation of office and mandate. It failed: rotation destroyed competence, and the apparatus came back in through the front door almost as soon as it had been thrown out through the back. Duration is not the harmful parameter – whoever shortens it only loses competence and leaves the displacement of the purpose untouched.

The harmful parameter is the **coupling of the means with self-preservation**. The resolution starts at two points:

First, the **purpose is made irreversible** – anchored, not privatisable, not alterable. The assets can no longer be turned back into a saleable form. That is the proven, feasible part (steward-ownership demonstrates it), and it fends off the route of expropriation and commercialisation.

Second – and this is the real inventive task that is still open – the **selection gradient must be reversed**: the organisation should continue to exist only as long as it serves the purpose, and should begin to falter as soon as it betrays it. Fidelity to the purpose must become the condition of survival.

Here a distinction between three kinds of continuity helps. The *institutional* – the organisation continues to exist – and the *legal* – asset lock, foundation ownership, unsaleability – can be secured well; that is the proven part just mentioned. Only the *semantic* remains open: that after eighty years the purpose still means the same thing. For the irreversibility of the form protects only the letter; it does not prevent the apparatus from keeping the shell and hollowing out the purpose from within. No robust mechanism against this has yet been demonstrated.

The reason is almost logical: every written purpose is language, language has to be interpreted, and the interpreter sits inside the organisation. A purpose council can establish betrayal – but who establishes whether the purpose council is still interpreting correctly? The recursion (*quis custodiet ipsos custodes*) ends only at a point of reference that lies outside and does not need to be interpreted.

The most promising lead points away from prevention and towards making deviation survivable. One does not make deviation impossible – that recurses into infinity – but self-correcting. The model is found not in company law but in the commons of free software. Two provisions protect the purpose there over decades: *copyleft* – an asset lock not on assets but on knowledge, which keeps the good open for ever – and the *right to fork*. If

the leadership drifts, no panel sits in judgement over “betrayal of purpose”; the community splits off the work and carries it on itself, and the captured version simply loses its people. When OpenOffice stagnated under Oracle, the community forked LibreOffice; the original withered without anyone having to pass judgement. People voted with their feet.

Applied to InnoCoop, this means: the external point of reference that cannot be interpreted away is in the end neither a text nor a guardian, but the very injustice against which the institution exists – the never-ending supply of inventors who have been filtered out. It lies outside, renews itself and cannot be interpreted away. Survival and authority are coupled to it: alongside the asset lock comes a right of the community of inventors to fork and to refound. If the foundation’s panel is captured, the inventors may found anew, and the purpose-bound assets follow, by a default rule, the active community – not the apparatus sitting on the empty shell.

This does not solve the problem, but it makes it smaller – and honesty requires naming the limit. Code can be forked because it is non-rival; copying costs nothing. Capital is rival. “The assets follow the fork” therefore does need a rule for *when* they follow, and establishing that is again a point open to attack. But it is a different, harder question than before: no longer “has the purpose been semantically betrayed?” (undecidable), but “is the active community moving away?” – difficult, but external and observable in behaviour. It is the difference between a judge ruling on a company’s failure and customers who stop buying. How to build this trigger cleanly and itself proof against capture remains the highest design task of the whole. We do not pass it on as solved but as sharply posed: the proven part – the irreversible form; the best-known lead – copyleft and fork as an external point of reference; and the open remainder – the trigger that the capital follows.

6. The statics: one principle, many interfaces

The iron law has several faces. Each demands the same basic movement – the separation that secures the purpose and decouples self-preservation – only applied at a different interface.

Compulsion to commercialise (the economic means). Everything that touches the market is drawn into business: pricing instead of judging, the safe instead of the risky. *Separation:* the irreversibility of the foundation form, which rules out later privatisation. – The American Battelle Institute stayed at the foundation pole and has lived for almost a hundred years; its German offshoot was converted into a GmbH and went under in insolvency. The form decides.

Compulsion to bureaucratise (the political means). Everything that seeks legitimacy or state money is drawn into procedure – an administration that sustains itself. *Separation:* from the state InnoCoop takes the legal form and its recognition, not its money and not its procedures. The state enables the form; it does not finance it.

The chimera (wanting to be both at once). The hybrid inherits not the virtues but the diseases of both poles – sluggish and profit-driven at the same time. *Separation:* not balancing, but leaning firmly on the associative pole and actively resisting the pull towards the other.

Oligarchy (the functionary class that becomes independent). This is the iron law in its purest form. *Separation:* the reversal of the selection gradient from Section 5 – not the naive shortening of duration, but the coupling of continued existence to fidelity to purpose.

7. The existence proofs

InnoCoop is no utopia; its components exist, only never together and never in this place.

Nikolaus Laing is the emblematic case – existence proof and requirements specification at once. For decades the German inventor lived from exactly the model InnoCoop has in mind: invent, realise, license. His tangential fan of 1956 is still found in almost every household today, his spherical motor pump continues to run – now at Grundfos – and, according to Der Spiegel in 1974, products based on his inventions achieved worldwide sales running into billions, carried by an invention factory with over a hundred scientists and well over a thousand patents. An independent institution can live from inventing and licensing; Laing demonstrated it. And much of what was dismissed as crankery on his part – heat recovery, the heat pump, radiative cooling – is standard today: the apparatus rejected as a whole what was right in its parts, instead of examining the parts.

And yet he failed in the end – and it is precisely this failure that writes InnoCoop's functional specification. It was not free inventing that failed – the same institution proved the opposite for decades – but the form of ownership, and at two points. Outwardly: a visionary, capital-hungry expansion led Laing into dependence on outside capital, and there capital applied its leverage until he lost control over his work. Inwardly: for many years the whole project depended essentially on the Laing couple – regardless of how far others were involved in particular respects. It was their work; it stood and fell with their persons, instead of resting in a community that would have outlasted it. The lesson is therefore twofold: to build a structure that can carry even the capital-hungry venture without surrendering control – and to place the work on a community of stakeholders instead of letting it hang on individual persons. What Laing lacked was not genius but duration – and duration is created neither by employment nor by individual ownership, but by the cooperative alone.

The **American Battelle Institute** is the proof of the capital engine. It carried xerography when a dozen companies had turned away the inventor Chester Carlson; out of the one invention it carried came Xerox, and its returns carried decades of research. That is the loop we start one stage earlier. The model here is the founding idea, not today's Battelle: the

organisation of today lives predominantly from government and defence-related contracts and is hardly any longer the independent institution for inventors it once was. We take the existence proof and the model – not the present-day form and not the money.

Steward-ownership – WALA, Alnatura and the cooperative forms – is the proof of the irreversible legal form: enterprises whose ownership is neither saleable nor inheritable but serves the purpose in trust. They show that the wall against privatisation holds.

What is missing is only the application of these proven parts to the earliest, most expensive, most failure-prone stage: the act of inventing itself.

8. Positioning: neither market nor state

InnoCoop is a new route alongside the familiar institutional ones – and alongside the largely failed route of the independent inventor as a lone fighter. In their basic forms, the familiar routes follow the forms that Franz Oppenheimer called the economic and the political means. InnoCoop, by contrast, is based on the associative form.

From this follows its anchoring, hard and simple: the start-up and core capital already comes from an invention. Governance lies with the community of inventors. The legal form is rooted and irreversible. The state is needed as the enabler of the form, not as a source of money.

This is no abstraction. The capital engine already exists, for this concept grows out of a real case. The polygon turning process – basic patent EP 0097 346 B1, filed in 1983 – changed the machining of polygonal shaft–hub connections and profiles. For over thirty years it has been running in series production: at SKF for rolling-bearing cage rings and at other companies such as Index-Traub, Weisser, DMG MORI and Wieland. From this process, firms have in some cases drawn considerable competitive advantages over decades.

The basic patents expired long ago; use was and is free and legal. Precisely therein lies the strength of the proposal: there is no legal claim, and therefore the contribution of the long-standing users is not an enforced back-claim but a voluntary act. Meanwhile development continues – on the basis of new property rights and developments, licences and income are possible within the framework of InnoCoop – further developments that reach far beyond the known state of the art.

From this follows the proposal: the companies that have in some cases been drawing considerable benefit from the process for decades make an appropriate share of these earnings available to the InnoCoop project – as a one-off or as a regular payment – and in return become its founding patrons and preferred licensees of the new and subsequent inventions. The start-up capital is not an achievement of the institution – it has not yet produced anything – but a lifetime's achievement of more than forty years that the inventor brings into the project: the process itself, its further development, the new property rights. The long-standing users join with a share of their earnings. Only from then on does the actual cycle begin, in which the institution feeds on the value that it itself produces in cooperation with the inventors of the future.

9. How the apparatus gets going

The cycle sustains itself as soon as inventions bring in appropriate licence income. Beyond the start-up and core capital outlined in the previous chapter, it makes sense to think about further appropriate sources of financing.

- **A first invention that can be licensed and realised quickly** as bootstrapping – a hit that starts the loop.
- **Crowdfunding** – not just money but a movement: citizens finance inventors, and public attention grows with the money.
- **Cooperative shares held by the inventors** themselves – the cooperative as a genuine cooperative with contributions from its members.

- **Existing foundations and philanthropy** – the steward-ownership network, foundations whose purpose is education or the promotion of inventors.
- **Income from the training stage** – TRIZ and mechatronics courses bring in their own revenue; the first link helps to finance itself.
- **Side income from tools** – licences for the simulation and profile software as a small, steady return flow.
- **The public sector only for the form** – charitable status and recognition of the legal form, not as a source of money; public funding at most in a targeted way for training, so that no bureaucracy creeps in.

None of these sources may violate the condition that stands above everything: whoever finances does not steer. Money yes – control no.

10. The first step

No congress, no appeal, no funding application. A conversation – with the first founding patron, from among those who have benefited longest and to a particular degree from the advantages of polygon technology. Not with an invoice, but with the invitation to be the first: preferred licensee of the further-developed invention and co-founder of the institution that holds it. Whoever co-founds first sets the standard; the others follow the precedent instead of negotiating it.

In parallel the entry stage takes shape: the training pilot, by which the first link of the cycle can prove itself on a small scale before the whole grows. And the legal form is built with existing law – as the foundation-owned companies have shown – so that InnoCoop does not have to wait for a law that does not yet exist.

The foundation, however, is the one sentence with which it stands or falls: to be the place where invention can become innovation – and where the inventor who brings forth the invention is supported to the best of its ability and no longer cast out.

Fourth Column – Education

The Education of the Inventing Practitioner

The fourth Column asks: how is the ability to invent formed – taught, practised, passed on?

A superstition hangs over inventing that is older and more stubborn than others in technology: the belief that inventing is a gift. You either have it or you don't. Altshuller's discovery refutes this superstition: inventing has a grammar, and whatever has a grammar can be taught and learned. But refuting the superstition is not yet an educational practice – it has to be set up institutionally, has to be tailored to people who are able to take in the teaching, and has to take place in an environment that actually needs what is taught.

This Column proposes where that could begin: in vocational training in mechatronics, not, to begin with, in academic engineering education. With the person we call the *inventing practitioner* – the person who keeps a hand on the workpiece, feels the contradiction where it becomes real, and can resolve it systematically because he has learned TRIZ as a method and design engineering as a language. Not the half-finished engineer, but a figure in his own right: the craftsman with the vocabulary of systematic invention.

This part is more of a draft than the others. It is a letter to two people we need in order to make it real: Michael Orloff, who can bring TRIZ to pedagogical maturity, and Dietrich Lembke, who embodies German design engineering in the Ilmenau tradition. It is an invitation to build the column with us – not the finished building. It ends in a question, not in an answer.

Dear Michael,

two columns of this book are already standing. The first asked what the inventor needs *within himself* – the prerequisites and the practice of inventing, the virtue of the right mean, which decays neither in too little nor in too much. The second asked what he needs *around himself* – an institution that carries the invention through its most vulnerable phase and does not throw the inventor who brings it forth out of the process. The third column, which we begin here, asks about what lies between the two and makes both possible in the first place: **How is the ability to invent formed? How can it be taught, practised, passed on?**

We are writing this part deliberately towards you. Not out of politeness, but because it lies in your territory like no other. What we present here is a beginning, not a finished building – written with the intention that you step into it, contradict it, deepen it. When you wrote to us that now was a good time to bring the book to completion, we took that for what it is: the call to stop merely sketching this third column and to build it with you.

1 · The Most Stubborn Error

A superstition hangs over inventing, older and more stubborn than any other in technology: the belief that inventing is a *gift*. The flash of inspiration, the spark, the genius – you either have it or you don't. This superstition is comfortable, because it relieves us of any effort to teach and learn inventing. Yet even a gift has to be trained.

It is precisely here that Altshuller's discovery stands as a standing refutation. By going through tens of thousands of patents not according to their subject but according to their *structure*, he found what the superstition of genius denies: that strong solutions carry recurring patterns, that behind the apparent flash there is a technical contradiction, and that the resolution of this contradiction follows a limited number of principles. Inventing – such is the whole boldness of this thought – has a grammar. And whatever has a grammar can be taught and learned.

You carry this line directly onward – from the Vorkuta camp, to which a letter to Stalin about the situation of inventors brought him for twenty-five years and from which only Stalin's death freed him, all the way to the academy in Berlin. And even there he did not give up as an inventor – with an irony one could hardly have invented: in the camp he found the colleagues he needed, and with them, at last, the time to think further about the approaches he had brought with him. What free life denies the inventor – the right companions and the time – was there in abundance at the place of utmost unfreedom. That is why this part cannot be written without you: we can report the idea, but you are one of the few who live it.

An example from an entirely different field makes this tangible. The drawing teacher Betty Edwards showed in *Drawing on the Right Side of the Brain* that drawing too – seemingly the pure gift – follows a learnable method: a particular way of perceiving what is seen, which those with a talent for drawing probably apply unconsciously. Whoever learns this way of seeing learns to draw; we have tried it, and it works. What her method does for drawing, TRIZ does for inventing: it makes conscious and teachable what the gifted do without naming it.

And yet it would be foolish to deny aptitude altogether. There are people who bring with them a basic inventive disposition and develop it further through the circumstances of their lives – just as someone finds fulfilment in drawing, gets into flow and drives the method to perfection. For them TRIZ is not a substitute for the aptitude but its amplifier: with the method they reach better results faster. But – and this is the real blow against the cult of genius – anyone who wants to can learn inventing, just as anyone can learn to draw. Whether someone finds his fulfilment in it and produces something extraordinary is decided by his disposition and by many circumstances. The method opens the door; how far someone goes through it is another question.

Let us hold on to the first sentence of this part, however plain it sounds: **Inventing is not a gift but a method. And what is method belongs in education.**

2 · Who Invents – the Round-the-Corner Thinker

If inventing is a method – who, then, is the person who practises it? It is worth defining him precisely, because the pedagogy we propose deals with a particular type.

The way a person approaches a problem can be described almost geometrically. There is the **straight-ahead thinker**: he sees a path and follows it, better, faster, more free of errors than anyone else. He optimises what exists. Without him nothing would get finished, no plant would run, no tram would move. There is the **lateral thinker**: he leaves the main direction, questions, opposes – but he stays on the same plane and defines himself by what he is against. And there is the **round-the-corner thinker**: he does not accept the space of the problem. He jumps, surfaces at a point that seemingly has nothing to do with the starting point, and finds solutions that did not even exist in the original problem space. He *invents* instead of optimising.

The inventor this book is about is the third. And here lies his undoing: systems are built by straight-ahead thinkers, and so they promote straight-ahead thinkers. The round-the-corner thinker is not even rejected – rejection would presuppose understanding. He is *not understood*, overlooked, dropped through every grid, because the grids are built for other shapes. Only once his idea has changed the world is he retrospectively declared a genius.

The same distinction bears another name if one looks at the result rather than at the mind: **incremental** versus **radical**. The incremental inventor improves what exists – that is Germany's world-champion discipline, the foundation of "Made in Germany". The radical inventor invents what does not exist and changes the rules. As long as the world changes slowly, the incremental one is enough. But at every paradigm shift – electric motor, software, AI – he loses, however good he is. He is the best coachman just as the automobile is being invented.

And here lies the reason why TRIZ stands at the centre of this part and is not merely a better method of perfecting: TRIZ *resolves the contradiction* instead of optimising it away. It is the grammar of radical, not of incremental, inventing. A pedagogy that produced only better straight-ahead thinkers would miss its goal – it would form the masters of what exists while the world invents the new elsewhere. The point is to educate *and to protect* the round-the-corner thinker, whom the system would otherwise re-educate into a straight-ahead thinker.

One more word, so that this does not turn into arrogance. Each of these styles of thinking has its place, and none is the best. One of us likes to put it this way, from his own experience: a good thing not everyone is like me – otherwise no tram would run. The inventor needs the straight-ahead thinker, who puts his idea into practice and keeps the operation running, as much as the latter needs him. The pedagogy we are looking for does not despise the path-follower; it only wants to prevent him from remaining the only one the system lets through.

3 · The German Separation

If inventing can be taught – where, then, is it taught? The German answer, honestly considered, is: hardly at all, and least of all where it would be most fruitful.

Germany maintains a separation that it considers natural and for which it pays dearly. On one side the hand – the dual vocational training system, the workshop, the practitioner who assembles, maintains, operates. On the other side the head – the university, the research department, the place where the inventive is presumed to reside. The mechatronics trainee, that is, precisely the person who stands closest to the real machine, where the contradictions appear first and undisguised, is prepared to keep existing systems running. The inventive ability is considered to be at home elsewhere.

That is the German separation: the hand and the inventive method are kept apart. The practitioner is not supposed to invent, the inventor is not

supposed to have his hand on the workpiece. And both are impoverished by it – the practitioner because he lacks the tool to recognise his daily contradiction as solvable; the academic inventor because he lacks the place where the contradiction becomes *real*.

Interlude in the First Person – How I Became One

Here one of the two of us steps out of the "we". For what follows is not an argument about a possibility but the testimony of a proof lived out. – H. L.

I can say of myself without exaggeration: I aspired to the ideal of the mechatronics engineer when the word did not yet exist and when it was still considered normal for everyone to stay only in the trade he had learned.

From my father I had the craftsmanship and everything to do with two-wheelers and engines. I did my apprenticeship as a heavy-current electrician at Reifenhäuser – today it would be called an energy-systems electronics technician. To that came plastics technology, process engineering, measurement and control engineering, and, growing into a new era, the first integrated circuits and digital technology. What the term "mechatronics" later tried to name – the fusion of mechanics, electronics and control – was never separated in my training. Not because a curriculum provided for it, but because a father, a company, a machine park and a colleague came together.

The most important thing was the machine park. For its time Reifenhäuser had an extensive and modern stock, with many numerically controlled machines. These machine tools fascinated me more than the plastics plants the company was known for. I spent many lunch breaks in front of them. And already in my second year of apprenticeship a colleague gave me a first lathe, which I set up in the cellar of my parents' house. From there, from those lunch breaks and that cellar, a straight line runs to everything that came afterwards.

My first inventions were used in plants. Others, at least as valuable, were rejected out of incomprehension. One of them was a contactless film-width

measurement – I did not just design it, I built it. Its fitness for purpose was therefore not a claim but stood there as a finished device; and precisely for that reason its rejection is the purest example of what I mean. It did not fail on its merits – the thing stood there finished – but on the incomprehension of those who had to decide.

And here lies what saved me from frustration: I found not only rejection but also recognition. That was no comfort for the soul, it was a calibration. Because some things were accepted, I knew that the rejection of the others was not down to me, nor to the quality of the thing, but to the ignorance of the decision-makers. This certainty spared me two wrong paths at once – the self-doubt that breaks you, and the self-exaltation that makes you bitter.

And the rejections taught me something else that I consider the most important of all: not to set my heart on any of my inventions. I commit myself to them with everything I have. But if it does not work, I can also let them lie. For an inventor must on no account become pig-headed, and he must not fool himself. The moment you can no longer examine your own cause because you love it too much is the moment the inventor turns into something else.

4 · The Inventing Practitioner

The proposal that follows from this diagnosis is simple and has a concrete beginning. In a first working paper for the Berufliches Fortbildungszentrum we put it like this: **TRIZ as the methodological backbone of an additional qualification “Inventive Problem Solving” in mechatronics training.**

Not to replace the craftsmanship, but to *complete* it. The inventing practitioner is not the mechatronics technician who assembles less, but the one who can in addition name a technical contradiction, resolve it systematically and carry the design of the solution through to maturity. And here we must be more precise than the first thought suggests: it is not two elements that form him, but three.

The first is **practice** – the hand on the workpiece, the embodied knowledge through which the contradiction becomes real in the first place. The second is **TRIZ** – the method that resolves the contradiction systematically instead of leaving it to the chance of a sudden idea. The third is **design engineering** – the systematic design methodology that turns the resolved contradiction into a fully calculated, reproducible solution ready for production.

Why the third element is indispensable becomes clear if one imagines it away. TRIZ without practice becomes formalism – the theorist who recites the procedure without ever having stood at the machine. But TRIZ and practice alone, without design engineering, lead only to **tinkering** – to the brilliant idea, the one-off in the cellar that works and yet never reaches readiness for series production. Only the three together make the complete inventing practitioner: the one who recognises the contradiction in the thing itself, resolves it systematically and carries its design through to readiness for production.

Systematic design theory is a German strength with more than one school. The one, more widespread, bears the names of Pahl and Beitz; the other goes back to Friedrich Hansen and the Ilmenau tradition. For this second line we have a witness in our own circle: Dietrich Lembke, who studied under Hansen in Ilmenau – it was through him that we first learned of this school at all, where previously we had known only Pahl/Beitz. Just as your voice, Michael, is indispensable for TRIZ, so Dietrich can contribute to design engineering and give it impetus. In this way each side of the triangle finds its bearer: practice in the testimony of the one who stood at the machine, TRIZ in you, design engineering in Dietrich. Which of the two schools tips the balance – or whether both are needed – is among the questions we still have to settle with our co-authors.

Why in vocational training of all places, and not only at university? Because this is where Germany’s real strength lies – the dual system, for which the world envies it. Implementing the systematic methods in this system would yield something no purely academic course can produce: an

inventing that is rooted in the hand, at the point where the problems really are. The inventing practitioner does not emerge in the seminar room but at the machine that does not do what it should – facing someone who has learned to read this not as a nuisance but as a contradiction.

One more thought, which points beyond this part and leads back to the second column. To find the three elements – practice, TRIZ, design engineering – united in *one* person is a stroke of luck, not the rule. More often they rest on different shoulders: one has his hand on the workpiece, another masters the method, a third calculates the design. Inventing then becomes the business of a group of people with different focuses and abilities.

And here lurks the real difficulty – not the technical one but the human one. Inventing in a group fails, as a rule, on the all-too-human: on the quarrel over the share and over the name, on the inability to subordinate oneself to the common cause, on the vanity that is stronger than the work. That is exactly why the organisation the second column proposes – the inventors' cooperative and foundation – is not a mere financing framework. It is the place where the collaboration of different people first becomes possible and lasting: a form that secures each person his share and his name and so defuses the all-too-human that would otherwise blow the group apart. What education produces in abilities, the cooperative brings together.

5 · The Link to Practice – What TRIZ Still Suffers From Today

Here we must name a weakness, and it lies not in TRIZ itself but in the way it is usually taught. TRIZ in the seminar room works on *abstracted* contradictions: you take a problem that is already formulated as a contradiction, apply the matrix or the forty principles and generate directions for solutions – clean, systematic, and detached from any real hardware. This forms people who master the formalism without possessing the embodied knowledge of *how things really jam, bind, fail, behave*. They can recite the procedure, but they have never stood at the machine that rattles.

And one must not reproach the teachers of the method with not having tried – they have struggled, often all their lives, for the link to practice. But they never managed to establish it *independently*, as a firm ground carried by the method itself. Where it came about at all, it arose only in individual companies, and there it always depended on the conditions that happened to prevail: a well-disposed boss, an open development department, a concrete product. If the condition fell away, the link to practice fell away with it. It was never structural, always borrowed – and that is not a reproach to the people but a finding about the matter.

That is the sore point: **The contradiction becomes real not on paper but in the thing itself.** It reveals itself first to the one who has his hands on the workpiece, and in a form that no textbook formulates in advance. The contactless film measurement from the interlude is precise evidence of this: the contradiction “the edge must be detectable, but is invisible” is in no matrix – it confronts only the person standing in front of the running, transparent film. And its resolution – the birefringence of the film itself, turned into a source of contrast between crossed polarisers – is a physical effect that only someone who knows the material will reach for. That is inventing from the embodied problem, not from the formalism. And it was, without the name being mentioned at the time, pure TRIZ.

From this follows the sharpest justification of this entire part. **TRIZ belongs in vocational training, and not only at university, precisely because vocational training is the place of the link to practice.** The dual apprenticeship supplies the hands that the TRIZ formalism lacks; it is the place where the contradiction appears as a real, jamming, recalcitrant thing. Connecting TRIZ not only with heads but also with hands cures exactly the disease from which TRIZ as a pure seminar method still suffers today. The formalism alone forms the theorist; the formalism at work on the thing forms the inventor.

And here the arc closes towards the yardstick that follows shortly. The abstract formalist without practice – who applies the method to problems he has never touched – stands closer to the charlatan than he would care to: he

produces the *appearance* of systematic inventing without its substance. The link to practice is the antidote, and that is why this is not our critique of TRIZ but the defence of genuine TRIZ against its seminar-style flattening.

Whether our diagnosis holds with this sharpness – that the TRIZ movement, despite all its efforts, never managed to establish an independent link to practice, and that where it arose it always depended on the conditions in individual companies – only two people who know it from close up can confirm or qualify with authority: you, dear Michael, and Dietmar Zobel. On this sentence stands or falls the sharpest justification for why TRIZ education must take place not only in the seminar room but also in the workshop. That is why we put it before you not as an assertion but for examination.

6 · The Next Step – the University

Vocational training is the foundation, not the limit. Because that is where practice lives, the inventing practitioner must *begin* there – but TRIZ and systematic design methodology unfold their full power only where research is done and ideas are developed further. In a subsequent step, therefore, TRIZ also belongs at university: not instead of vocational training but building on it, just as the inventing practitioner can rise from journeyman to engineer without losing his hand.

For what is missing today is precisely this engineer who comes from practice. One only has to look at the situation: today there are so many mechanical engineers with doctorates who have never operated a machine – let alone being capable of building one themselves. That is the German separation at its end point: the engineer who researches and writes about the machine without ever having had his hand on it. The path we describe – from vocational training upwards to university – reverses this. It would once again produce engineers who know the machine because they stood at it before they thought about it.

And Germany does not have to invent the university for this first. It already stands – in Nuremberg. The **University of Technology Nuremberg (UTN)**, founded in 2021, is the first newly founded state university in Bavaria in over four decades and set out as Germany's first dedicated AI university. It explicitly sees itself as a living laboratory of a new university thinking: no rigid faculties, but interdisciplinary departments; no classic degree programmes, but ones put together in an interdisciplinary and modular way.

For our approach this is ideal ground – for two reasons. First, **mechatronics** has a rank of its own there: "Mechatronic Engineering" is one of the departments, with its own test hall and the declared aim of concentrating on next-generation manufacturing technologies – exactly the field of the inventing practitioner. Second, by doing without faculty walls, UTN is open to something a traditional university finds hard to accommodate: an education that does not distribute practice, TRIZ and design engineering among separate chairs but brings them together. Where the classic university perpetuates the German separation institutionally, UTN is built to overcome it.

That **artificial intelligence** forms the second focus there, alongside mechatronics, opens a further door that we only hint at here: how systematic inventing after Altshuller relates to AI – tool, partner or competitor – is among the great open questions, and it would have its natural ground in a place that houses both. But that is a chapter of its own – and yet the answer is already present here on a small scale: this book came about in a cooperation between human and AI and is thus itself an example of the future possibilities that we see as symbiosis.

For now the finding is enough: the place where the second step could be taken already exists – and it lies on our doorstep.

That this path holds is not a postulate but a finding – and you have seen it with your own eyes. Since you were called to China in 2008 to teach TRIZ there, a country has systematically built up an inventive culture of its own and anchored the method all the way into its training. In the same period Germany has gone on celebrating itself as the land of inventors without having produced inventions of weight for decades.

We do not write this as a lament – the lament would be cheap and would help no one. We write it as what it empirically is: proof that anchoring the method in *education* works, and that the window for Germany is not closed if it acts. China did not wait for the genius. It taught the grammar of inventing.

And the look towards China sharpens something else as well – it confirms our triangle from a side we would not have expected. China too, in training large numbers of engineers, ran into the wall we have just named: the abstract method alone did not carry, and the engineers found it hard to grasp the concrete problem and to translate the solution into real innovation. The Chinese answer was a further development that explicitly couples the inventive process to the *design process* – and that, as the specialist literature shows, draws among other things on German design theory. That is, by a different route and out of a different need, the same insight that carries our triangle: TRIZ alone remains formalism; TRIZ and practice alone remain tinkering; only the connection with systematic design methodology becomes disciplined invention. Two cultures, independently of each other, arriving at the same insight – one could hardly wish for a stronger argument for the soundness of the approach.

[Here is the place for Michael's eyewitness account: how was TRIZ brought into training in China back then – and how do things stand today? Does the state anchoring still hold, or has it sunk into routine? Has the coupling to the design process really closed the gap in the link to practice? What the publications show as structure, only he can tell as lived reality.]

8 · The Right Thing in the Right Place

The look towards China is easily misunderstood – as a call for more state. For what worked there was ordered by the state: the method carried from above into training, programmes, centres, plans. If one draws this conclusion, one runs into a contradiction that an attentive reader notices at once: elsewhere – in the great inventions that the West itself produced and then gave up – it was precisely the excess of state, of regulation, of permits that suffocated, and the freedom to act without permission that carried.

The contradiction is none once it is grasped correctly. It is not about more or less state. It is about everyone doing the right thing in the right place in the right way.

Inventing requires **freedom** – the room to act without permission. Where this freedom is lacking, the results show it. That is the first, indispensable function, and it belongs to the individual and to free exchange, not to the authorities. It is what the first column of this book describes under the attitude of the inventor.

But freedom alone does not carry the invention over the two walls at which the individual fails – the wall of incomprehension and the ditch before readiness for series production. For that an **institution** is needed. Not the coordinating, regulating state that wants to prescribe to every wave where it is to move – but the voluntary cooperative that protects the inventor and carries the idea across the ditch. That is the second column, InnoCoop: the function is fulfilled, but by the right route – through cooperation, not through compulsion.

And thirdly it needs a **ground** on which the inventor is formed in the first place. That is the legitimate task of the state: not to coordinate invention, but to create the ground – vocational training, the university, teaching the grammar of inventing. Throw a stone into the water and let the waves do what waves do; do not prescribe to every wave its path. That is the third column, this one here. And where the market fails because of its short time

horizon – with the generational investments that pay off only over decades – there the state may and must also carry out what no one else carries out. The education of the inventor is such an investment too: it pays off not in the next legislative period but in the next generation.

By this yardstick everything becomes clear. China fulfils the functions – and thereby beats the indifference of the one who fulfils none; but it often fulfils them by the wrong route, through compulsion. Germany does the wrong thing in place of *all* the right things: it regulates and debates instead of leaving freedom alone, creating the ground, protecting the inventor. The question is never "more or less?" – it is always: the right thing, in the right place, in the right way.

So the three columns of this book come together under a single principle. The freedom of the inventor, the cooperative that carries him, and the education that brings him forth – each is a right function in the right place. None replaces the other, and none tolerates being mixed with what does not belong to it.

9 · The Yardstick – and the Danger

Here we must be careful, and no one knows that better than you. For a method of inventing, badly taught, turns into its own opposite: a cult. Into a list of forty principles rattled off without understanding; into a collection of maxims that give the appearance of system and betray the substance. It is this caricature – TRIZ as a checklist, as a Western seminar product – that has brought the genuine method into disrepute in the West.

In *HÄRETIKER*, Dietmar Zobel draws in the preface exactly the line that matters: the distinction between the genuine pioneer and the charlatan. This distinction is not ornament; it is the load-bearing principle of any pedagogy of inventing. For the inventing practitioner must be formed by *genuine* TRIZ, not by its imitation – otherwise we produce users of checklists and call them inventors.

And that is why your voice in this part is not decorative but structural. Only someone who stands in Altshuller's direct line can keep the pedagogy honest – can guarantee that at the end of training there stands someone who *understands* the contradiction, and not someone who recites a procedure. The Meta-Algorithm of Invention (MAI), the worthy goal, the early formative figure that Schliemann carried all his life: that is the stuff a genuine education of the inventor is made of, and you bring it with you.

10 · The Goal – Back at the Top

At the end of this path stands a particular person: the engineer with practical experience who has learned in the course of his training to think inventively, to solve problems, to design perfectly – and to turn his ideas into products and services fit for the world market. This last link is as decisive as the first ones: an invention that is not carried through to a product fit for the world market remains a fragment.

For here lies the touchstone against which everything has to be measured. What newly emerges in Germany must once again become *fit for the world market* – with the declared aim of being back at the top. That is no modest programme, nor is it meant to be. Everything that is not at least fit for the world market should be forgotten as soon as possible: not out of harshness, but because the scarce strength of a country that has squandered its lead must not be wasted on mediocrity. It is not about inventing for the sake of inventing, but about inventions the world needs – and that also find their way to it.

Whether this goal is attainable is decided not by intention but by the education that brings forth the inventing practitioner. And with that we come to the question that only one person can answer with us.

11 · The Invitation

Let us sum up the arc of this book as it now lies before us. The first part asked what the inventor needs within himself. The second, what he needs around himself. This third asks how the ability is formed and passed on.

That is why we present it as a beginning and not as something finished. We have defined the inventor – the round-the-corner thinker –, made the diagnosis – the German separation –, named the proposal – the inventing practitioner –, touched the sore point – the missing link to practice from which TRIZ still suffers today –, shown the proof – the Chinese mirror –, placed the whole under one principle – the right thing in the right place – and marked the yardstick: the distinction between the genuine and the imitation. What is missing is what we cannot bring in on our own – and for that we need not one voice, but two.

Up to this point this part has been a letter to you, Michael. At its end, however, it must widen, because two of the three sides of the triangle do not belong to us. One is yours: the concrete pedagogical form of TRIZ. How do you lead a trainee from the first sensed contradiction to its structured resolution? What does the exercise look like through which the inventing practitioner is formed without becoming a checklist user? Where does the method stop being a tool and begin to become an attitude? That is your life's theme, not ours.

The other side calls in a second person: Dietrich Lembke, who learned under Friedrich Hansen in Ilmenau. Dietrich, to you goes the question of design engineering – the third side, without which the brilliant idea would remain tinkering. How does systematic design theory fit together with TRIZ and practice? How do you teach the step from the resolved contradiction to the fully calculated solution ready for production? And which school carries – Friedrich Hansen's Ilmenau school, that of Pahl and Beitz, or both?

These are your questions, each in its place in the triangle. We have built the framework into which your answers belong. Complete the book with us.

Afterword – From Idea to Impact

What becomes of the idea in the world: the inventor's responsibility and the path from the workshop to impact. No longer a course of instruction, but an outlook – written from the inventor's point of view, for the inventor.

17. Transform or Outsource

The first question of the outlook: the inventor's responsibility for the impact of what he does. Why ills can only be redistributed – and why outsourcing them is the comfortable but dishonest path.

I. Why responsibility belongs to practice

With the ninth layer, the deed, the idea enters the world. And at the very moment it does so, something arises that the inventor would gladly push away from himself: a responsibility for what it does there. It is tempting to think that this is no longer his business. *I invent; what the world makes of it is for others to answer for.* This sentence is comfortable, and it is wrong. For the inventor has not handed the world just anything, but precisely this, shaped this way and no other, with these properties and these consequences. He has chosen, and whoever chooses is responsible.

That is why responsibility does not belong only to the consequences of inventing, but to the practice itself. It is the final form of the innovation mesotes of which the previous chapter spoke – that right measure which, in every layer, sought the right relation between too little and too much. Only now this measure is no longer directed inwards, at the quality of the work, but outwards, at its impact. And it has its root in the first of the engineer's three sentences.

II. The sum of ills and the two paths

The first sentence ran: the sum of all ills is constant. You do not solve a problem, you transform it; every solution creates new difficulties, and the total quantity of ills stays the same – only their distribution can be changed. For inner practice this meant finding the right distribution: which ills do I want to have? For the impact on the world it means something sharper. For if the ill cannot be abolished but only shifted, then there are two very different ways of "solving" a problem.

The first is transformation. You bring the ill into a form that you yourself can bear and are willing to answer for. You take the price named by the second sentence onto your own account. The second is outsourcing. You push the ill to where it no longer appears in your own accounts – onto society, onto the environment, onto future generations, into other countries, into later times. In both cases the sum stays the same. The ill has not disappeared. But in the second case it no longer stands in your own balance sheet, but in someone else's, which nobody reads.

This is the decisive distinction of this chapter, and the inventor must be able to make it for every piece of work. Am I transforming the ill, or outsourcing it? Am I bearing the price, or passing it on? From the inside the two paths often look the same – both "solve" the problem, both "work" – and that is exactly why the distinction is so easy to overlook and so important to make.

III. The grammar that pushes towards outsourcing

The inventor rarely makes this choice in a vacuum. As soon as his idea leaves the workshop and enters a company, a market, a financing arrangement, it falls into a language that has already prejudged the choice. This language asks about the margin, the market share, the payback, the lead over the competition. It is precise and powerful. But it has no word for the costs that arise outside its own balance sheet – except as an irksome footnote that one tries to keep small in order to escape regulation.

There is no need to think of this as malice, and it does not help the inventor to see it as a conspiracy. It is a grammar. A language that asks only about its own advantage will not even perceive the outsourcing of ills as outsourcing, but as efficiency. What burdens the other and relieves me appears in it simply as profit. That is exactly where the danger lies: not that someone maliciously shifts the ills, but that a whole way of thinking no longer even sees the shifting.

For the inventor, a simple watchfulness follows from this. He must know that the environment his idea enters will push him towards outsourcing without calling it that – that the question “does it pay?” almost always conceals the question “for whom does it pay, and at whose expense?” Inventive and engineering thinking are all too often applied today not in a problem- and solution-oriented way, but in a profit-oriented one; and profit-oriented application is, by its very nature, the outsourcing of ills. Whoever sees through this is not yet free of it – but he can choose instead of being driven.

IV. The illusion of neutral technology

Against this responsibility stands an old, comfortable story: that of neutral technology. According to it, what is invented is value-free in itself; only its use makes it good or bad, and the inventor is not responsible for its use. A knife cuts bread or throats – how is that the knife’s fault? So speaks whoever wants to be rid of responsibility.

But the story is false, and the inventor knows it better than anyone, because he made the decisions himself. Every technical decision contains a hidden decision about values. Whoever builds a machine so that it runs only with proprietary parts has decided about ownership and dependence before a single customer has used it. Whoever designs a product so that it breaks after two years and cannot be repaired has decided about the consumption of the world. Whoever chooses a solution that is cheap because it discharges its waste into a river has poisoned the river before the sums add up. These decisions are built into the design, not just into the use. The engineer is not a

neutral craftsman but a co-shaper of conditions – of power, dependence, durability, consumption. The question is therefore never only *whether* something works, but for whom it works and at whose expense. Whoever entrenches himself behind “I only build” denies the choice he has long since made.

V. The honest man’s prisoner’s dilemma

It would be dishonest, and unfair to the inventor, to stop here and moralise. For the responsibility this chapter demands is costly, and it is especially so for someone who stands alone. Whoever takes the external costs upon himself – chooses the durable rather than the cheap material, the clean rather than the dirty solution, fair rather than exploitative sourcing – makes his product more expensive, while the competitor who outsources the costs offers it more cheaply. The honest one loses on price, and the market rewards not him but whoever shifts the ills most skilfully. This is the form the prisoner’s dilemma takes in the inventor’s life: what is better for all is riskier for the individual.

This dilemma is real, and this book does not resolve it; its resolution lies in the structures that determine which behaviour pays off, and that is the subject of another volume. All that can happen here is to name the dilemma, so that the inventor faces it with open eyes. For it makes a difference whether one outsources the ills because one sees through the compulsion and still sees no other way, or whether one does so without even noticing. The first is a tragic situation; the second is a delusion. And sometimes, more often than the compulsion would have us believe, there is a third way – a solution that is honest *and* viable – and finding it is the finest task of the inventor who has learnt to tell the two paths apart.

VI. What genuine innovation would be

From all of this there emerges, almost of its own accord, a picture of what genuine innovation would be – not as a rule, but as a series of questions the inventor puts to his own work before he lets it out of his hands. Am I

transforming the ill or outsourcing it? Who bears the costs that do not appear in my accounts – and would I bear them if they did? Is my solution thought through over its whole lifetime, from manufacture to decay, or only up to the sale? And, the question from the sixteenth chapter, now turned outwards: which ills am I loading onto others, and do I want to be the person who does that?

This is not bureaucracy, nor a checklist to be ticked off. It is the innovation mesotes carried out into the world: the same keeping of measure that in the workshop decided between too little and too much now decides between the ill I bear and the one I shift onto others. The inventor who asks in this way will not find the pure solution every time – the prisoner's dilemma is too powerful, and the price is sometimes too high. But he will know what he is doing. And, more often than the cynic believes, he will find ways of really solving the problem instead of merely shifting it – for that is precisely the highest form of inventing: not to pass an ill on, but to make it smaller than it was.

VII. What this means for the theory

Responsibility is not an ingredient tacked onto inventing after the fact. It is the innovation mesotes in its final, outward-facing form. The inventor who skilfully outsources an ill has missed his measure, however clever his device may be – he has solved a problem by creating a bigger one and pushing it out of sight. The inventor who transforms it has hit the measure, even if it costs him more. The one is skill without wisdom; the other is the art this whole book has been about, carried through to the end.

With this the ethical aspect of the outlook is outlined – the first of the two great questions that open up beyond the workshop. The second is of a different kind. It does not ask whether the idea enters the world *well*, but *whether* it does so at all. For between the tested design and real impact lies a landscape full of chasms, in which most inventions sink before they are of use or harm to anyone. The most famous of these chasms bears a sombre

name – the valley of death – and it and its siblings are the subject of the next chapter.

Note on sources: This chapter takes further Dr. Lückert's first sentence (the sum of all ills is constant). The distinction between transformation and outsourcing, the grammar of profit maximisation and the illusion of neutral technology are developed by the authors in the essay "Die ausgelagerten Übel" ("The Outsourced Ills", 2026).